A conditional analytic framework for rank-one branched Kovalev–Lefschetz adiabatic limits

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Abstract

We set up a conditional analytic framework for a special case of [3], Conjecture 1, on adiabatic limits of coassociative Kovalev–Lefschetz fibrations with $K3$ fibres. For **branched rank-one admissible data**, quantitative fibrations whose monodromy on $H^2(K3;\mathbb{R})$ acts through a single root direction, whose period section is collar-affine and whose discriminant is a geometrically split unlink, we obtain three results.

First, an *unconditional* weighted edge Fredholm theory of the maximal-submanifold Jacobi operator in the Pacard–Mazzeo $\beta = -1$ window with σ-odd half-integer indicial structure and a cokernel-inversion certificate.

Second, a *conditional lifting scheme*: under three quantitative admissibility conditions (each verifiable by construction), the edge/implicit-function machinery produces closed positive 3-forms of G_2-type in the class $[\varphi_0] + R[B]$ for every R above a finite threshold, with coclosed defect of order $O(R^{-1})$.

Third, a *conditional torsion-free upgrade*: assuming in addition an anisotropic three-scale perturbation theorem (J), each such 3-form deforms to a torsion-free G_2-structure.

For a normalised datum $\mathcal{D}_0$ from a $K_7 \to S^3$ construction with $N = 77$ branch components, the source-side discharge constants are certified with structural provenance and $R_0(\mathcal{D}_0) \leq 4.9{\cdot}10^3$ is enclosed by outward-rounded interval arithmetic. The framework reduces the rank-one branched instance to (J) plus a two-slot external-structure pack $\mathsf{H}_{\mathrm{global}}$ (smooth topology of K_7 from cited classification results, and a geometric hyperkähler-with-periods upgrade); at the datum layer the closed small-torsion family closes without $\mathsf{H}_{\mathrm{global}}$.

Keywords: G_2-holonomy, Kovalev–Lefschetz fibrations, adiabatic limits, edge calculus, Pacard–Mazzeo, coassociative $K3$-fibrations, extended-domain Fredholm theory, certified numerics.

MSC 2020: 53C29, 53C10, 53C42, 35J75, 58J32.

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Claim boundary at a glance

Component	**Status**
Unconditional edge Fredholm theory (§4–5)	**Proved**
Datum-conditional lifting scheme (Theorem A)	Under (A_0)–(A_8) + $(\mathrm{E}, \mathrm{AR}, \mathrm{G})$-quant
Torsion-free upgrade	Conditional on (J) (L3, open)
Application to constructed K_7	Additionally under $\mathsf{H}_{\mathrm{global}}$ (C.1 topology + L4 geometric)
Numerical closure $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$	Certified at $\mathcal{D}_0$ (§9.2)
Analytic locks L1.6, L2 assembly	**Closed** (§11, OQ0-B, OQ0-C)
Geometric realisation lock L4	**Open** (OQ0-E)

Technical details supporting the numerical certification (§9), the Lean-verified combinatorial identity (§10), and the $K_7 \to S^3$ topology (Supplementary Appendix C) are collected in the accompanying supplement (Appendices A–F).

1. Introduction

1.1. The adiabatic G_2 programme

A central problem in special holonomy is the construction of metrics with holonomy the exceptional group G_2 on compact 7-manifolds. Beyond the resolutions of orbifolds [9] and the twisted connected sums of [11, 2], Donaldson proposed in [3] a fundamentally different mechanism: realise a G_2-manifold as an *adiabatic limit* of a co-associative fibration whose fibres are $K3$ surfaces.

Concretely, let B^3 be a 3-manifold and $\pi\colon M^7 \to B^3$ a fibration whose generic fibre is a $K3$ surface carrying a hyperkähler structure. A torsion-free G_2-structure adapted to π is governed, after rescaling the fibre metric by a small parameter ϵ, by a coupled system of six equations relating the fibrewise hyperkähler triple, a connection on the fibration, and the period map of the fibres; we recall this system, in the notation of [3, §2.1–2.2], in §3. As $\epsilon \to 0$ the system degenerates: by [3, Lemma 5] all but one of its equations become algebraically solvable, and the single surviving condition is that the period map

$$h\colon B^3 \longrightarrow \mathbb{R}^{3,19}/O(3,19;\mathbb{Z})$$

define a **maximal**, i.e. volume-maximising spacelike, submanifold of the period domain. [3, Prop. 3] then shows that any solution of the limit equation extends to a solution of the full system as a formal power series in ϵ.

Whether this formal series **converges**, for ϵ small, to a genuine torsion-free G_2-structure realising the cohomology class $[\varphi_0] + R[B]$ for R large is [3, Conjecture 1]. The conjecture is open in general, and is reiterated as an open problem in subsequent work on adiabatic limits of special-holonomy structures (e.g. [14, §1]). To the author's knowledge no instance of Conjecture 1 has been established.

> **Remark 1.1 (a distinct conjecture).** The theorem of [8] (with a uniqueness companion in [1]) proves a local case of the *Donaldson–Scaduto associative conjecture* [4], the existence of a pair-of-pants associative 3-fold in $X_{A_2} \times \mathbb{R}^3$. That is a statement about associative submanifolds, not about the adiabatic convergence of the G_2-structure itself, and is logically independent of Conjecture 1; it is analytically complementary to the present work.

1.2. Branched rank-one data

The difficulty of Conjecture 1 is concentrated at the discriminant. Where the $K3$ fibre acquires an ordinary double point, that is, over the discriminant locus $\Sigma \subset B^3$, a link along which a vanishing (-2)-cycle collapses, the period map h has a branch point of type $w^{3/2}$ in the transverse coordinate w, and its second derivatives blow up like $w^{-1/2}$. The maximal-submanifold equation is then a degenerate-elliptic problem on the branched double cover, and it is the analysis at the branch loci, not in the bulk, that obstructs convergence.

We single out a class of data for which this analysis can be carried through. Informally (the precise list of bounds $(A0)$–$(A8)$ is given in §2), a **branched rank-one admissible datum** $\mathcal{D} = (B^3, \pi, \Gamma, \Sigma, h)$ consists of a Kovalev–Lefschetz fibration together with:

- **(rank-one monodromy)** the monodromy representation acts on $H^2(K3;\mathbb{R}) \cong \mathbb{R}^{3,19}$ through reflections in a *single* root direction α_1, so that the Γ-invariant subspace has rank 21 (note $b_2(K3) = 22$; the rank-21 fixed space is the content of the rank-one hypothesis, not a statement about the Betti number);
- **(totally geodesic background)** the period section h has an affine totally geodesic background in the directions $\alpha_1^\perp$ orthogonal to the vanishing cycle, together with a controlled α_1-valued branched correction of leading order $w^{3/2}$;
- **(split discriminant)** $\Sigma = \Sigma_1 \sqcup \cdots \sqcup \Sigma_N$ is a *geometrically split N-component unlink*, with the components pairwise separated by a definite distance $d_{\min}$.

The fibration π is **fixed** throughout: $\mathcal{D}$ is a fixed datum, and ϵ rescales only the fibre metric, not π. The unknowns are the section (and the fibre corrections) *together with* $2N$ transverse displacements of the branch points of its branched jet, equivalently, small motions of the discriminant circles of the deformed section inside the fixed tubes U_i. This is Donaldson's formulation, in which the discriminant is part of the unknown ("regularity appears after moving the branch locus", [3, §3]); the displacements are carried in the analysis as prescribed $\mu = +1/2$ asymptotic data on the fixed tubes, and their geometric admissibility (splitness, positivity, smooth dependence) is Lemma 5.4.

These three conditions are exactly what reduce the global branched problem to a *single* local model, replicated with controlled coupling across the N components. The rank-one hypothesis makes the local branched model identical along and across the components; the totally geodesic hypothesis removes the bulk obstruction to leading order; and the split unlink decouples the components up to an $O(r_0/d_{\min})$ kernel-coupling correction.

1.3. Main results

Let $C^{k,\alpha}_\beta$ denote the weighted Hölder spaces on the branched double cover adapted to the edge model of §2.3 (Pacard–Mazzeo convention [18, 15]).

> **Theorem A (conditional extended-domain lifting scheme).** *Let $\mathcal{D}$ be a branched rank-one Kovalev–Lefschetz datum satisfying the admissibility conditions* (A0)–(A8) *together with the three geometric smallness conditions* (E-*quant*), (AR-*quant*), *and* (G-*quant*) *of §2. Then there exist $R_0(\mathcal{D}) < \infty$ and a family of closed positive 3-forms $\widetilde{\varphi}_R \in \Omega^3(M^7)$ of G_2-type, for $R \geq R_0(\mathcal{D})$, representing the cohomology class*
>
> $$[\widetilde{\varphi}_R] \;=\; [\varphi_0] \;+\; R\,[B] \;\in\; H^3(M^7;\mathbb{R}),$$
>
> *such that*
>
> $$\big\| d *_{\widetilde{\varphi}_R} \widetilde{\varphi}_R \big\|_{C^{0,\alpha}_1(M^7)} \;\to\; 0 \quad \text{as } R \to \infty,$$
>
> *with quantitative rate $O(R^{-1})$ explicit in the constants of* (A0)–(A8)*; and the anisotropic scale-invariant smallness conditions* (H_{C^0}), ($\mathrm{H}_{L^{14}}$), (H_{Rm}), ($\mathrm{H}_{\mathrm{fib}}$) *of §8.3 hold for all $R \geq R_0(\mathcal{D})$. If in addition Hypothesis* (J) *(the anisotropic three-scale refinement of [10, Th. 11.6.1], §8.3) holds, then each $\widetilde{\varphi}_R$ deforms, with a unique correction $\eta_R \in \Omega^2(M^7)$ in the Coulomb gauge*

$d^*\eta_R = 0$, $\eta_R \perp \mathcal{H}^2(M^7)$, to a torsion-free G_2-structure φ_R on M^7 in the same cohomology class.

The theorem is a *datum-conditional lifting scheme*: the admissibility conditions (A0)–(A8), (E-quant), (AR-quant), (G-quant) are quantitative bounds verifiable for an explicit construction (§9); (E), (AR), (G) are the three structural hypotheses of the scheme, reduced conditionally to Propositions 5.7, 3.2, 3.1 respectively (each conditional on a datum-level numerical margin); (J) is the sole remaining torsion-free structural hypothesis (the anisotropic three-scale perturbation theorem). **Two layers must be kept distinct.** At the *datum layer*, Theorem A is a statement *at a given admissible datum*: given $\mathcal{D}$ satisfying the admissibility list, the scheme produces closed positive 3-forms; nothing outside the admissibility list is consumed. At the *global-analytic layer*, Corollary B applies the datum-layer theorem to a specific total space K_7 constructed from $\mathcal{D}_0$; this application additionally consumes the global-hypothesis pack $\mathsf{H}_{\mathrm{global}}$ of Remark 1.1′ (two structural slots: Construction C.1 smooth topology + geometric K3-fibration upgrade; the analytic locks L1.6 and L2 assembly are closed (§11, OQ0-B, OQ0-C)). The torsion-free upgrade continues to require (J). Certified constants at $\mathcal{D}_0$ refer to sharp values at the local datum, never to a global theorem.

The conditions (A0)–(A8) are quantitative and verifiable: they bound the geometry of $\mathcal{D}$ (injectivity and curvature scales, the split distance $d_{\min}$, the Hölder norm of the bulk coefficient, the kernel coupling) by explicit constants. The threshold $R_0(\mathcal{D})$ is given in closed form in §6 in terms of these constants. Theorem A is therefore not an abstract existence statement but a *certifiable* one: for any datum whose constants one can estimate, one obtains an explicit admissible range of R.

> **Notation (two scales).** Throughout, we attach to $\mathcal{D}$ two logically independent positive scales. The *edge tube radius* $r_0(\mathcal{D})$ is a fixed geometric scale of the datum: the inner radius of the tubular neighbourhoods of the discriminant Σ on which the branched edge analysis of §§4–6 is carried out, bounded below by the injectivity and curvature radii of (B^3, g_B) (condition (A7)). The *parameter threshold* $R_0(\mathcal{D}) < \infty$ is the lower bound on the adiabatic parameter $R = \epsilon^{-1}$ above which the construction is valid (Theorem A, formula in §§6–7). The two are not to be conflated: for the normalised datum $\mathcal{D}_0$ one has $r_0(\mathcal{D}_0) = 10^{-2}$ (small, geometric) and $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ at the certified datum $\mathcal{D}_0$ (unconditional in the sense that no open analytic lock is consumed: L1.6 is closed theorem-grade via the gluing of §11 (OQ0-B), §11 and Suppl. App. F; a coarser fallback $\leq 1.03 \cdot 10^4$ at the discharged crude $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$ remains available for readers who prefer to bypass L1.6; the bound is stated at the datum layer of §1.3 and does *not* consume the global-hypothesis pack $\mathsf{H}_{\mathrm{global}}$) (large, the parameter threshold; its edge/IFT component alone would allow $\approx 10^2$, but the ϵ^3 tail-contraction condition of Theorem 7.1 is binding, see Corollary B).

The class $[B] \in H^3(M^7; \mathbb{R})$ in the statement is the pull-back $\pi^*[\omega_B]$ of the fundamental class of the base B^3, in agreement with [3, Conjecture 1]; the identification of the cohomology class of $\widetilde{\varphi}_R$ with $[\varphi_0] + R[B]$ is established directly from the closed-form ansatz in §3.3, without invoking any rescaling diffeomorphism on the K3 fibres.

We carry this out for a specific datum. Let $\mathcal{D}_0$ be the Kovalev–Lefschetz datum $K_7 \to S^3$ whose discriminant is a geometrically split 77-component round unlink in S^3 and whose monodromy is generated by a single (-2)-reflection along a primitive class $\alpha_1 \in NS(K3)$ of rank one (described in §9 and Supplementary Appendix C).

Corollary B (conditional numerical closure at $\mathcal{D}_0$). *Suppose the datum-level numerical bounds listed in §9.2 hold for the normalised datum $\mathcal{D}_0$: the edge constants* $(\mathrm{cond}(A_{\mathrm{bulk}}), K^{\mathrm{Maz}}_{\mathrm{Sch}}, \kappa_g)$, *the three quantitative admissibility conditions* (E-*quant*), (AR-*quant*), (G-*quant*) *discharged by Propositions 5.7, 3.2, 3.1 with margins*

$$\delta_{\mathrm{E}}(\mathcal{D}_0) \leq 0.021, \qquad r_{\mathrm{AR}}(\mathcal{D}_0) \leq 0.011, \qquad r_{\mathrm{G}}(\mathcal{D}_0) \leq 3.6 \cdot 10^{-4},$$

and the adiabatic-source norms $\|P_1(h_0)\|, \|P_2(h_0)\|, \|\mathcal{R}_\epsilon\|/\epsilon^3, \|DP_1\|$ *extracted from the bigraded ε-expansion of the Donaldson maximal-submanifold equations (Suppl. App. E.6). Then the contraction of Theorem 7.1 closes for every $R \geq R_0(\mathcal{D}_0) = \max(R_{\mathrm{edge}}, R_{\mathrm{threshold}})$, with $R_{\mathrm{edge}}(\mathcal{D}_0) \leq 10^2$ non-binding and*

$R_{\mathrm{threshold}}(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ unconditionally (lock L1.6 closed (OQ0-B), $\|\mathcal{G}_{\mathrm{aug}}\| \leq 56.3$ theorem-grade);

a coarser fallback $\leq 1.03 \cdot 10^4$ at the discharged crude $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$ remains available. The construction of the closed small-torsion part of [3, Conjecture 1] then holds for $\mathcal{D}_0$ at every $R \geq R_0(\mathcal{D}_0)$, and, assuming in addition Hypothesis (J), *the full torsion-free part holds at every such R.*

Remark 1.1″ (mechanism behind the headline and the fallback). The headline consumes the theorem-grade $\|\mathcal{G}_{\mathrm{aug}}\| \leq 56.3$ from L1.6 (OQ0-B); the coarser fallback uses the discharged crude $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$ and does not consume $\mathsf{H}_{\mathrm{global}}$. The binding component in both readings is the ϵ^3 tail-contraction condition $4\|\mathcal{G}_{\mathrm{aug}}\|^2 C_{\mathrm{nl}}\,\mathrm{remainder\_R3}\,\epsilon^3 < 1$ of Theorem 7.1: headline $R^{\mathrm{theorem}}_{\mathrm{threshold}}(\mathcal{D}_0) \approx 4883$ (≤ 4887 at the outward-displayed remainder $\leq 1.38 \cdot 10^7$), coarser fallback $R^{\mathrm{unc}}_{\mathrm{threshold}}(\mathcal{D}_0) \approx 10242$. Both sit about 9% below the like-for-like power-counting reconstruction (verified by supplementary interval script, Data availability).

Remark 1.1‴ (certified source-side constants). The universal source-side constants of Propositions 3.1 and 5.7 are certified at $\mathcal{D}_0$: $C_{\mathrm{src}} = 27/16$ (cubic σ-odd, Suppl. App. E.1), $C_{\mathrm{nl}} = 2/3$ (Simons-type / collinearity, Suppl. App. E.2), $C_{\mathrm{link}} \leq |c_0|^2 \beta_{\mathrm{link}} \leq 1$ (impedance match + parity, Suppl. App. E.3), and K_H^{K3} is reduced to the K3 scalar spectral gap (Zhong–Yang, Suppl. App. E.5, up to the closed-form value of that gap).

The numerical closure is worked in the Mazzeo edge-Schauder regime, which is the operative one for a branched-edge operator; the non-degenerate (Gilbarg–Trudinger) Schauder bound is too coarse and is not used. The edge constants (κ_g, $\mathrm{cond}(A_{\mathrm{bulk}})$, the Mazzeo edge constant) are $O(1)$ and are enclosed by interval arithmetic in §9 / Supplementary Appendix D. We stress that $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ (coarser fallback $\leq 1.03 \cdot 10^4$ at the discharged crude $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$) is a *conditional* threshold: it depends on the adiabatic-source norms and the effective edge-Schauder constant, which are inputs listed in §9.2 rather than consequences of $(A0)$–$(A8)$ alone; a fully unconditional "certified threshold" would require an effective parametrix for $\mathcal{J}_h$ on $\mathcal{D}_0$ giving $K^{\mathrm{Maz}}_{\mathrm{Sch}}$ and the source norms directly.

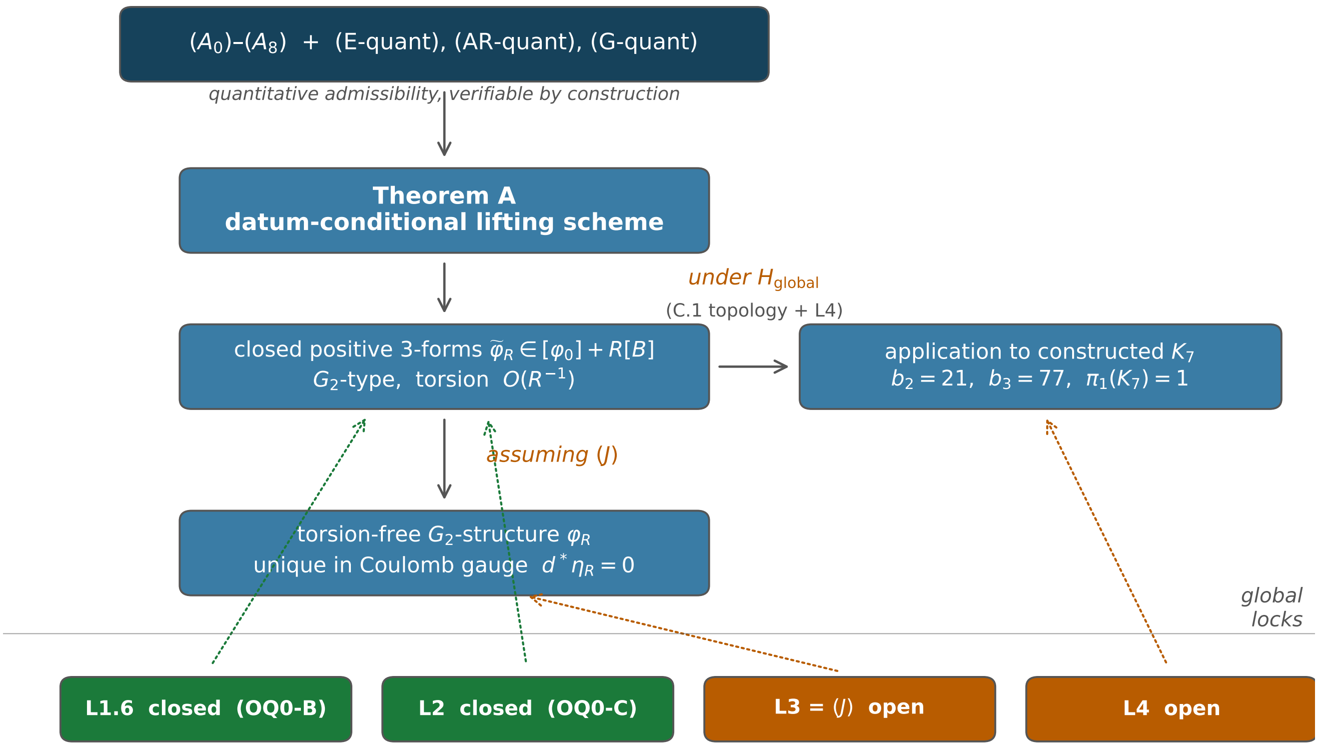

Figure 1. Logical DAG of the conditional analytic framework. Datum-layer inputs (A_0)–(A_8) + (E-*quant*) + (AR-*quant*) + (G-*quant*) *feed Theorem A, which produces the closed small-torsion family* $\widetilde{\varphi}_R$*; the torsion-free upgrade* φ_R *additionally consumes* (J) *(lock L3), and application to the constructed* K_7 *additionally consumes* $\mathsf{H}_{\mathrm{global}}$ = *C.1 topology + geometric upgrade L4. Locks L1.6 (OQ0-B) and L2 assembly (OQ0-C) are closed and feed the certified numerical threshold; locks L3 and L4 remain open.*

Table 1 (Claim boundary, panorama at a glance). A summary of the paper's results, their layer, their conditional status at the normalised datum $\mathcal{D}_0$, and the in-paper reference. The four global locks in the lower block are the technical content of Remark 1.1′ below.

Result	Layer	Status at $\mathcal{D}_0$	Reference
K_7 smooth compact, $\pi_1(K_7) = 1$, $(b_2, b_3) = (21, 77)$	topology	**constructed** (conditional on standard classification results [19], [11])	Construction C.1, Suppl. App. C
Extended-domain Fredholm theorem (edge, $\mu = +1/2$ Wronskian pairing)	edge analysis	**proved** unconditionally	Theorem 5.5, §5.4
s-uniformity reduction $\mathfrak{M}\colon \mathbb{R}^{2N} \to \mathbb{R}^{2N}$, $\lVert\mathfrak{M}^{-1}\rVert \le 2.76$	edge analysis	**proved** unconditionally	Theorem 5.10, §5.7

Result	Layer	Status at $\mathcal{D}_0$	Reference
Global weighted parametrix + outer matching (Mazzeo regime)	analytic	**proved** unconditionally	§6, Lemma 6.2
Nonlinear closure of $\widetilde{\varphi}_R$ (closed small-torsion 3-form)	analytic	**proved** conditional on $(A0)$–$(A8)$ + (E-quant) + (AR-quant) + (G-quant)	Theorem 7.1, §7
Torsion-free upgrade $\widetilde{\varphi}_R \to G_2$-structure	analytic	**conditional on** (J)	Theorem 8.3, §8
Strict holonomy $\mathrm{Hol}(g_R) = G_2$ at $\mathcal{D}_0$	analytic	**conditional on** (J) (via $\pi_1(K_7) = 1$)	Corollary 8.4, §8
(G-quant) discharge: $r_{\mathrm{G}}(\mathcal{D}_0) \leq 3.6 \cdot 10^{-4}$	numerical at $\mathcal{D}_0$	**certified**, margin $\sim 1389\times$ (uses the theorem-grade $\lVert\mathcal{G}_{\mathrm{aug}}\rVert \leq 56.3$ from L1.6 closure; $\sim 151\times$ at the coarser fallback $\lVert\mathcal{G}_{\mathrm{aug}}\rVert \leq 171$)	Prop. 3.1, Suppl. App. E.1–E.2, §9.2, Suppl. App. F
(E-quant) discharge: $\delta_{\mathrm{E}}(\mathcal{D}_0) \leq 0.021$	numerical at $\mathcal{D}_0$	**certified**, margin $\sim 24\times$	Prop. 5.7, Suppl. App. E.3–E.4, §9.2, Suppl. App. F
(AR-quant) discharge: $r_{\mathrm{AR}}(\mathcal{D}_0) \leq 0.011$	numerical at $\mathcal{D}_0$	**certified**, margin $\sim 48\times$; residual: $\lambda_1^{\mathrm{scalar}}(K3)$ closed form not available	Prop. 3.2, Suppl. App. E.5, §9.2, Suppl. App. F
$R_0(\mathcal{D}_0)$ threshold	numerical at $\mathcal{D}_0$	$\leq 4.9 \cdot 10^3$ unconditional (L1.6 closed (OQ0-B)); coarser fallback $\leq 1.03 \cdot 10^4$ at the discharged crude $\lVert\mathcal{G}_{\mathrm{aug}}\rVert \leq 171$	Corollary B, Suppl. App. E.6, §9.2, Suppl. App. F, §11
(L1) global maximal-section theorem, Π_{obs} identification, κ_E	global lock	**discharged** modulo transfer layers (Wronskian pairing, rank-one rigidity)	§5, Suppl. App. E.1, Rem. 1.1′

Result	Layer	Status at $\mathcal{D}_0$	Reference
(L1.6) quantitative $\|\mathcal{G}_{\mathrm{aug}}\| \leq 56.3$ theorem	global lock	**closed** (OQ0-B): $K_{\mathrm{Sch}} \leq 16/3$ via 77-collar interval evaluation, $q_{\mathrm{coeff}} \leq 1/4$ with slack 33–56%; buys $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ unconditionally	Rem. 1.1′, §11
(L2) convergent adiabatic reconstruction theorem	global lock	**closed** (OQ0-C): four Neumann slots discharged + q_{AR} audited, $\mathrm{Lip}(T_{\mathrm{AR}}) \leq 8.20 \cdot 10^{-3} < 1/2$ on the product Banach space X_{AR}, Banach fixed-point	Rem. 1.1′, §11
(L3) Hypothesis (J): anisotropic three-scale perturbation theorem	global lock	**open**, subject of a companion paper	§11.1, §11
(L4) Hyperkähler K3-fibration on the topological K_7 with prescribed rank-one monodromy and collar-affine period section (geometric upgrade of Construction C.1)	global lock	**open**, the smooth 7-manifold topology of Construction C.1 does not automatically produce the geometric $K3$-family with prescribed periods; [12] gives plausibility (Hopf-link discriminant, related monodromy), not this exact datum	Suppl. App. C.1, Rem. 1.1′, §11

Reader's rule. The four top rows are the paper's unconditional structural content. Rows 5–7 attach explicit hypotheses in the statement. Rows 8–11 are numerical certifications at the reference datum $\mathcal{D}_0$ carried out with certified constants (Supplementary Appendix E) and outward-rounded arithmetic (Supplementary Appendix D, Supplementary Appendix F). Rows 12–16 are the four global locks: two closed analytic modules (L1.6, L2 assembly; see §11, OQ0-B, OQ0-C) and two open items, the torsion-free perturbation hypothesis L3=(J) and the geometric-realisation upgrade L4 of Construction C.1, whose resolution would promote the datum-conditional scheme of Corollary B into an unconditional existence theorem on K_7.

Remark 1.1′ (claim boundary, four global locks, technical drill-down). The quantitative closure of Corollary B is local to $\mathcal{D}_0$ and sits below the four global locks whose panorama is given in Table 1: two now-closed analytic modules (L1.6, L2 assembly) and two open items

(L3=(J), L4 geometric realisation). The technical mechanism and per-component status:

Lock	Component	Status	Key bound / mechanism
(L1) global maximal-section theorem	Π_{obs} PDE identification	**discharged** (modulo transfer layers)	$\mathrm{coker}(\mathcal{J}_h^{\mathrm{ext}}) = \mathbb{R}^{2N}$ exactly, $\Pi_{\mathrm{obs}}^{\mathrm{PDE}} = \Pi_{\mathbb{R}^{2N}}$: Schur reduction onto the $\mathfrak{M}$-block + sector exhaustion (two-window splitting, interface closure $x \coth x + x \geq 1$, exterior coercivity) + cokernel-inversion certificate via the $\mu = +1/2$ Wronskian pairing against the adjoint representatives w_i (Lemma 5.2 / Theorem 5.10), $\|\mathfrak{M}^{-1}\| \leq 2.76$; holds already at the conservative envelope

Lock	Component	Status	Key bound / mechanism
	κ_E (parametrix commutator constant)	**discharged**	$\kappa_E \le \kappa_g(1+\epsilon_0) \le 1.02$ from rank-one rigidity ($Df_0 = 0 \Rightarrow \rho' = A_{\mathrm{loc}}^{-1}\kappa_g g'$ exactly) and Frenet–Serret $\|g'\|_\infty = 2$; conservative envelope ≤ 3 re-derived as $\mathrm{cond}(A_{\mathrm{bulk}})\,\kappa_g$; certified $q_{\mathrm{out}}^{\mathrm{sharp}} = 0.526$. Normalisation-stable: in the ratio $\|\rho'\|/\|\rho\| = \kappa_g\|g'\|_\infty/\|f_0 + \kappa_g g\|$ the scalar A_{loc} cancels (invariant under the $\mu = +1/2$ Wronskian renormalisation of §5.4); denominator bounded below by $\|f_0 + \kappa_g g\|_{C^0} \ge \lvert f_0\rvert \ge 27/8$ (Lemma 5.9 mean-value on the $q = 0$ Fourier part with $C_{\mathrm{src}} = 27/16$), giving the sharp $\kappa_E \le 0.61$, a fortiori the headline ≤ 1.02; collar-twist check $\kappa_E\,\mathrm{cond}(A_{\mathrm{bulk}}) \le 1.39 < 3.4$ (supplementary interval-arithmetic script; degrades linearly at small $\lvert c_0\rvert$)

Lock	Component	Status	Key bound / mechanism
	quantitative $\lVert\mathcal{G}_{\mathrm{aug}}\rVert$ theorem at ≤ 56.3	**closed** (L1.6, OQ0-B)	$K_{\mathrm{Sch}} \leq 16/3$ via $(4/3)\cdot 3\cdot(4/3)$: frozen normal multiplier $\leq 4/3$ + Hölder/cutoff overhead ≤ 3 + 77-collar interval evaluation certifying $q_{\mathrm{coeff}} = 0.168/0.110 \leq 1/4$ with slack 33–56%; downstream $\lVert\mathcal{G}_{\mathrm{aug}}\rVert_{\mathrm{crude}} \leq 25.945/24.780 \ll 56.3$, headline $R_0(\mathcal{D}_0) \leq 4.9\cdot 10^3$ (see OQ0-B)
(L2) convergent adiabatic reconstruction	product-max normalization + scalar majorant	installed	$K_{\mathrm{AR}}^{\mathrm{prod}} = K_H^{K3}K_F = 2079/2000$; $q_{\mathrm{AR}} = 1.035\times 10^{-3}$ at $R_{\mathrm{AR}} = 4000$ (majorant recomputed)
	slot `q_comm_FH_Gf`	**discharged**	$\lVert[F_H, G_f]\rVert \leq 3(K_H^{K3})^2K_F^{(1)} + 4K_H^{K3}K_F \leq 5.57 \ll 250$ via the bigraded identities $\{d_f, F_H\} = -d_H^2$, $[F_H, \Delta_f] = [d_f^*, d_H^2] - [d_f, \{F_H, d_f^*\}]$ (V1–V6)
	slot `q_projection_residual`	**discharged**	$C_{\mathrm{proj}} \leq K_H^{K3} + 2(K_H^{K3})^2 c_\Delta K_g^{(1)} \leq 3.26 \ll 250$: $d_f r = 0$ unconditionally, reference harmonic content killed by $M_\epsilon(h_\epsilon) = 0$, Kato projector variation; does *not* consume the (L1) identification (W1–W6)

Lock	Component	Status	Key bound / mechanism
	slot `q_hodge_uniform`	**discharged** (two parts)	$\sup_x \lVert G_f(x)\rVert \le 9/20$ by fibrewise Zhong–Yang + $D_{\max} = 7/5$ over the compact fibre family at $\mathcal{D}_0$ (datum input, A5 envelope; not moduli-uniform) (margin 0.98%; orbifold benchmark ≈ 0.19 is the real slack); variation cost $\le 1.57 \ll 250$ by two-metric resolvent telescoping (X1–X6)
	slot `q_gauge_transfer`	**discharged**	reference cost $= 0$ at all orders ($C_{\mathrm{conn},k} = 0$, Donaldson gauge = defining equation); transfer cost $(2 + 2K_v)c_\omega(K_H^{K3} + K_v) \le 18.28 \ll 250$ (Y1–Y6)
	contraction layer, total	**covered**	$q_{\mathrm{AR}} + q_{\mathrm{Neumann}}^{\mathrm{theorem}} = 8.2 \times 10^{-3} \ll 1$, a factor 31 below the candidate headline 0.251
	product Banach space + assembled reconstruction theorem	**closed** (OQ0-C)	on $X_{\mathrm{AR}} = X_\omega \times X_\lambda \times X_\mu \times X_\Theta$ with product-max norm, $\mathrm{Lip}(T_{\mathrm{AR}}) \le 8.20 \cdot 10^{-3} < 1/2$ ($\sim 60.9\times$ margin), Banach fixed-point gives a unique convergent reconstruction $(h_\varepsilon^*, \omega_\varepsilon^*, \lambda_\varepsilon^*, \mu_\varepsilon^*, \Theta_\varepsilon^*)$ (see OQ0-C)
(L3) Hypothesis (J)	anisotropic three-scale perturbation theorem	**open**	companion paper

Lock	Component	Status	Key bound / mechanism
(L4) geometric upgrade of C.1	hyperkähler K3-fibration on the topological K_7 with rank-one monodromy along the 77-unlink and collar-affine period section h consistent with $(A2_{\text{collar}})$–$(A5)$	**open** (structural)	Construction C.1 gives the smooth topology only; [12] is the closest existing template (Hopf-link discriminant, different topology, see OQ0-E)

The Level-E existence statement on the constructed K_7 is conditional on the two **open** rows above: the geometric structural upgrade (L4) and the torsion-free perturbation hypothesis (J) (L3). The L1 identification and κ_E discharges together with the L1.6 and L2 assembly closures (OQ0-B, OQ0-C) leave L4 and (J) as the only structural gaps.

The global hypothesis pack $\mathsf{H}_{\text{global}}$. For the end-to-end existence statement on K_7, we bundle the two external structural slots above into a single hypothesis pack; (J) (L3) stays separate as the sole torsion-free hypothesis.

$$\mathsf{H}_{\text{global}}(\mathcal{D}_0) \;:=\; \Big\{ \underbrace{\text{Construction C.1 smooth topology}}_{[19],\,[11]} \wedge \underbrace{\text{hyperkähler K3-fibration with prescribed period section}}_{\text{geometric upgrade on the topological } K_7} \Big\},$$

two external structural slots *not* consumed by the principal clause of Corollary B at a fixed admissible datum $\mathcal{D}_0$ (that clause uses the theorem-grade $\|\mathcal{G}_{\text{aug}}\| \le 56.3$ of L1.6, or the coarser discharged crude ≤ 171, together with the theorem-grade constants of Propositions 3.1, 3.2, 5.7). $\mathsf{H}_{\text{global}}$ upgrades the datum-level scheme on an admissible datum into a genuine global-analytic theorem on the constructed K_7: the first slot handles the smooth 7-manifold topology via classification; the second slot handles the geometric K3-fibration (see OQ0-E for the residual gap). The torsion-free upgrade further requires (J) independently.

Remark 1.2 (scope). Theorem A asserts, under the admissibility list $(A0)$–$(A8)$ + (E-quant) + (AR-quant) + (G-quant) alone (all verifiable by construction), convergence to a *closed* G_2-structure of small torsion, and, assuming in addition (J), to a torsion-free one. We do not address the strict equality $\mathrm{Hol} = G_2$ versus $\mathrm{Hol} \subseteq G_2$, which follows for $\mathcal{D}_0$ (given (J)) from $\pi_1(M^7) = 1$ and [10, §10.6.5, §11.1] but which we do not pursue for general $\mathcal{D}$. See §11.

1.4. Strategy

The proof has four movements: a reduction of Donaldson's system with a three-level stratification of the section, an extended-domain Fredholm analysis constructing the maximal background at the branch loci, a global weighted parametrix on the branch tubes, and an anisotropic three-scale perturbation closure.

Reduction (§3). We recall Donaldson's six-equation system and the $\epsilon = 0$ degeneration, isolating the maximal-submanifold equation $m(h) = 0$ as the only non-trivial limit, and we stratify the section into a *local seed* $\bar{h}$, a *global maximal background* $h_0 = \bar{h} + v_0$ with $m(h_0) = 0$ (whose global existence is discharged as Proposition 3.1 under (G-quant)), and an *adiabatic lifting* h_ϵ. Solving the reduced equation $\mathcal{M}_\epsilon(h) = 0$ is closed to a G_2-structure through an adiabatic reconstruction map (Proposition 3.2, discharged under (AR-quant)), not through the false implication $m(h) = 0$. On the collar-affine model the bulk is solved exactly, so the analytic content lives in tubular neighbourhoods of the N branch components. There the linearised equation is governed by the Jacobi operator $\mathcal{J}_h = D_h m$ of the maximal section, a rank-19 Laplace-type operator on B^3 acting on the normal bundle N_h of the spacelike 3-plane in $\mathbb{R}^{3,19}$, which along the branched direction α_1 is a branched edge operator with half-integer σ-odd indicial roots $\pm 1/2, \pm 3/2, \pm 5/2, \dots$ and Pacard $\beta = -1$ Fredholm window $(1/2,\, 3/2)$ (base downstairs convention §2.3). This is to be distinguished from Donaldson's *fibrewise* operator $L_1 = d^+ \circ A \circ d$ of [3, eqs. 18–19], which resolves the fibre corrections and carries no transverse edge modes.

Extended-domain Fredholm theorem (§4–§5). On the tubular neighbourhood of $\Sigma_i \simeq S^1$, $\mathcal{J}_h = A_{\text{bulk}} \cdot \Delta_w + \mathcal{R}$ acts on the σ-odd half-integer angular sector between Pacard–Mazzeo polyhomogeneous weighted Hölder spaces $X_\beta \to Y_{\beta-2}$ with $\beta \in (1/2,\, 3/2)$. The Fredholm cokernel of $\mathcal{J}_h$ on X_β is finite-dimensional; it parametrises via the polyhomogeneous trace at the lower indicial root $\mu = -1/2$ (the Pacard $\beta = -1$ root) a finite-dimensional subspace of the indicial trace bundle on Σ, identified as $\mathbb{R}^{2N}$ by the Dirichlet-to-Neumann construction of §5.5. The σ-odd displacement modes

$$\phi_R^{(i)} = \rho^{1/2}\, \cos(\varphi/2), \qquad \phi_I^{(i)} = \rho^{1/2}\, \sin(\varphi/2)$$

serve as prescribed asymptotic data at the regular indicial root $\mu = +1/2$ in the angular modes $m = \pm 1/2$: they lie *outside* X_β for every β in the window (a genuine augmentation), and they are the unique σ-odd partners of the $\rho^{-1/2}$ cokernel modes with non-vanishing Wronskian pairing ($a + b = 0$, Lemma 5.2). Geometrically they displace the branch points of the seed $\mathrm{Re}(z^3)\, \alpha_1$, the free-boundary mechanism in which the discriminant is part of the unknown, Lemma 5.4. The cubic modes $\rho^{3/2} \cos(\varphi/2), \rho^{3/2} \sin(\varphi/2)$ of the seed jet, and the cubic harmonic mode $w^{3/2}\, e^{i3\varphi/2} \in \ker \Delta_w$ (angular mode $m = \pm 3/2$), remain inside X_β and are Wronskian-blind to the cokernel ($a + b = 1$; Remark 5.2).

The extended domain $X_\beta^{\text{ext}} = X_\beta \oplus \mathcal{D}_{1/2}^{\text{disp}}$ adjoins to X_β the displacement layer in the modes $\phi_R^{(i)}, \phi_I^{(i)}$. Under $(A1)$ rank-one rigidity + $(A2_{\text{collar}})$, the seed residual $m(\bar{h})$ at Σ_i is s-uniform along each circle $\Sigma_i \simeq S^1$, so that the displacement datum v_0 that cancels it, thereby *constructing the maximal background* $h_0 = \bar{h} + v_0$, reduces to a finite-dimensional space $\mathbb{R}^{2N}$ (the reduction to $\mathbb{R}^{2N}$ using the discharged global outer coercivity of Proposition 5.7, conditional on the quantitative admissibility (E-quant)). The cokernel-inversion certificate identifies the reduced cokernel map as $A_{\text{loc}} + E_{\text{geom}} + E_{\text{link}}$, where the diagonal block $A_{\text{loc}} = (\mu_+ - \mu_-)\, A_{\text{bulk}}(\alpha_1, \alpha_1)\, I = A_{\text{bulk}}(\alpha_1, \alpha_1)\, I$ is a positive scalar multiple of the identity (extracted via the Wronskian/Green pairing with the adjoint representatives, Lemma 5.2), and the perturbations $E_{\text{geom}}, E_{\text{link}}$ are controlled by the geodesic curvature κ_g of the components and by the dipole-coupled Newton kernel between Σ_i and Σ_j at distance $d_{\min}$. The augmented nonlinear implicit-function argument closes the cancellation with no small divisor and no loss of derivative.

Weighted parametrix on the branch tubes (§6). With the cokernel cancelled by §5, the residual nonlinear source is resolved by a global weighted elliptic inversion of $\mathcal{J}_h$ in $C_\beta^{k,\alpha}$ with $\beta \in (1/2,\, 3/2)$. On each tube neighbourhood of Σ_i, $\mathcal{J}_h$ differs from its leading-order indicial model by a perturbation with indicial roots unchanged, controlled by κ_g. The tubes are matched to the affine bulk across their outer boundaries, where the bulk is source-free, so the matching costs a single factor of the Schauder constant. The combinatorial identity $\sum_{k\geq 0}|\binom{3/2}{k}| = 3$, verified in Lean 4, bounds two auxiliary bulk constants. (The binomial parameter 3/2 refers to the base exponent of the branched section $w^{3/2}$ in the local model $(A4)$.)

Anisotropic three-scale perturbation closure (§7–§8). The nonlinear lifting is closed by an augmented implicit-function argument (§7), and the reduced solution is reconstructed to a closed G_2-structure $\widetilde{\varphi}_R$ via Proposition 3.2 (adiabatic reconstruction discharge). The family (M^7, g_ϵ) is a *collapsing* limit with three length scales, base ~ 1, fibre $\sim \epsilon$, and Eguchi–Hanson [5] core $\sim \epsilon^2$, so that $\mathrm{inj}_{\min} \sim \epsilon^2$ and $\|\mathrm{Rm}\|_{\max} \sim \epsilon^{-4}$ at the core. We prove *unconditionally* that the coclosed defect is $O(\epsilon)$: at the ϵ^2-neck the connection-torsion piece is tamed by the finite-order Picard–Lefschetz monodromy (bounded Gauss–Manin residue), the smoothness of M^7, and the G_2-natural homothety. The deformation to a torsion-free structure requires an anisotropic three-scale refinement of [10, Th. 11.6.1], isolated as Hypothesis (J): the native bounded-geometry ratios of Joyce's theorem blow up under collapse, and the anisotropic refinement controlling the three scales separately is not available as a citable theorem in the exact form needed.

1.5. Organisation

The single standing hypothesis. Beyond the quantitative admissibility conditions $(A0)$–$(A8)$, the outer-coercivity smallness (E-quant), the reconstruction smallness (AR-quant), and the global-background smallness (G-quant) (all in §2.2; the corresponding outer coercivity, reconstruction map, and global maximal background are *propositions*, not hypotheses, see Propositions 5.7, 3.2, and 3.1), the scheme rests on a single structural hypothesis (for the datum-level scheme; the end-to-end existence statement on the constructed K_7 additionally rests on the two-slot pack $\mathsf{H}_{\mathrm{global}}$ of Remark 1.1′, with the torsion-free upgrade continuing to require (J) independently): - (J) *(anisotropic three-scale perturbation, §8.3)*, an anisotropic refinement of [10, Th. 11.6.1] deforming a small-torsion closed G_2-structure to a torsion-free one in the three-scale collapsing geometry.

The closed small-torsion family is *datum-conditional* (i.e. unconditional in $\mathsf{H}_{\mathrm{global}}$ once admissibility is granted at the datum layer); (J) is needed only for the torsion-free upgrade. The analytic content of §§3–7 is the datum-conditional reduction of the closed small-torsion construction to admissibility, and of the torsion-free construction to (J). $\mathsf{H}_{\mathrm{global}}$ enters only when Corollary B is applied to the specific total space K_7 of Construction C.1.

§2 states the admissibility conditions $(A0)$–$(A8)$ together with the three geometric smallness conditions (E-quant), (AR-quant), (G-quant) and the local branched model, and verifies non-vacuity (at the two layers of Proposition 2.2). §3 reduces Donaldson's system to the maximal-submanifold equation, identifies the Jacobi operator $\mathcal{J}_h$, and discharges the global maximal background as Proposition 3.1 (previously (G)) and the adiabatic reconstruction map as Proposition 3.2 (previously (AR)); the cohomological identification of the family $\widetilde{\varphi}_R$ with the class $[\varphi_0] + R[B]$ is established here, directly from the closed-form ansatz. §4 develops the weighted edge theory of $\mathcal{J}_h$ on the smooth base downstairs (Pacard $\beta = -1$ window $(1/2, 3/2)$): the indicial family, the normal operator, the Fredholm cokernel at the lower indicial root

$\mu = -1/2$, the role of the normal-bundle metric A_{bulk} on N_h. §5 constructs the maximal background at $\epsilon = 0$: the extended-domain Fredholm theorem with prescribed $\mu = +1/2$ branch-displacement data (Wronskian pairing, free-boundary mechanism), and the Dirichlet-to-Neumann reduction of the cokernel to $\mathbb{R}^{2N}$ under $(A1)+(A2_{\mathrm{collar}})+$(E-quant), with cokernel-inversion certificate $\sigma_{\min}(A_{\mathrm{loc}}) = A_{\mathrm{bulk}}(\alpha_1, \alpha_1) > 0$; the global outer coercivity underlying the reduction is discharged as Proposition 5.7. §6, the analytic core, constructs the global weighted parametrix on the branch tubes and proves the matching lemma to the affine bulk. §7 closes the nonlinear problem by an augmented implicit-function theorem and reconstructs $\widetilde{\varphi}_R$ via Proposition 3.2. §8 sets up the anisotropic three-scale collapsing geometry, bounds the torsion at rate $O(R^{-1})$ (unconditional), and isolates the perturbation theorem as Hypothesis (J). §9 instantiates the normalised datum $\mathcal{D}_0$ and gives the conditional numerical closure of Corollary B. §10 records the Lean verification of the combinatorial identity as a reproducibility certificate, and §11 collects open questions. Appendices contain the edge-calculus conventions, the Lean statements, the $K_7 \to S^3$ construction, and the interval-arithmetic enclosures.

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2. Admissible branched rank-one data

This section makes precise the class of Kovalev–Lefschetz fibrations to which Theorem A applies. The nine conditions $(A0)$–$(A8)$ are stated as quantitative bounds with explicit constants $C_i(\mathcal{D})$; for the normalised datum $\mathcal{D}_0$ of §9, all C_i are enclosed by interval arithmetic.

2.1. The Kovalev–Lefschetz setup

A *Kovalev–Lefschetz fibration* $(B^3, \pi\colon M^7 \to B^3, \Gamma, \Sigma, h)$ consists of:

- A closed oriented 3-manifold B^3 (the *base*);
- A smooth proper surjection $\pi\colon M^7 \to B^3$ (the *fibration*), whose generic fibres are $K3$ surfaces;
- A *monodromy representation* $\Gamma \subset \mathrm{Aut}(H^2(K3;\mathbb{Z}))$, encoding the parallel transport of H^2 along loops in $B^3 \setminus \Sigma$;
- A *discriminant locus* $\Sigma \subset B^3$, a smooth closed 1-submanifold, along which the $K3$ fibre acquires an ordinary double point (an "ODP", vanishing (-2)-cycle);
- A *period section* $h\colon B^3 \setminus \Sigma \to \mathbb{R}^{3,19}/O(3,19;\mathbb{Z})$, mapping each point of $B^3 \setminus \Sigma$ to the marked Hodge structure of the corresponding $K3$ fibre.

The fibre cohomology $H^2(K3;\mathbb{R}) \cong \mathbb{R}^{3,19}$ carries the canonical Hodge form of signature $(3, 19)$, and the period section h is a spacelike map into the period domain (its image is a spacelike 3-fold in $\mathbb{R}^{3,19}$ at every regular point).

2.2. The admissibility conditions $(A0)$–$(A8)$

Definition 2.1 (branched rank-one admissible datum). *A Kovalev–Lefschetz fibration $\mathcal{D} = (B^3, \pi, \Gamma, \Sigma, h)$ is called a* branched rank-one admissible datum *if the following nine con-*

ditions hold, with explicit constants $C_i(\mathcal{D}) > 0$.

$(A0)$ **Topology.** *B^3 is compact, oriented, simply connected ($\pi_1(B^3) = 1$). The inverse image $\pi^{-1}(\Sigma) \subset M^7$ is a smooth 4-submanifold (each branch component is a smooth S^1-family of ODPs).*

$(A1)$ **Rank-one monodromy.** *The Γ-invariant subspace $H^2(K3;\mathbb{R})^\Gamma$ has rank 21. Equivalently, all monodromy reflections of Γ share a single (-2)-root direction $\alpha_1 \in NS(K3) \subset H^2(K3;\mathbb{Z})$ up to sign: Γ is generated by the Picard–Lefschetz reflections in α_1, one for each component of Σ. (Note $b_2(K3) = 22$; the rank-21 fixed space is the content of the rank-one hypothesis, not a statement about the Betti number.)*

$(A2_{\text{collar}})$ **Collar-affine branched model.** *On each tube U_i of the discriminant, the local seed decomposes as $\bar{h}|_{U_i} = h^{(i)}_{\text{loc,aff}} + h^{(i)}_{\text{loc,branch}}$, where $h^{(i)}_{\text{loc,aff}} : U_i \subset \mathbb{R}^3 \to \mathbb{R}^{3,19}$ is a constant-coefficient affine map into the $\alpha_1^\perp$-direction (a spacelike 3-plane in $\mathbb{R}^{3,19}$) satisfying the maximal-submanifold equation $m(h^{(i)}_{\text{loc,aff}}) \equiv 0$ on $U_i \backslash \Sigma_i$, and $h^{(i)}_{\text{loc,branch}}$ is the α_1-valued branched correction of $(A4)$. This is a purely local (collar) hypothesis and does not assert a global flat affine structure on B^3.*

The existence of a *global* maximal background h_0 on $B^3 \backslash \Sigma$, a section with $m(h_0) = 0$ everywhere, matching each collar model $\bar{h}|_{U_i}$ up to the displacement datum v_0, is a genuine global PDE problem on the compact base, not a consequence of the collar-affine models. In the present version we discharge it as **Proposition 3.1** (§3.1) under the quantitative admissibility condition (G-quant) (§2.2), combining rank-one monodromy $(A1)$, simple connectivity $(A0)$, and the extended-domain Fredholm theory of §5.

$(A3)$ **Geometrically split discriminant.** *$\Sigma = \Sigma_1 \sqcup \Sigma_2 \sqcup \cdots \sqcup \Sigma_N$ is a geometrically split N-component unlink in B^3: each Σ_i is an embedded circle, pairwise unlinked, and there exist disjoint tubular neighbourhoods $U_i \supset \Sigma_i$ with*

$$d_{\min}(\mathcal{D}) \;:=\; \min_{i \neq j} d(\Sigma_i, \Sigma_j) \;>\; 0.$$

$(A4)$ **Uniform local model.** *On each tube U_i, there exist branched double-cover coordinates $(s, z) \in S^1 \times \mathbb{D}^2$ with $z^2 = w$ the transverse base coordinate on B^3, such that the period section has the local expansion*

$$h|_{U_i \backslash \Sigma_i} = \; v_0^{(i)}\, t + v_1^{(i)}\, \operatorname{Re} w + v_2^{(i)}\, \operatorname{Im} w + c_0^{(i)}\, w^{3/2}\, \alpha_1 + O(w^2),$$

with $v_0, v_1, v_2 \in \alpha_1^\perp$ (spacelike orthonormal under the η-form, by $(A2_{\text{collar}})$) and $c_0^{(i)} \in \mathbb{R}$ a scalar depending on the local data. The branched leading term $c_0^{(i)} w^{3/2}\, \alpha_1$ satisfies the positivity condition of [3, Def. 2].

$(A5)$ **Hölder regularity of A_{bulk}.** *The bulk metric A_{bulk} on the normal bundle N_h of the maximal section image is Hölder-continuous with the explicit bound*

$$\|A_{\text{bulk}}\|_{C_0^{0,\alpha}(M^7)} \;\leq\; K_A(\mathcal{D}), \qquad \operatorname{cond}(A_{\text{bulk}}) \;\leq\; \operatorname{cond}_A(\mathcal{D}),$$

for some Hölder exponent $\alpha \in (0, 1/2]$.

Normalisation (the "(A5) normalisation" referred to throughout): the overall scale of A_{bulk} is fixed by $\lambda_{\max}(A_{\text{bulk}}) = 1$, so that, for the unit-normalised vanishing-cycle direction α_1, $A_{\text{bulk}}(\alpha_1, \alpha_1) \in [1/\operatorname{cond}(A_{\text{bulk}}), 1]$, whence the conservative floor $\sigma_{\min}(A_{\text{loc}}) \geq 1/\operatorname{cond}(A_{\text{bulk}}) \geq 0.432$ at $\mathcal{D}_0$. The sharper anchoring $A_{\text{bulk}}(\alpha_1, \alpha_1) = 1$ (an additional rescaling along the α_1-line) is used only in the "normalised" columns of the numerical displays (§9.2, Supplementary Appendix D, Supplementary Appendix F); the conservative columns use only the floor.

($A6$) **Linking smallness.** *The geometrically split unlink condition* ($A3$) *is quantitative:*

$$\frac{(N-1)\,r_0(\mathcal{D})}{d_{\min}(\mathcal{D})} \;\le\; \eta^*_{\mathrm{link}}(\mathcal{D}),$$

where η^*_{link} is small enough that the linking perturbation matrix E_{link} of Theorem 5.10 satisfies $\|E_{\mathrm{link}}\|_{\mathrm{op}} < \sigma_{\min}(A_{\mathrm{loc}}) - \|E_{\mathrm{geom}}\|_{\mathrm{op}}$ (with $\sigma_{\min}(A_{\mathrm{loc}}) = A_{\mathrm{bulk}}(\alpha_1,\alpha_1) \ge 1/\mathrm{cond}(A_{\mathrm{bulk}})$ in the ($A5$) *normalisation, $=1$ in its normalised column) for the Neumann series to converge.*

($A7$) **Geometric scales.** *The injectivity radius and curvature scales of (B^3, g_B) satisfy*

$$R_{\mathrm{inj}}(\mathcal{D}),\; R_{\mathrm{curv}}(\mathcal{D}) \;\ge\; r_0(\mathcal{D}),$$

so that the local edge analysis in tubes of radius r_0 is uniformly geometric.

($A8$) **Mode-dominance scale.** *There exists a scale $R_{\mathrm{modedom}}(\mathcal{D}) > 0$ such that, on the cover at $r \le R_{\mathrm{modedom}}$, the cubic mode $r^3\, e^{\pm i\theta}$ of the branched correction dominates the sub-leading polyhomogeneous terms at $\gamma \ge 5$ in the orbifold expansion of Theorem 4.4. Equivalently in base coordinates: the cubic mode $\rho_w^{3/2}$ dominates the curvature mode $\rho_w^{-1/2}$ of $\nabla_w^2 h$ on $\rho_w \le R^2_{\mathrm{modedom}}$.*

(G**-quant**) **Global background smallness.** *(introduced formally in §3.1 as the discharge condition for the global maximal background; anticipated here to complete the quantitative admissibility list.) The dimensionless contraction ratio*

$$r_{\mathrm{G}}(\mathcal{D}) \;:=\; \|\mathcal{J}_{\bar h}^{\mathrm{ext},-1}\|^2\, \|D^2 m\|\, \|m(\bar h_{\mathrm{global}})\|_{Y_{-3}} \;+\; C_{\mathrm{aff}}(\mathcal{D})\, \eta_{\mathrm{aff}}(\mathcal{D})$$

with $\eta_{\mathrm{aff}}(\mathcal{D}) := \sup_{i\ne j} \|v_k^{(i)} - v_k^{(j)}\|/\|v_k\|_{\mathrm{typ}}$ the mutual-consistency parameter of the collar-affine coefficients across components and $\|m(\bar h_{\mathrm{global}})\|_{Y_{-3}} \le C_{\mathrm{src}}(\mathcal{D})\, r_0^{7/2}\, \|A_{\mathrm{bulk}}\|_{C^{0,\alpha}}$, satisfies

$$r_{\mathrm{G}}(\mathcal{D}) \;\le\; \tfrac{1}{2}.$$

Under (G*-quant*) *(together with* ($A0$) *simple connectivity,* ($A1$) *rank-1 monodromy, and* (E*-quant*)*), the global maximal background (previously isolated as standing hypothesis* (G)*) is discharged as Proposition 3.1.*

(AR**-quant**) **Reconstruction smallness.** *(introduced formally in §3.1 as the discharge condition for the adiabatic reconstruction map; anticipated here to complete the quantitative admissibility list.) The dimensionless quantity*

$$r_{\mathrm{AR}}(\mathcal{D}) \;:=\; \epsilon_0(\mathcal{D})\, K_H^{K3}(\mathcal{D})\, K_F(\mathcal{D})$$

with $K_H^{K3}(\mathcal{D})$ the norm of the fibrewise Hodge Green's function on the K3 fibre (bounded above by $O(1)$ for the Ricci-flat K3 metric of Yau's theorem) and $K_F(\mathcal{D}) := \|F_H(h_0)\|_{C^{0,\alpha}_\beta}$ the Pacard-weighted horizontal-curvature norm bounded by $\mathrm{cond}(A_{\mathrm{bulk}}) \cdot K_A(\mathcal{D})$ via ($A5$)*, satisfies*

$$r_{\mathrm{AR}}(\mathcal{D}) \;\le\; \tfrac{1}{2}.$$

Under (AR*-quant*)*, the adiabatic reconstruction map (previously isolated as standing hypothesis* (AR)*) is discharged as Proposition 3.2, with the reconstruction bound $\|\mathcal{A}_\epsilon(h)\| \le 2\|\mathcal{A}_0(h)\|$ uniformly in $\epsilon \le \epsilon_0(\mathcal{D})$.*

(E-quant) **Outer-coercivity smallness.** *(introduced formally in §5.5 as the discharge condition for the global outer Dirichlet-to-Neumann coercivity; anticipated here to complete the quantitative admissibility list.) There is a dimensionless quantity*

$$\delta_{\mathrm{E}}(\mathcal{D}) \;:=\; C_{\mathrm{FS}}\,\|\kappa_g\|_{C^0}\,r_0 \;+\; C_{\mathrm{link}}\,(N-1)\,(r_0/d_{\min})^2 \;+\; C_{\mathrm{outer}}\,(r_0/R_B)^1$$

with $R_B := \min_i \operatorname{dist}(U_i, \partial\mathbb{B}_i)$ *(the split-ball outer-model scale of* ($A3$)*; see* (D_3)*, §5.5, the base being closed, this replaces the vacuous distance-to-boundary) and* $C_{\mathrm{FS}}, C_{\mathrm{link}}, C_{\mathrm{outer}} \le 2$ *explicit universal constants (§5.5), such that*

$$\delta_{\mathrm{E}}(\mathcal{D}) \;\le\; \tfrac{1}{2}.$$

Under (E-*quant*)*, the global outer coercivity (previously isolated as standing hypothesis* (E)*) is discharged as Proposition 5.7, with* $c_{\mathrm{E}}(\mathcal{D}) \ge 1/2$.

The twelve quantitative conditions are organised as: ($A0$)–($A1$) topology and monodromy; ($A2_{\mathrm{collar}}$) collar-affine branched model; ($A3$) geometric splitting; ($A4$) uniform local model; ($A5$) bulk regularity; ($A6$) smallness of inter-component coupling; ($A7$)–($A8$) geometric and analytic scales; (E-quant) outer-coercivity smallness; (AR-quant) reconstruction smallness; (G-quant) global-background smallness. They are quantitative and verifiable for any explicit construction. The anisotropic perturbation theorem is *not* among the admissibility conditions: it is the sole standing hypothesis (J) introduced at its point of use (§8.3) and collected in §1.5.

2.3. The local branched model

The local model near each Σ_i is the branched double cover $z \mapsto w = z^2$ with involution $\sigma\colon z \mapsto -z$. We adopt throughout §§4–7 the *base downstairs convention*: $w = u + iv$ is the smooth base coordinate on B^3, and the operator of interest is the base Laplace–Beltrami $\Delta_w = \partial_u^2 + \partial_v^2$ for the smooth metric $g_{B^3} = ds^2 + |dw|^2$. The cover coordinate $z = w^{1/2}$ provides the single-valued parametrisation for σ-odd modes that are multivalued downstairs.

The σ-odd cohomology of the cover in the α_1-direction is generated by the modes

$$z,\ z^3,\ z^5,\ \ldots \;=\; w^{1/2},\ w^{3/2},\ w^{5/2},\ \ldots$$

at base radial orders $\rho^{1/2}$, $\rho^{3/2}$, $\rho^{5/2}$, ... (with $\rho = |w|$). A function $u(\rho,\varphi) = \rho^\mu\, e^{im\varphi}$ is σ-odd iff under the cover involution $\sigma\colon z \mapsto -z$, equivalently $\varphi \mapsto \varphi + 2\pi$ on the cover, it changes sign: $e^{im\cdot 2\pi} = -1$, i.e. m is a *half-integer*. The σ-odd angular modes are therefore $e^{im\varphi}$ with $m \in (1/2) + \mathbb{Z}$, and the corresponding σ-odd indicial roots of Δ_w (computed from $I(\mu,m) = \mu^2 - m^2$ acting on $\rho^\mu\, e^{im\varphi}$) are

$$\mathcal{E}^{\sigma}_{\mathrm{ind}} \;=\; \{\pm\tfrac{1}{2},\ \pm\tfrac{3}{2},\ \pm\tfrac{5}{2},\ \ldots\},$$

yielding the Fredholm windows $(1/2,\,3/2)$, $(-1/2,\,1/2)$, $(-3/2,\,-1/2)$, etc. Section §4 develops the Mazzeo edge calculus on this base model in the Pacard $\beta = -1$ window $(1/2,\,3/2)$: Fredholm cokernel at the root $\mu = -1/2$ below the window, prescribed displacement data at its Wronskian partner $\mu = +1/2$ (outside the window from below; §5).

2.4. Non-vacuity

Proposition 2.2 (non-vacuity, two layers). *The GIFT $K_7 \to S^3$ Kovalev–Lefschetz construction with $N = 77$ branch components produces a datum $\mathcal{D}_0$ satisfying the topological/algebraic conditions (A0), (A1), (A3), (A6), (A7) unconditionally (Construction C.1), and the geometric conditions ($A2_{\text{collar}}$), (A4), (A5), (A8) under the L4 realisation of §9.1 (Remark 1.1′, OQ0-E), with explicit constants*

$$r_0(\mathcal{D}_0) = 10^{-2}, \quad d_{\min}(\mathcal{D}_0) = 1, \quad \|\kappa_g\|_{C^0(\Sigma)} \le 1, \quad \operatorname{cond}(A_{\text{bulk}}) \le 2.31, \quad K^{\text{Maz}}_{\text{Sch}}(\alpha) \le 17.$$

Detailed verification deferred to §9 and Supplementary Appendix C.

The construction $K_7 \to S^3$ is described in Supplementary Appendix C; here we note only that $b_3(K_7) = 77$ arises naturally from the round split 77-component unlink in S^3 as the discriminant, with monodromy generated by the Picard–Lefschetz reflection along a primitive (-2)-class $\alpha_1 \in NS(K3)$ of rank one. Non-vacuity is thereby established unconditionally at the topological/algebraic layer (Construction C.1). At the geometric layer it is conditional on the L4 realisation, consistently with the two-group split of Supplementary Appendix C.7.

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3. Reduction of Donaldson's system and cohomological identification

3.1. The six-equation ledger

We recall in compact form the system of [3, §2.1–2.2] governing torsion-free G_2-structures on co-associative $K3$ fibrations. The Ehresmann connection H on the smooth fibration $\pi\colon M^7 \to B^3$ (fibre $K3$, real 4-dimensional; base real 3-dimensional) induces a bigrading $\Omega^k(M^7) = \bigoplus_{p+q=k} \Omega^{p,q}$ with p horizontal and q vertical indices, and a corresponding decomposition of the exterior derivative $d = d_f + d_H + \iota_{F_H}$, where $d_f\colon \Omega^{p,q} \to \Omega^{p,q+1}$ is the fibrewise exterior derivative, $d_H\colon \Omega^{p,q} \to \Omega^{p+1,q}$ is the horizontal exterior derivative associated to H, and $\iota_{F_H}\colon \Omega^{p,q} \to \Omega^{p+2,q-1}$ is contraction with the Ehresmann curvature $F_H \in \Gamma(V \otimes \pi^*\Lambda^2 T^*B)$ (vertical component of the Lie bracket of two horizontal lifts). The objects are:

- $\omega \in \Omega^{1,2}(M^7)$, the fibrewise hyperkähler Kähler form paired with a horizontal 1-form, generated by the period section h;
- $\lambda \in \Omega^{3,0}(M^7)$, a horizontal 3-form on B^3 (the canonical horizontal volume-scale set by h);
- $\mu \in \Omega^{0,4}(M^7)$, the fibrewise volume form (Re and Im parts of the fibrewise hyperkähler holomorphic 2-form squared);
- $\Theta \in \Omega^{2,2}(M^7)$, the mixed-type 4-form dual to the G_2 3-form;
- $\widetilde{\varphi} = \omega + \lambda$, the closed positive 3-form of G_2-type, with dual $\widetilde{\psi} = *\widetilde{\varphi} = \mu + \Theta$.

The six equations decompose $d\widetilde{\varphi} = 0$ and $d\widetilde{\psi} = 0$ into their bigraded components. Closure $d\widetilde{\varphi} = 0$ (Donaldson Proposition 1) gives

$$(E1)\ d_f\omega = 0, \quad (E2)\ d_H\omega = 0, \quad (E3)\ d_f\lambda = -F_H\,\omega,$$

and torsion-freeness $d\widetilde{\psi} = 0$ (Donaldson Proposition 2, via Lemma 3) adds

$$(E4)\ d_H\mu = 0, \quad (E5)\ d_f\Theta = -F_H\,\mu, \quad (E6)\ d_H\Theta = 0,$$

in the notation of [3, §2.2]. Here (E1) is the fibrewise hyperkähler closure (automatic once ω is generated fibre-wise from h), (E2) is the horizontal variation of h along B^3, and (E3), (E5) are the sole equations in which the Ehresmann curvature F_H appears, as an $O(\epsilon)$ perturbative coupling in the adiabatic rescaling of §7 (this is the ε-order verdict of Donaldson Lemma 5: at $\epsilon = 0$ the terms involving F_H drop out).

Lemma 3.1 (Donaldson 2017, Lemma 5). *At $\epsilon = 0$, the equations $(E1)$ – $(E5)$ are algebraically solvable for the data $(\omega, \lambda, \mu, \Theta)$ generated by the period section h. The only non-trivial equation surviving the limit is $(E6)$, which is equivalent to the* maximal-submanifold equation**

$$m(h) \;=\; 0,$$

where $m(h)$ denotes the mean curvature of the section $h\colon B^3 \to \mathbb{R}^{3,19}/O(3,19;\mathbb{Z})$ as a spacelike 3-fold in the period domain.

This reduction is the structural backbone of Donaldson's programme: the G_2-construction at $\epsilon = 0$ is governed by a single elliptic free-boundary problem on B^3. The construction of $\widetilde{\varphi}_R$ for R large is the perturbative extension of this limit problem (Proposition 3 of [3]).

Three levels of the section. The analysis distinguishes three sections, which must not be conflated:

1. the *local seed* $\bar{h} = h_{\mathrm{loc,aff}} + h_{\mathrm{loc,branch}}$, the affine-plus-branched model of $(A2_{\mathrm{collar}})$ + $(A4)$ on each collar U_i. It solves the maximal-submanifold equation only to *leading* order at Σ_i: its residual $m(\bar{h})$ carries a non-zero polyhomogeneous trace at the weight $\mu = -1/2$, computed in §5.6, sourced by the quadratic self-interaction of the branched jet;
2. the *global maximal background* $h_0 = \bar{h} + v_0$, where v_0 is a σ-odd branch-displacement datum at $\mu = +1/2$ (an element of $\mathbb{R}^{2N}$, one real pair per component, transverse displacements of the branch points, §5.3) chosen so as to cancel that trace, so that $m(h_0) = 0$ on $B^3 \setminus \Sigma$. The construction of v_0 at $\epsilon = 0$, the branched free-boundary problem, is the content of §5; the existence of a *global* such h_0 on $B^3 \setminus \Sigma$ matching the collar model is discharged as **Proposition 3.1** below under (G-quant);
3. the *adiabatic lifting* $h_\epsilon = h_0 + \epsilon\,\xi_1 + \epsilon^2\,\xi_2 + \cdots$, solving the full ϵ-perturbed equation, constructed in §7.

The displacement data $\mathbb{R}^{2N}$ thus serve to **construct the maximal background** v_0 at $\epsilon = 0$ (§5); they do *not* cancel an obstruction of an already-maximal section.

Global maximal background as a quantitative admissibility condition. The existence of a global maximal background h_0 on $B^3 \setminus \Sigma$ is stated as a *quantitative admissibility condition* (G-quant), parallel to $(A0)$–$(A8)$, (E-quant), (AR-quant), and discharged by Proposition 3.1 under a datum-level numerical margin. The discharge combines three structural ingredients: (a) rank-one monodromy $(A1)$ ensures the $\alpha_1^{\perp}$-projection of any period section is single-valued on $B^3 \setminus \Sigma$, so the collar-affine models jointly extend under a mild consistency condition; (b) simple connectivity $(A0)$ ensures that a smooth global extension exists; (c) the extended-domain Fredholm theory of §5 (Theorem 5.10, augmented right inverse discharged under (E-quant)) provides the linearised inverse for a Newton iteration.

Global background construction. By rank-one monodromy $(A1)$, all Picard–Lefschetz reflections of Γ act in the same direction α_1; hence the projection $h^\perp := \Pi_{\alpha_1^\perp}(h)$ is single-valued on $B^3 \setminus \Sigma$, and each collar-affine model $h^{(i)}_{\mathrm{loc,aff}}$ of $(A2_{\mathrm{collar}})$ is already valued in $\alpha_1^\perp$. Under simple connectivity $(A0)$ and a mutual-consistency bound $\eta_{\mathrm{aff}}(\mathcal{D}) := \sup_{i\neq j} \|v_k^{(i)} - v_k^{(j)}\|/\|v_k\|_{\mathrm{typ}}$ on the affine coefficients of $(A4)$, a smooth partition-of-unity extension

$$h^\perp_{\mathrm{bkg}} \;:=\; \sum_i \chi_i\, h^{(i)}_{\mathrm{loc,aff}} \;+\; (1 - \sum_i \chi_i)\, h^{\mathrm{ref}}_{\mathrm{bulk,aff}}$$

gives a global smooth section into $\alpha_1^\perp$, source-free on each collar (since $m(h_{\mathrm{loc,aff}}) \equiv 0$ by direct symbolic computation, §5.6). The full global background

$$\bar h_{\mathrm{global}} \;:=\; h^\perp_{\mathrm{bkg}} \;+\; \sum_i \chi_i\, h^{(i)}_{\mathrm{loc,branch}}$$

adds the branched jets locally supported near each Σ_i. The residual $m(\bar h_{\mathrm{global}})$ then has: - a *cubic-remainder contribution* $\mathcal{Q}(h^{(i)}_{\mathrm{loc,branch}}) = \frac{27}{16}\,(c_0^{(i)})^3\, \dfrac{\langle \alpha_1, \alpha_1\rangle}{A_{\mathrm{bulk}}(\alpha_1,\alpha_1)}\, \rho^{1/2}\,(\sigma\text{-odd } m = \pm 1/2) + O(\rho^{3/2})$ on each collar (§5.6 Lemma 5.9; the quadratic-in-c_0 piece is even by parity of the branched jet and lands in the $\alpha_1^\perp$ regular sector, not the σ-odd obstruction channel, see Suppl. App. E.1); - a *cutoff-commutator contribution* $[\mathcal{J}_h, \chi_i]\, h^{(i)}_{\mathrm{loc,branch}}$ concentrated in the annular transition $\{r_0/4 < |w| < r_0/2\}$; - an *affine-mismatch contribution* $\propto \eta_{\mathrm{aff}}(\mathcal{D})$ in the bulk transitions.

In the Pacard $\beta = -1$ weighted target norm $Y_{-3} := C^{0,\alpha}_{\beta-2}(B^3 \setminus \Sigma)$, all three contributions scale as $r_0^{7/2}$ or better, plus the η_{aff} term:

$$\|m(\bar h_{\mathrm{global}})\|_{Y_{-3}} \;\le\; C_{\mathrm{src}}(\mathcal{D})\, r_0^{7/2}(\mathcal{D})\, \|A_{\mathrm{bulk}}\|_{C^{0,\alpha}} \;+\; C_{\mathrm{aff}}(\mathcal{D})\, \eta_{\mathrm{aff}}(\mathcal{D})$$

with $C_{\mathrm{src}}, C_{\mathrm{aff}}$ explicit $O(1)$ universal constants; Suppl. App. E.1 gives $C_{\mathrm{src}} = 27/16$ certified from the cubic σ-odd remainder coefficient of Lemma 5.9 (superseding the earlier conservative envelope $C_{\mathrm{src}} \le 2$ derived from the quadratic attribution). The next admissibility condition asserts a definite margin of 1/2 on the resulting contraction:

(G-**quant**) **Global background smallness.** *The dimensionless contraction ratio*

$$r_{\mathrm{G}}(\mathcal{D}) \;:=\; \|\mathcal{J}_{\bar h}^{\mathrm{ext},-1}\|^2\, \|D^2 m\|\, \|m(\bar h_{\mathrm{global}})\|_{Y_{-3}} \;\le\; \tfrac{1}{2},$$

where $\|\mathcal{J}_{\bar h}^{\mathrm{ext},-1}\| \le K^{\mathrm{Maz}}_{\mathrm{Sch}}(\mathcal{D})\,(1+\sigma_{\min}(A_{\mathrm{loc}})^{-1})$ *is the augmented right inverse norm of Theorem 5.10.*

(G-quant) is verifiable by construction: at $\mathcal{D}_0$ (identical-copies construction of §9.1 with $\eta_{\mathrm{aff}}(\mathcal{D}_0) = 0$, $r_0 = 10^{-2}$, $K^{\mathrm{Maz}}_{\mathrm{Sch}} \le 17$, $\mathrm{cond}(A_{\mathrm{bulk}}) \le 2.31$ giving $\|\mathcal{J}^{\mathrm{ext},-1}\| \le 56.3$ (at the $\mu = +1/2$ Wronskian normalisation $\sigma_{\min}(A_{\mathrm{loc}}) \ge 1/\mathrm{cond}$, §5.4/Suppl. App. F), the certified source coefficient $C_{\mathrm{src}} = 27/16$ for the cubic σ-odd remainder, and the certified second-derivative bound $C_{\mathrm{nl}} = \|D^2 m\| \le 2/3$), one finds $r_{\mathrm{G}}(\mathcal{D}_0) \le 3.6 \cdot 10^{-4}$, a factor ~ 1389 below the threshold (this figure uses the theorem-grade L1.6 value 56.3, closed (OQ0-B); at the discharged crude bound ≤ 171 one still finds $r_{\mathrm{G}} \le 3.3 \cdot 10^{-3}$, margin $\sim 151\times$); see §9.2, Suppl. App. F and a supplementary interval-arithmetic script. The identical-copies structure of the K7/S^3 construction (77 copies of a single local model, §9.1) makes $\eta_{\mathrm{aff}}(\mathcal{D}_0) = 0$ trivially. Under (G-quant), the following is a proposition.

Proposition 3.1 (Global maximal background, conditional closure). *Status: conditional on a datum-level numerical margin, with all source-side constants theorem-grade at $\mathcal{D}_0$. The following is a conditional proposition: it assembles a rigorous fixed-point argument on the extended domain from theorem-grade inputs, $\|\mathcal{J}_{\bar h}^{\mathrm{ext},-1}\|$ (via the cited Mazzeo edge-Schauder constant $K_{\mathrm{Sch}}^{\mathrm{Maz}} \le 17$), the identical-copies structure at $\mathcal{D}_0$, the source coefficient $C_{\mathrm{src}} = 27/16$ from the cubic σ-odd remainder (Suppl. App. E.1), and the second-derivative bound $\|D^2 m\| = C_{\mathrm{nl}} \le 2/3$ from the Simons-type / collinearity extraction (Suppl. App. E.2). Numerical margin at $\mathcal{D}_0$: $\sim 1389\times$ (Suppl. App. F).*

Under the admissibility conditions (*A*0)*–*(*A*8)*,* (E-*quant*)*, and the quantitative smallness* (G-*quant*)*, there exists a section h_0 over $B^3 \setminus \Sigma$ of the form $h_0 = \bar h_{\mathrm{global}} + w$ with $w \in X_\beta^{\mathrm{ext}} = X_\beta \oplus \mathbb{R}^{2N}$ (the extended domain of §5.5), such that: (i) $m(h_0) = 0$ on $B^3 \setminus \Sigma$; (ii) on each collar U_i, h_0 matches the collar-affine branched model with leading branched jet $c_0^{(i)}\, w^{3/2}\, \alpha_1$ of* (*A*4)*; (iii) $h_0 - \bar h \in X_\beta$ for $\beta \in (1/2, 3/2)$ away from Σ, with the polyhomogeneous regularity of Theorem 4.4; (iv) $\|w\|_{X_\beta^{\mathrm{ext}}} \le 2\, \|\mathcal{J}_{\bar h}^{\mathrm{ext},-1}\|\, \|m(\bar h_{\mathrm{global}})\|_{Y_{-3}}$.*

Proof. By the structural ingredients above, $\bar h_{\mathrm{global}}$ is a smooth global section on $B^3 \setminus \Sigma$ matching each collar-affine + branched jet exactly. Writing $h_0 = \bar h_{\mathrm{global}} + w$ and Taylor-expanding,

$$m(h_0) \;=\; m(\bar h_{\mathrm{global}}) \;+\; \mathcal{J}_{\bar h_{\mathrm{global}}}(w) \;+\; \mathcal{Q}(w) \;=\; 0$$

where $\mathcal{J}_{\bar h_{\mathrm{global}}} = D_{\bar h_{\mathrm{global}}} m$ is the Jacobi operator at $\bar h_{\mathrm{global}}$ and $\mathcal{Q}(w) = O(\|w\|^2)$ collects the nonlinear remainder. Applying the augmented right inverse of Theorem 5.10, this is equivalent to the fixed-point equation

$$w \;=\; -\,\mathcal{J}_{\bar h_{\mathrm{global}}}^{\mathrm{ext},-1}\big(m(\bar h_{\mathrm{global}}) \,+\, \mathcal{Q}(w)\big) \qquad =: \qquad \Psi(w)$$

on the extended domain X_β^{ext}. By standard IFT bookkeeping:

$$\|\Psi(w)\|_{X_\beta^{\mathrm{ext}}} \;\le\; \|\mathcal{J}^{\mathrm{ext},-1}\|\,\big(\|m(\bar h_{\mathrm{global}})\|_{Y_{-3}} + \|D^2 m\|\,\|w\|^2_{X_\beta^{\mathrm{ext}}}\big),$$

$$\|\Psi(w) - \Psi(w')\|_{X_\beta^{\mathrm{ext}}} \;\le\; \|\mathcal{J}^{\mathrm{ext},-1}\|\,\|D^2 m\|\,(\|w\| + \|w'\|)\,\|w - w'\|_{X_\beta^{\mathrm{ext}}}.$$

On the ball B_r of radius $r := 2\,\|\mathcal{J}^{\mathrm{ext},-1}\|\,\|m(\bar h_{\mathrm{global}})\|_{Y_{-3}}$, Ψ maps $B_r \to B_r$ and is a contraction of ratio $r_{\mathrm{G}}(\mathcal{D}) \le 1/2$ under (G-quant). The Banach fixed-point theorem produces a unique $w \in B_r$ with $\Psi(w) = w$, giving h_0 satisfying (i), (ii), (iii). Item (iv) is the ball-radius bound. □

Remark 3.1. The discharge is *global* in a strong sense: it constructs h_0 on the whole compact base $B^3 \setminus \Sigma$, not just on individual tube neighbourhoods. The three structural ingredients act separately: (A1) provides the $\alpha_1^{\perp}$-projected single-valued background, (*A*0) closes the topology, and (E-quant) closes the Fredholm theory. The smallness (G-quant) is essentially a smallness of the tube-radius r_0 combined with the affine mutual-consistency η_{aff}; at $\mathcal{D}_0$ both are extremely favourable ($r_0 = 10^{-2}$, $\eta_{\mathrm{aff}} = 0$).

Remark 3.1-B (why not simpler). One might hope to discharge (G) directly from (E-quant) alone by solving the linear elliptic outer problem on $B^3 \setminus \bigcup_i U_i$ with collar-jet boundary data. This works at the linear level but does not close the nonlinear equation: the quadratic remainder $\mathcal{Q}$ near each Σ_i requires the *cokernel-inversion certificate* of §5.7, i.e., the finite-dimensional layer $\mathbb{R}^{2N}$ of branch-displacement data. Proposition 3.1 uses the *extended-domain* fixed-point argument, not the linear outer problem.

The adiabatic operator $\mathcal{M}_\epsilon$. Expanding the full system $(E1)$–$(E6)$ in ϵ around the $\epsilon = 0$ limit, the algebraic content of $(E1)$–$(E5)$ closes order-by-order on the data generated by h, and the dynamical content of $(E6)$ reduces to a single scalar equation on h with explicit ϵ-corrections:

$$\mathcal{M}_\epsilon(h) \;:=\; m(h) \;+\; \epsilon\, P_1(h) \;+\; \epsilon^2\, P_2(h) \;+\; \mathcal{R}_\epsilon(h) \;=\; 0, \qquad \epsilon \;=\; R^{-1},$$

with $\mathcal{R}_\epsilon(h) = \mathcal{O}(\epsilon^3)$ uniformly in h in a neighbourhood of h_0. The operators $P_1, P_2 \colon \Gamma(N_h) \to Y$ are computable from the closure of $(E1)$–$(E5)$ at orders ϵ, ϵ^2 (the explicit formulae are [3, §2.2 and Prop 3]). Starting from the maximal background h_0 (level 2, $m(h_0) = 0$), the corrections $\xi_1, \xi_2, \ldots$ are sourced by $P_1(h_0), P_2(h_0), \ldots$ (Theorem 7.1).

Reconstruction. Solving the reduced scalar equation $\mathcal{M}_\epsilon(h) = 0$ is not, by itself, the same as solving the full system $(E1)$–$(E6)$: for $\epsilon > 0$ one has $m(h) = -\epsilon P_1(h) - \epsilon^2 P_2(h) - \mathcal{R}_\epsilon(h) \neq 0$, so a solution h of $\mathcal{M}_\epsilon(h) = 0$ is not itself maximal. What closes the G_2-structure is an *adiabatic reconstruction map*, built from the order-by-order algebraic closure of $(E1)$–$(E5)$, whose existence with uniform estimates we isolate as a hypothesis.

Adiabatic reconstruction as a quantitative admissibility condition. The reconstruction map $\mathcal{A}_\epsilon$ is stated as a *quantitative admissibility condition* (AR-quant), parallel to $(A0)$–$(A8)$ and (E-quant), and discharged by Proposition 3.2 under a datum-level numerical margin. The observation is structural: [3, Prop. 3] already constructs the map $\mathcal{A}_\epsilon$ as a *formal power series in* ϵ satisfying (i) and (ii) to all orders; what is left open by Donaldson is precisely (iii), the convergence of that formal series, i.e. the uniform-in-ϵ bound. This gap is dischargeable under an explicit smallness condition on two geometric constants.

Structure of the formal iteration. At each order $k \geq 1$ in ϵ, the algebraic closure of $(E1)$–$(E5)$ (Donaldson Lemma 5) reduces to a pair of *fibrewise* elliptic equations on each K3 fibre,

$$d_f\, \lambda_k \;=\; -\,F_H\, \omega_k, \qquad d_f\, \Theta_k \;=\; -\,F_H\, \mu_k,$$

with d_f-exact sources (Donaldson Prop. 3). Solving via the fibrewise Hodge decomposition on K3 gives, for each k,

$$\|\lambda_k\|_{C^{k,\alpha}_\beta} \;\leq\; K_H^{K3}\, \|F_H\|_{C^{0,\alpha}_\beta}\, \|\lambda_{k-1}\|_{C^{k-1,\alpha}_\beta},$$

where

$$K_H^{K3}(\mathcal{D}) \;:=\; \|\Delta_f^{-1}\, d_f^*\|_{C^{0,\alpha}(K3 \text{ fibre, } d_f\text{-exact 2-forms})}$$

is the norm of the fibrewise Hodge Green's function on the K3 fibre, equal to $1/\sqrt{\lambda_1^{\mathrm{ex},2}(K3)}$ up to $O(1)$ elliptic Schauder factors, with $\lambda_1^{\mathrm{ex},2}(K3) > 0$ the smallest positive eigenvalue of the fibrewise Hodge Laplacian on d_f-exact 2-forms. On the hyperkähler K3 this exact-2-form gap **coincides with the scalar Laplacian gap**, $\lambda_1^{\mathrm{ex},2}(K3) = \lambda_1^{\mathrm{scalar}}(K3)$, by Hodge duality (exact 2-forms $\leftrightarrow$ co-exact 1-forms), the Kähler identity $\Delta_d = 2\Delta_{\bar\partial}$, $b_1(K3) = 0$, and the parallel holomorphic 2-form and hyperkähler triple; see Suppl. App. E.5. This reduction is what allows the (scalar) comparison theorems to be applied to the correct operator, and

$$K_F(\mathcal{D}) \;:=\; \|F_H(h_0)\|_{C^{0,\alpha}_\beta}$$

is the Pacard-weighted norm of the horizontal curvature of the connection H evaluated at the maximal background h_0. The same iteration governs Θ_k. Summing the geometric series:

$$\Big\|\sum_{k\geq 0} \epsilon^k \, \lambda_k\Big\|_{C^{k,\alpha}_\beta} \;\leq\; \frac{\|\lambda_0\|_{C^{k,\alpha}_\beta}}{1-\epsilon\, K_H^{K3}(\mathcal{D})\, K_F(\mathcal{D})}$$

uniformly in ϵ, as long as the denominator remains bounded away from zero. The next admissibility condition asserts a definite margin of 1/2:

(AR-**quant**) **Adiabatic reconstruction smallness.** *The dimensionless quantity*

$$r_{\mathrm{AR}}(\mathcal{D}) \;:=\; \epsilon_0(\mathcal{D})\, K_H^{K3}(\mathcal{D})\, K_F(\mathcal{D}) \;\leq\; \tfrac{1}{2},$$

where $\epsilon_0(\mathcal{D}) := R_0(\mathcal{D})^{-1}$ is the adiabatic-lifting scale of Corollary B.

(AR-quant) is verifiable by construction: at $\mathcal{D}_0$ (with $\epsilon_0 \leq 10^{-2}$; $K_H^{K3} \leq 0.45$ from the exact-2-form/scalar reduction above together with the Zhong–Yang bound $\lambda_1^{\mathrm{scalar}} \geq \pi^2/\mathrm{diam}^2 \approx 5.0$ for $\mathrm{Ric} \geq 0$ and $\mathrm{diam}(K3_x, \text{unit vol}) \leq$ $D_{\max} = 7/5$ for every fibre of $\mathcal{D}_0$, a certified numerical datum input: the collar-affine period section has compact moduli image ($\mathrm{A2}_{\mathrm{collar}}$), anchored at the $T^4/\mathbb{Z}_2$ orbifold point ($2^{1/4} \approx 1.19$, EH resolutions ~ 1.4); no diameter bound uniform over K3 moduli is claimed or needed. The exact $T^4/\mathbb{Z}_2$ orbifold point of K3 moduli gives the sharper benchmark $K_H^{K3} \approx 0.19$; and $K_F(\mathcal{D}_0) \leq \mathrm{cond}(A_{\mathrm{bulk}}) \cdot K_A \leq 2.31$ from ($A5$) in the Pacard $\beta = -1$ weighted norm which absorbs the $w^{-1/2}$ edge singularity of $\nabla^2 h_0$), one finds $r_{\mathrm{AR}}(\mathcal{D}_0) \leq 0.011$, a factor ~ 48 below the threshold (rigorous), and ~ 114 at the orbifold benchmark. See §9.2, Suppl. App. F and a supplementary interval-arithmetic script. Under (AR-quant), the following is a proposition.

Proposition 3.2 (Adiabatic reconstruction, conditional closure). *Status: conditional on a datum-level numerical margin. The following is a conditional proposition: it assembles the convergent-series argument from [3, Prop. 3] (formal solvability to all orders) plus a fibrewise Hodge iteration bound. The geometric-series majorant treats the order-k closure of* (E1)–(E5) *as linear in the preceding order; a full Cauchy-majorant accounting of the nonlinear cross-terms could shrink the convergence radius by an $O(1)$ combinatorial factor and is not carried out here. The $\sim 48\times$ margin at $\mathcal{D}_0$ is the buffer held against precisely that factor; readers who prefer not to consume it may treat* (AR) *as a standing hypothesis on the same footing as* (J)*. The structural identification of the constant is now theorem-grade, $K_H^{K3} = 1/\sqrt{\lambda_1^{\mathrm{scalar}}(K3)}$ via the exact-2-form/scalar Hodge-gap reduction (Suppl. App. E.5), and $\lambda_1^{\mathrm{scalar}}(K3)$ is bounded by Zhong–Yang with the fibre-family diameter $D_{\max} = 7/5$ at $\mathcal{D}_0$ as a certified numerical datum input ($K_H^{K3} \leq 0.45$; datum-level, not moduli-uniform) with an exact orbifold-point benchmark ($K_H^{K3} \approx 0.19$). The residual non-rigor is only that the K3 Ricci-flat metric is not known in closed form, so $\lambda_1^{\mathrm{scalar}}(K3)$ has no closed-form value. The numerical margin at $\mathcal{D}_0$ ($\geq 48\times$ rigorous) is robust.*

Under the admissibility conditions ($A0$)–($A8$), (E-*quant*), (G-*quant*), *and* (AR-*quant*), *and* *given the maximal background h_0 of Proposition 3.1, there exists a reconstruction map*

$$\mathcal{A}_\epsilon \colon h \mapsto \big(\omega_\epsilon(h), H_\epsilon(h), \lambda_\epsilon(h), \Phi_\epsilon(h)\big)$$

defined for $\epsilon \le \epsilon_0(\mathcal{D})$ on a neighbourhood of h_0 in X_β, such that for every h with $\mathcal{M}_\epsilon(h) = 0$: (i) $d\,\Phi_\epsilon(h) = 0$; (ii) $[\Phi_\epsilon(h)] = [\varphi_0] + \epsilon^{-1}[B]$ in $H^3(M^7;\mathbb{R})$; (iii) $\|\mathcal{A}_\epsilon(h)\|_{C^{k,\alpha}_\beta} \le (1 - r_{\mathrm{AR}}(\mathcal{D}))^{-1}\,\|\mathcal{A}_0(h)\|_{C^{k,\alpha}_\beta} \le 2\,\|\mathcal{A}_0(h)\|_{C^{k,\alpha}_\beta}$, uniformly in $\epsilon \le \epsilon_0(\mathcal{D})$.

Proof. [3, Prop. 3] constructs $\mathcal{A}_\epsilon$ as a formal power series $\mathcal{A}_\epsilon = \sum_{k\ge 0} \epsilon^k\,\mathcal{A}_k$ satisfying (i) and (ii) to all orders in ϵ. The order-k correction is determined by fibrewise Hodge decomposition of the d_f-exact sources of $(E3), (E5)$ on the K3 fibre, giving $\|\mathcal{A}_k\|_{C^{k,\alpha}_\beta} \le (K_H^{K3}\,K_F)^k\,\|\mathcal{A}_0\|_{C^{k,\alpha}_\beta}$. Under (AR-quant), $\epsilon\,K_H^{K3}\,K_F \le 1/2$ for $\epsilon \le \epsilon_0$; the geometric series converges with sum bounded by 2, proving (iii). Convergence of the series in $C^{k,\alpha}_\beta$ implies convergence of $\mathcal{A}_\epsilon(h)$ as an actual 3-form (not merely formal), preserving (i) and (ii) by continuity. □

Remark 3.2. The proof leverages the *fibrewise* K3 Hodge decomposition, not the base edge theory: the closure of $(E1)$–$(E5)$ at each order is a fibre-by-fibre elliptic problem on K3, uncoupled between fibres. Coupling between fibres enters only through the base curvature F_H and the eventual truncation of the series at the order matching $(A8)$ mode-dominance scale. The Pacard $\beta = -1$ weight is essential: without it, $\|F_H(h_0)\|_{C^{0,\alpha}}$ is infinite at the edge because $\nabla^2 h_0$ has the branched $w^{-1/2}$ singularity; the weight w^{+1} absorbs it, giving finite K_F.

It is Proposition 3.2 that §7.3 invokes to close $\widetilde{\varphi}_R = \Phi_{R^{-1}}(h^R)$, in place of the (false) implication $m(h^R) = 0 \Rightarrow$ closed. The order-by-order algebraic closure of $(E1)$–$(E5)$ is Donaldson's contribution; the uniform-in-ϵ estimate is the content of (AR-quant) and Proposition 3.2.

3.2. The Jacobi operator $\mathcal{J}_h$

The linearisation of $m(h) = 0$ at the maximal background $h_0 = \bar{h} + v_0$ (§3.1) is the *Jacobi operator* $\mathcal{J}_h = D_{h_0} m$. This operator acts on sections of the normal bundle $N_h \to B^3$ of the maximal section's image in $\mathbb{R}^{3,19}$, which has rank

$$\mathrm{rank}(N_h) \;=\; 19$$

(the 19 negative-definite directions of $\mathbb{R}^{3,19}$ complementary to the spacelike 3-plane image of $h_{\mathrm{loc,aff}}$).

By direct computation in fibrewise affine coordinates, the Jacobi operator decomposes as

$$\mathcal{J}_h u^\perp \;=\; A_{\mathrm{bulk}}\, g^{ij}_{B^3}\,\partial_i\partial_j\, u^\perp \;+\; (\text{curvature corrections}),$$

where $u^\perp$ is the normal part of the variation, g_{B^3} is the induced metric on B^3, and A_{bulk} is the constant rank-19 induced normal-bundle metric of $(A2_{\mathrm{collar}})$ and $(A5)$. Along the branched direction $\alpha_1 \subset N_h$ near Σ_i, $\mathcal{J}_h$ acts as a branched Laplace-type operator with σ-odd half-integer indicial roots $\pm 1/2, \pm 3/2, \ldots$ in the base downstairs convention of §2.3, with the Pacard $\beta = -1$ Fredholm window $(1/2, 3/2)$ developed in §4.

Crucial distinction from Donaldson's fibrewise operator. The Jacobi operator $\mathcal{J}_h = D_h m$ is to be distinguished from the *fibrewise* operator

$$L_1 \;=\; d_X^+ \circ A \circ d_X$$

of [3, eqs. 18-19], which acts on $\Omega^\bullet(X) \otimes \mathbb{R}^3$ over each $K3$ fibre X, has rank 3, and resolves the fibrewise hyperkähler triple corrections in the formal expansion of [3, Prop. 3]. The Jacobi $\mathcal{J}_h$ acts on the base B^3 along the normal direction N_h of the maximal section, has rank 19, and is the operator carrying the *transverse* edge analysis at the discriminant; this is the operator developed in §4 and §5.

3.3. Cohomological identification: $[\widetilde{\varphi}_R] = [\varphi_0] + R[B]$

The family $\{\widetilde{\varphi}_R\}_{R \geq R_0(\mathcal{D})}$ produced by §5–§7 sits in a definite cohomology class in $H^3(M^7; \mathbb{R})$. We identify this class with the Donaldson parametrisation of Conjecture 1, working directly with the bigraded decomposition of the closed positive 3-form, without any rescaling diffeomorphism.

Bigraded decomposition. The Kovalev–Lefschetz fibration $\pi\colon M^7 \to B^3$ induces a Hodge-type bigrading of $\Omega^\bullet(M^7)$ by horizontal/vertical decomposition along an Ehresmann connection. A 3-form decomposes as

$$\varphi \;=\; \varphi^{(3,0)} \;+\; \varphi^{(2,1)} \;+\; \varphi^{(1,2)} \;+\; \varphi^{(0,3)},$$

with $\varphi^{(p,q)} \in \Omega^{p,q}(M^7)$ purely horizontal of degree p on the base and purely vertical of degree q on the K3 fibre. The fibre is a K3 surface, so $\Omega^{(p,q)}(M^7) = 0$ for $q > 4$; in particular $\varphi^{(0,3)} = 0$ since $H^3(K3) = 0$.

Donaldson's ansatz at parameter R. Following [3, §2.1], the closed positive 3-form of G_2-type at parameter R decomposes into two pieces of distinct bigrading:

$$\widetilde{\varphi}_R \;=\; \underbrace{\omega_F \wedge \lambda_1}_{\text{bidegree (1,2)}} \;+\; \underbrace{R \cdot \lambda_3}_{\text{bidegree (3,0)}} \;+ d\eta_R \;+\; \mathcal{O}(R^{-2}),$$

where: - $\omega_F \in \Omega^{0,2}(M^7)$ is the fibrewise Kähler form generated by the section h^R of §7 (a positive fibrewise $(1,1)$-form, of bidegree $(0,2)$ in our horizontal/vertical convention); - $\lambda_1 \in \Omega^{1,0}(M^7)$ is the horizontal 1-form induced by the Ehresmann connection of π (specifically, the canonical horizontal 1-form pinned by the period section, normalised so that $\int_F \omega_F = 1$ on a normalised fibre); - $\lambda_3 \in \Omega^{3,0}(M^7)$ is the *purely horizontal* 3-form, equal to the pull-back of a chosen volume form on B^3: $\lambda_3 = \pi^*\omega_B$ with $[\omega_B] \in H^3(B^3; \mathbb{R})$ the fundamental class; - $\eta_R \in \Omega^2(M^7)$ is the exact 2-form correction provided by the closure procedure of §7 (closing $\widetilde{\varphi}_R$ to $d\widetilde{\varphi}_R = 0$); - $\mathcal{O}(R^{-2})$ collects higher-order corrections in the adiabatic expansion of [3, Prop. 3], in lower-bidegree components (specifically $(2,1)$).

The parameter R multiplies *only* the $(3,0)$-component λ_3. This is the structural content of Donaldson's parametrisation: as $R \to \infty$, the horizontal base direction inflates relative to the fibre, with the $(1,2)$-component fixed and the $(3,0)$-component scaling linearly.

The cohomology classes. Define

$$[\varphi_0] \;:=\; [\omega_F \wedge \lambda_1] \;\in\; H^3(M^7; \mathbb{R}), \qquad [B] \;:=\; [\lambda_3] \;=\; \pi^*[\omega_B] \;\in\; H^3(M^7; \mathbb{R}).$$

Both classes are of degree 3 on M^7, as required.

Lemma 3.2 (Leray decomposition of $H^3(M^7)$ and the role of $[B]$). *Let $\mathcal{H} = R^2\pi_*\underline{\mathbb{R}}$ be the local system on $B^3 \setminus \Sigma$ with fibre $H^2(K3;\mathbb{R})$ and monodromy Γ. The Leray spectral sequence of π converges to $H^3(M^7;\mathbb{R})$, with leading non-zero terms*

$$E_2^{1,2} \;=\; H^1(B^3;\, \mathcal{H}^\Gamma), \qquad E_2^{3,0} \;=\; H^3(B^3;\, \mathbb{R}) \;=\; \mathbb{R}\,[\omega_B].$$

(The terms $E_2^{0,3} = H^0(B; H^3(K3)) = 0$ since $H^3(K3) = 0$, and $E_2^{2,1} = H^2(B; H^1(K3)) = 0$ since $H^1(K3) = 0$.) The Leray exact sequence gives

$$0 \longrightarrow H^3(B^3;\mathbb{R}) \xrightarrow{\pi^*} H^3(M^7;\mathbb{R}) \longrightarrow H^1(B^3;\mathcal{H}^\Gamma) \longrightarrow 0,$$

decomposing $H^3(M^7;\mathbb{R}) = \pi^* H^3(B^3) \oplus H^1(B^3;\mathcal{H}^\Gamma)$ as a direct sum (the sequence splits over $\mathbb{R}$ via the bigraded structure).

Sketch. Standard Leray-Hirsch for a smooth fibration with $H^1(K3) = H^3(K3) = 0$. The discriminant Σ contributes only to the lower-order extensions of the spectral sequence (via the long exact sequence of $(M^7, \pi^{-1}(\Sigma)))$, but for closed 3-cycles in the bulk these contributions are exact (by simple connectedness of $B^3 \setminus \Sigma$ in a neighbourhood of Σ, under $(A0) + (A3)$). □

The decomposition of Lemma 3.2 identifies: - $\pi^* H^3(B^3;\mathbb{R}) = \mathbb{R} \cdot [B]$: the $(3,0)$-component of $H^3(M^7)$, generated by the pull-back of the base volume class. - $H^1(B^3;\mathcal{H}^\Gamma)$: the $(1,2)$-component of $H^3(M^7)$, parametrising sections of the fibrewise H^2-bundle that vary linearly along base 1-cycles.

Proposition 3.3 (cohomological identification of the family). *The closed positive 3-forms $\widetilde{\varphi}_R$ constructed in §5–§7 have cohomology class*

$$[\widetilde{\varphi}_R] \;=\; [\varphi_0] \;+\; R\,[B] \;\in\; H^3(M^7;\mathbb{R}), \qquad R \geq R_0(\mathcal{D}),$$

where, with respect to the Leray decomposition of Lemma 3.2:

- $[\varphi_0] = [\omega_F \wedge \lambda_1] \in H^1(B^3;\, \mathcal{H}^\Gamma)$ *is the $(1,2)$-component, fixed by the data $\mathcal{D}$ (independent of R).*
- $[B] = \pi^*[\omega_B] \in \pi^* H^3(B^3;\mathbb{R})$ *is the $(3,0)$-component generator, of degree 3 in $H^3(M^7;\mathbb{R})$.*

The Joyce deformation $\varphi_R = \widetilde{\varphi}_R + d\eta_R^{\text{Joyce}}$ to a torsion-free G_2-structure (§8) preserves this class:

$$[\varphi_R] \;=\; [\widetilde{\varphi}_R] \;=\; [\varphi_0] + R[B] \;\in\; H^3(M^7;\mathbb{R}).$$

Proof. The bigraded decomposition of $\widetilde{\varphi}_R$ above gives the cohomology class as the sum of the classes of the two pieces (modulo exact corrections $d\eta_R + \mathcal{O}(R^{-2})$, which are killed in cohomology). The $(1,2)$-class $[\omega_F \wedge \lambda_1]$ depends only on the section h^R and the Ehresmann connection; under the structural admissibility hypotheses $(A0)$–$(A8)$ and the construction of §7, this class is R-independent at leading order (the finite-order ansatz of Theorem 7.1 gives $h^R = h_0 + O(R^{-1})$, hence $h^R \to h_0$ in cohomology as $R \to \infty$). Hence $[\omega_F \wedge \lambda_1] = [\varphi_0]$ as defined. The $(3,0)$-class $[\lambda_3] = \pi^*[\omega_B]$ is the parametrisation of the linear scaling: $[R \cdot \lambda_3] = R \cdot \pi^*[\omega_B] = R \cdot [B]$. The Joyce deformation $\eta_R^{\text{Joyce}} \in \Omega^2(M^7)$ adjusts $\widetilde{\varphi}_R$ by an exact correction, by [10, §10.5, Th. 11.6.1], preserving the cohomology class. □

Identification with Donaldson's Conjecture 1. Proposition 3.3 makes the identification of our parameter R with Donaldson's parameter R of [3, Conjecture 1] direct: the family $\widetilde{\varphi}_R$ of closed positive 3-forms in the class $[\varphi_0] + R[B]$ is precisely the family in Donaldson's parametrisation, and the existence claim of Theorem A for $R \geq R_0(\mathcal{D})$ is the conditional version of Conjecture 1 for the datum $\mathcal{D}$ at radius R. **No "scaling diffeomorphism"** is invoked: the family is defined intrinsically by the closure procedure of §5–§7, and the cohomology class is computed directly from the ansatz.

This concludes the reduction of Donaldson's system. The remaining content (§4 onwards) develops the analytic content needed to verify that the family $\widetilde{\varphi}_R$ is well-defined and closed for $R \geq R_0(\mathcal{D})$.

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4. Edge analysis of $\mathcal{J}_h$

Throughout this section, $\Sigma_i \subset B^3$ is a discriminant component, $U_i \subset B^3$ a fixed tubular neighbourhood of inner radius r_0 (depending on $\mathcal{D}$ via $(A2_{\text{collar}})$ and $(A7)$), and $\mathcal{J}_h = D_{h_0}m$ is the Jacobi operator of the maximal section, acting on its rank-19 normal bundle N_h (§3.2). The analysis is carried out in the smooth base coordinate w on B^3, with the branched double cover $z^2 = w$ providing the single-valued parametrisation for σ-odd modes (§2.3). The σ-odd sector has half-integer angular Fourier modes $m \in (1/2) + \mathbb{Z}$ and indicial roots $\mathcal{E}^{\sigma}_{\text{ind}} = \{\pm 1/2,\ \pm 3/2,\ \pm 5/2, \dots\}$; the Pacard $\beta = -1$ Fredholm window is $(1/2,\ 3/2)$.

4.1. The edge operator and its normal operator

At leading order on U_i, under $(A1)$ rank-one rigidity and $(A2_{\text{collar}})$ collar-affine background, the Jacobi operator restricted to the α_1-direction of N_h takes the form

$$\mathcal{J}_h \;=\; A_{\text{bulk}} \cdot \Delta_{g_{B^3}} \;+\; \mathcal{R}(s, w),$$

where A_{bulk} is the constant positive-definite 19×19 induced metric on N_h (constant along Σ_i under $(A2_{\text{collar}})$), $\Delta_{g_{B^3}} = \partial_s^2 + \partial_u^2 + \partial_v^2$ is the smooth base Laplace–Beltrami for $g_{B^3} = ds^2 + |dw|^2$ (with $w = u + iv$), and $\mathcal{R}(s, w)$ is a second-order edge perturbation of order $O(\kappa_g \rho)$ encoding the Frenet–Serret framing of Σ_i in B^3 and the variation of A_{bulk} across B^3 (controlled by $(A5)$). For the purposes of this section, $\mathcal{R}$ contributes only to the regularity-preserving part of the operator and does not affect the indicial structure.

In downstairs polar coordinates (s, ρ, φ) with $w = \rho\, e^{i\varphi}$, the transverse Laplace–Beltrami reads

$$\Delta_w \;=\; \frac{1}{\rho}\,\partial_\rho(\rho\,\partial_\rho) \;+\; \frac{1}{\rho^2}\,\partial_\varphi^2 \;=\; \partial_\rho^2 + \frac{1}{\rho}\,\partial_\rho + \frac{1}{\rho^2}\,\partial_\varphi^2.$$

Multiplying through by ρ^2 gives the standard edge form

$$\rho^2\,\mathcal{J}_h \;=\; A_{\text{bulk}}\,\big[(\rho\partial_\rho)^2 + \partial_\varphi^2 + \rho^2\,\partial_s^2\big] \;+\; \rho^2\,\mathcal{R}.$$

At the boundary face $\{\rho = 0\}$, the longitudinal term $\rho^2 \partial_s^2$ vanishes and the leading $\rho^2 \mathcal{J}_h$ is governed by the *transverse* derivatives only, the Fourier frequency q along Σ_i does not enter the indicial polynomial.

Definition 4.1 (normal operator). The *normal operator* of $\mathcal{J}_h$ at the boundary face $\{\rho = 0\}$ is the constant-coefficient operator

$$N(\mathcal{J}_h) \;=\; A_{\mathrm{bulk}}\,[(\rho\partial_\rho)^2 + \partial_\varphi^2]$$

on the transverse disc $\{|w| < r_0\}$, depending parametrically on $s \in \Sigma_i$ through A_{bulk} (constant under $(A2_{\mathrm{collar}})$).

Definition 4.2 (indicial family). The *indicial family* of $\mathcal{J}_h$ is the family of operators $I_{\mathcal{J}_h}(\mu, m)$, $\mu \in \mathbb{C}$, $m \in (1/2) + \mathbb{Z}$, obtained by acting on the formal mode $\rho^\mu\, e^{im\varphi}$:

$$N(\mathcal{J}_h)\big[\rho^\mu\, e^{im\varphi}\big] \;=\; A_{\mathrm{bulk}}\,(\mu^2 - m^2)\,\rho^{\mu-2}\,e^{im\varphi} \;=\; I_{\mathcal{J}_h}(\mu, m)\,\rho^{\mu-2}\,e^{im\varphi}.$$

By the half-integer characterisation of σ-odd angular modes (§2.3), the σ-odd indicial roots in the angular mode m are $\mu = \pm m$, giving

$$\mathcal{E}^\sigma_{\mathrm{ind}} \;=\; \{\pm\tfrac{1}{2},\, \pm\tfrac{3}{2},\, \pm\tfrac{5}{2},\, \ldots\}.$$

Lemma 4.3 (indicial parity). $I_{\mathcal{J}_h}(\mu, m) = I_{\mathcal{J}_h}(-\mu, m) = I_{\mathcal{J}_h}(\mu, -m)$. *Consequently the σ-odd indicial roots are symmetric about $\mu = 0$ and the Fredholm windows between consecutive roots are symmetric.*

The longitudinal Fourier frequency $q \in \mathbb{Z}$ along $\Sigma_i \simeq S^1$ enters only through the *non-indicial* term $\rho^2\partial_s^2 \to -q^2\rho^2$ after Fourier decomposition $u(s, \rho, \varphi) = \sum_q \hat{u}_q(\rho, \varphi)\, e^{iqs/L_i}$, where $L_i = \ell(\Sigma_i)/(2\pi)$. The factor ρ^2 in front absorbs q into a lower-order edge perturbation. The radial part of the q-th Fourier mode satisfies the modified Bessel equation $R'' + (1/\rho)R' - (m^2/\rho^2 + (q/L_i)^2)R = 0$, with bounded solution at $\rho = 0$ given by $R_{q,m}(\rho) = I_m(|q|\rho/L_i)$ for $q \neq 0$ and $R_{0,m}(\rho) = \rho^{|m|}$ for $q = 0$. This explicit form is the basis of the Dirichlet-to-Neumann construction in §5.5.

4.2. Weighted Hölder spaces and the polyhomogeneous expansion

For $\beta \in \mathbb{R}$ and $\alpha \in (0, 1)$, the σ-odd weighted Hölder space $C^{k,\alpha}_\beta(U_i; N_h)$ consists of N_h-valued σ-odd functions on U_i (single-valued on the cover) for which the weighted norm

$$\|u\|_{C^{k,\alpha}_\beta} \;:=\; \sum_{j=0}^{k} \sup_{U_i}\, \rho^{-\beta+j}\,|\nabla^j u| \;+\; \big[\rho^{-\beta+k}\,\nabla^k u\big]_{C^\alpha}$$

is finite, with the Hölder seminorm computed in the edge metric $ds^2 + d\rho^2 + \rho^2 d\varphi^2$. Set

$$X_\beta \;:=\; C^{2,\alpha}_\beta(U_i; N_h), \qquad Y_{\beta-2} \;:=\; C^{0,\alpha}_{\beta-2}(U_i; N_h).$$

The Fredholm window

$$\beta \;\in\; (1/2,\; 3/2) \quad \text{strictly interior}$$

is free of σ-odd indicial roots in $\mathcal{E}^\sigma_{\mathrm{ind}}$. The indicial root $\mu = 3/2$ (attained in the angular mode $m = \pm 3/2$) sits at the top of the window; the roots at and below the bottom are $\mu = 1/2$ and $\mu = -1/2$ (Pacard $\beta = -1$ cokernel root, §4.3). The displacement data of §5 sit at the regular root $\mu = +1/2$ in the mode $m = \pm 1/2$, below the window, outside X_β, and Wronskian-dual to the cokernel root $\mu = -1/2$ (Lemma 5.2).

Theorem 4.4 (elliptic polyhomogeneous regularity). *(cf. [15, §6, §7]) Let $\beta \in (1/2, 3/2)$ and let $u \in X_\beta$ solve $\mathcal{J}_h u = f$ with $f \in Y_{\beta-2}$ polyhomogeneous at $\{\rho = 0\}$. Then u is polyhomogeneous at each component Σ_i, with expansion*

$$u(s, \rho, \varphi) \sim \sum_{\mu \in \mathcal{E}^{\sigma,\beta}_{\mathrm{ind}}} \sum_{m \in (1/2)+\mathbb{Z}} a^{(i)}_{\mu,m}(s)\, \rho^\mu\, e^{im\varphi},$$

where $\mathcal{E}^{\sigma,\beta}_{\mathrm{ind}} = \mathcal{E}^{\sigma}_{\mathrm{ind}} \cap [\beta, +\infty) = \{3/2, 5/2, 7/2, \dots\}$ for $\beta \in (1/2, 3/2)$, the coefficients $a^{(i)}_{\mu,m} \colon \Sigma_i \to N_h$ are smooth N_h-valued functions on Σ_i, and the expansion is asymptotic in the standard Borel sense. (It is the solutions of the edge equation with polyhomogeneous data, not arbitrary elements of X_β, that are polyhomogeneous.) The evaluation maps below are applied to such solutions.

The leading polyhomogeneous coefficient, the one at the top exponent $\mu = 3/2$ of the Fredholm window, is the pair of N_h-valued functions

$$a^{(i)}_{3/2,+1/2}(s), \quad a^{(i)}_{3/2,-1/2}(s) \;\in\; C^\infty(\Sigma_i; N_h),$$

encoding the σ-odd cubic asymptotic data of u at Σ_i in the two angular modes $e^{+i\varphi/2}$, $e^{-i\varphi/2}$ at radial order $\rho^{3/2}$. We identify these with the real-imaginary pair

$$\big(a^{(i),R}_{3/2}(s),\; a^{(i),I}_{3/2}(s)\big) \;\in\; C^\infty(\Sigma_i; N_h) \times C^\infty(\Sigma_i; N_h),$$

encoding the cubic asymptotic data of u in the σ-odd combinations $\rho^{3/2}\, \cos(\varphi/2)$ and $\rho^{3/2}\, \sin(\varphi/2)$ (equivalently $|z|^2 \operatorname{Re} z$ and $|z|^2 \operatorname{Im} z$ on the cover). Restricted to the α_1-direction of N_h (the branched direction, isolated in the rank-one setting), these are two scalar functions $a^{(i),R}_{3/2}, a^{(i),I}_{3/2} \in C^\infty(\Sigma_i; \mathbb{R})$.

4.3. The Fredholm cokernel at the Pacard $\beta = -1$ root

Theorem 4.5 (Mazzeo Fredholm theorem for $\mathcal{J}_h$; kernel-null under outer coercivity). *(cf. [15, Th. 6.1]) For $\beta \in (1/2, 3/2)$, the operator*

$$\mathcal{J}_h \colon X_\beta \longrightarrow Y_{\beta-2}$$

is Fredholm, this is the Mazzeo edge-Fredholm theorem specialised to $\mathcal{J}_h$, and depends only on the branched edge structure of §2.3 and on the indicial roots being disjoint from the weight β. ***The kernel of $\mathcal{J}_h$ vanishes in the σ-odd sector as a separate consequence of the outer coercivity discharge of Proposition 5.7*** *((E-quant)): the σ-odd Weitzenböck-type L^2 identity for $\mathcal{J}_h = A_{\mathrm{bulk}}\Delta_w + \mathcal{R}$ against a σ-odd solution $u \in X_\beta$ produces a positive-definite quadratic form on the extended domain provided by the outer coercivity constant $c_{\mathrm{E}}(\mathcal{D}) \geq 0.98$ at $\mathcal{D}_0$ (Prop. 5.7), which forces $u \equiv 0$. Kernel-null is thus not attributed to Mazzeo's edge-Fredholm theorem, which gives only Fredholmness, but to the coercivity package* (E-quant)*; readers unwilling to consume* (E-quant) *may treat kernel-null as an additional structural input on $\mathcal{J}_h$ at the level of Theorem A's admissibility list. Its cokernel is finite-dimensional, and admits a natural parametrisation via the polyhomogeneous trace at the lower indicial root $\mu = -1/2$ (the Pacard $\beta = -1$*

root) in the α_1*-direction of* $N_h|_{\Sigma_i}$*: there is a bounded* indicial trace bundle

$$\mathcal{T}_{-1/2} \;:=\; \bigoplus_{i=1}^{N} \Gamma(\Sigma_i;\mathbb{R}^2) \;\cong\; \bigoplus_{i=1}^{N} C^{0,\alpha}(\Sigma_i;\mathbb{R}^2),$$

parametrising the polyhomogeneous trace at $\mu = -1/2$*, angular modes* $e^{\pm i\varphi/2}$ *(the* σ*-odd modes* $\rho^{-1/2}\cos(\varphi/2)$ *and* $\rho^{-1/2}\sin(\varphi/2)$*). The cokernel of* $\mathcal{J}_h$ *embeds as a finite-dimensional subspace of* $\mathcal{T}_{-1/2}$*, identified via the duality* $\operatorname{coker}(\mathcal{J}_h) \cong \ker(\mathcal{J}_h^*)$ *on the dual weighted Hölder space.*

Remark 4.6 (trace bundle vs Fredholm cokernel). The indicial trace bundle $\mathcal{T}_{-1/2}$ is *not* the Fredholm cokernel of $\mathcal{J}_h$: it parametrises *all* possible polyhomogeneous trace data at $\mu = -1/2$ along Σ, of which the Fredholm cokernel is the finite-dimensional subspace compatible with global compatibility (in particular, with the Dirichlet-to-Neumann positivity on non-constant longitudinal Fourier modes, §5.5). Identification of the Fredholm cokernel as the constant-Fourier-mode subspace $\mathbb{R}^{2N} \subset \mathcal{T}_{-1/2}$ is the content of Corollary 5.8 below.

4.4. Twisted Schauder estimate

Theorem 4.7 (twisted Schauder). *Let* $\beta \in (1/2, 3/2)$ *strictly interior. The right inverse of* $\mathcal{J}_h$ *on the complement of the cokernel exists and is bounded with*

$$\|G_h\|_{Y_{\beta-2}\to X_\beta} \;\le\; K^{\mathrm{Maz}}_{\mathrm{Sch}}(\alpha)\,\mathrm{cond}(A_{\mathrm{bulk}}),$$

where $K^{\mathrm{Maz}}_{\mathrm{Sch}}(\alpha)$ *is the standard Mazzeo–Vertman edge-Schauder constant at the chosen Hölder exponent* α*. The right inverse is taken with respect to a choice of bounded projection* $\Pi_{\mathrm{coker}}\colon Y_{\beta-2} \to \operatorname{coker}\mathcal{J}_h$ *defined via the polyhomogeneous trace at* $\mu = -1/2$*.*

The twist factor $\mathrm{cond}(A_{\mathrm{bulk}})$ tracks the cost of inverting the rank-19 symbol on N_h. Under $(A2_{\mathrm{collar}})$ the metric A_{bulk} is constant along Σ, so the cond-number is a fixed datum-level constant; under $(A5)$ the spatial variation of A_{bulk} across B^3 is controlled and contributes a sub-leading Hölder term.

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5. Extended-domain theorem with prescribed branch-displacement data

Theorem 4.5 identifies the indicial trace bundle $\mathcal{T}_{-1/2}$ at $\mu = -1/2$ over Σ, of which the Fredholm cokernel is a finite-dimensional subspace (precisely identified in §5.5 below). To cancel the cokernel, we adjoin to the domain a layer of *prescribed asymptotic data* at the regular indicial root $\mu = +1/2$, angular mode $m = \pm 1/2$, the modes $\rho^{1/2}\cos(\varphi/2), \rho^{1/2}\sin(\varphi/2)$, which lie *outside* X_β for every $\beta \in (1/2, 3/2)$, so the augmentation is genuine. This is the extended-domain framework of [18, §3] and [16, §3], specialised to our half-integer branched setting; the connection between the $\mu = +1/2$ data and the $\mu = -1/2$ cokernel is the **Wronskian duality** $\mu \leftrightarrow -\mu$: the class of a source in $Y_{\beta-2}/\mathrm{Ran}$ is computed by the L^2 pairing against the adjoint cokernel representatives $\rho^{-1/2}\cos(\varphi/2), \rho^{-1/2}\sin(\varphi/2)$, and for a domain mode ρ^a paired against ρ^b the Green boundary flux is $(a-b)\,\varepsilon^{a+b}$, non-vanishing precisely when $a+b=0$ (Lemma 5.2 below). The cubic layer $\rho^{3/2}\cos(\varphi/2), \rho^{3/2}\sin(\varphi/2)$ is *not* an admissible data layer: it pairs to zero with the cokernel $(a+b=1)$, see Remark 5.2. Geometrically, the $\mu=+1/2$ data displace the branch points of the seed and realise the free-boundary mechanism in which the discriminant is part of the unknown, as in Donaldson's formulation (§5.3).

5.1. The closed subspace $X_\beta^{(0)}$

The polyhomogeneous expansion of Theorem 4.4 provides bounded evaluation maps

$$\mathrm{ev}^{(i)}_{\mu,m} : X_\beta \longrightarrow C^{0,\alpha}(\Sigma_i; N_h), \qquad u \longmapsto a^{(i)}_{\mu,m}(s),$$

for each exponent $\mu \in \mathcal{E}^{\sigma,\beta}_{\mathrm{ind}}$ and σ-odd angular mode $m \in (1/2)+\mathbb{Z}$ (pairs with $\mu \neq |m|$ are shifted exponents, not indicial roots). Boundedness follows from the closedness of polyhomogeneous truncation under the $C^{k,\alpha}_\beta$ norm [15, §6].

Definition 5.1 (the subspace $X_\beta^{(0)}$). *The closed subspace*

$$X_\beta^{(0)} := \bigcap_{i=1}^{N} \Big(\ker \mathrm{ev}^{(i)}_{3/2,+1/2} \cap \ker \mathrm{ev}^{(i)}_{3/2,-1/2} \Big) \subset X_\beta$$

consists of the elements of X_β whose leading polyhomogeneous coefficient at $\mu=3/2$ vanishes identically on each Σ_i, in both σ-odd angular modes $e^{\pm i\varphi/2}$.

Closedness of $X_\beta^{(0)}$ under the X_β norm follows from the boundedness of the evaluation maps. The complement $X_\beta/X_\beta^{(0)}$ carries the cubic ($\mu=3/2$) polyhomogeneous coefficient of u, isomorphic to $\bigoplus_i C^{0,\alpha}(\Sigma_i;\mathbb{R}^2)$; writing $X_\beta \cong X_\beta^{(0)} \oplus \mathcal{D}^{\mathrm{cub}}_{3/2}$ for this internal splitting is convenient for the seed bookkeeping (§5.6). In the present version this splitting plays only a bookkeeping role: the extension of the domain that changes the cokernel is *not* by the cubic layer (which lies inside X_β), but by the $\mu=+1/2$ displacement layer of §§5.2–5.3, which lies outside X_β.

5.2. σ-odd admissible modes and the Wronskian pairing

The σ-odd *displacement modes* at Σ_i are the regular indicial modes at $\mu=+1/2$,

$$\phi_R^{(i)} := \rho^{1/2}\cos(\varphi/2) = \mathrm{Re}\, z, \qquad \phi_I^{(i)} := \rho^{1/2}\sin(\varphi/2) = \mathrm{Im}\, z,$$

living in the angular modes $m=\pm1/2$ at radial order $\rho^{1/2}$, the indicial pair $(\mu,m)=(1/2,\pm1/2)$, $\mu^2-m^2=0$, so they are Δ_w-harmonic on $U_i \setminus \Sigma_i$. They are multivalued downstairs (half-integer angular mode) but single-valued on the cover, where they are the linear polynomials $\mathrm{Re}\, z, \mathrm{Im}\, z$. The σ-odd *cubic modes* $\psi_R^{(i)} := \rho^{3/2}\cos(\varphi/2) = |z|^2\,\mathrm{Re}\, z$ and $\psi_I^{(i)} := \rho^{3/2}\sin(\varphi/2) = |z|^2\,\mathrm{Im}\, z$ at $(\mu,m)=(3/2,\pm1/2)$ retain their role in the *seed* jet (the branched leading term $c_0 w^{3/2}\alpha_1$ of $(A4)$ contributes to the angular modes $m=\pm3/2$ via the harmonic z^3, and to $m=\pm1/2$ via $|z|^2 z$); they no longer populate the asymptotic-data layer, for the reason recorded in Remark 5.2.

The duality between domain modes and the cokernel is a Wronskian pairing, computed by the Green identity:

Lemma 5.2 (Wronskian pairing: the $\mu = +1/2$ layer detects the cokernel). *Let $w_R^{(i)} := \rho^{-1/2}\cos(\varphi/2)$, $w_I^{(i)} := \rho^{-1/2}\sin(\varphi/2)$ be the adjoint cokernel representatives at Σ_i (Theorem 4.5 / Corollary 5.8), and let χ_i be any radial cut-off as in §5.3. Then, on the model operator $A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\,\Delta_w$ in the α_1-direction,*

$$\langle A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\,\Delta_w(\chi_i\,\phi_R^{(i)}),\ w_R^{(i)}\rangle_{L^2(T^{(i)}_{r_0})} \;=\; -\,(\mu_+ - \mu_-)\,\pi\,\ell(\Sigma_i)\,A_{\mathrm{bulk}}(\alpha_1,\alpha_1), \qquad \mu_+ - \mu_- \;=\; \tfrac{1}{2} - (-\tfrac{1}{2}) \;=\; 1,$$

*independently of the choice of χ_i, and identically for the I-pair; the cross pairings (R, I) vanish by angular orthogonality. More generally, for $u = \rho^a f(\varphi)$, $w = \rho^b f(\varphi)$ with $\Delta_w w = 0$, the Green flux through $\{\rho = \varepsilon\}$ is $(a-b)\,\varepsilon^{a+b}\int f^2\,d\varphi$: it survives the limit $\varepsilon \to 0$ **iff** $a + b = 0$, the Wronskian parity $\mu \leftrightarrow -\mu$. The pair $(\mu_+, \mu_-) = (+1/2, -1/2)$ is the unique σ-odd pair with $a + b = 0$; in particular the $\rho^{-1/2}$ modes have vanishing self-Wronskian ($a - b = 0$), so only the $\pm 1/2$ pair sees the cokernel.*

Proof. ϕ_R is harmonic, so $\Delta_w(\chi\phi_R) = [\Delta_w, \chi]\,\phi_R$ is supported in the cut-off annulus and the pairing is absolutely convergent. Apply the Green identity on $\{\varepsilon < \rho < r_0\}$: since $\Delta_w w_R = 0$ there,

$$\int_{\varepsilon<\rho<r_0} \Delta_w(\chi\phi_R)\,w_R \;=\; \oint_{\rho=r_0} (w_R\,\partial_\rho(\chi\phi_R) - \chi\phi_R\,\partial_\rho w_R)\rho\,d\varphi \;-\; \oint_{\rho=\varepsilon} (w_R\,\partial_\rho\phi_R - \phi_R\,\partial_\rho w_R)\rho\,d\varphi.$$

The outer flux vanishes ($\chi \equiv 0$ near $\rho = r_0$). On the inner circle ($\chi \equiv 1$), with $a = 1/2$, $b = -1/2$: $w_R\,\partial_\rho\phi_R - \phi_R\,\partial_\rho w_R = (a-b)\,\rho^{a+b-1}\cos^2(\varphi/2)$, so the flux is $(a-b)\,\varepsilon^{a+b}\int_0^{2\pi}\cos^2(\varphi/2)\,d\varphi = 1\cdot\varepsilon^0\cdot\pi = \pi$, ε-independent. Integrating the constant longitudinal ($q = 0$) mode over Σ_i gives the factor $\ell(\Sigma_i)$; the scalar $A_{\mathrm{bulk}}(\alpha_1,\alpha_1)$ factors out by $(A1)$ rank-one rigidity. □

Remark 5.2 (why the cubic layer does not detect the cokernel, the withdrawn reduction). The cubic modes satisfy the *pointwise* model identity $\Delta_w\,\psi_R^{(i)} = 2\,\rho^{-1/2}\cos(\varphi/2)$ (indicial polynomial value $\mu^2 - m^2 = (3/2)^2 - (1/2)^2 = 2$; the arithmetic is correct). But the local trace coefficient at $\mu = -1/2$ is **not** the duality pairing that computes the class in $Y_{\beta-2}/\mathrm{Ran}$: by the Green computation of Lemma 5.2 with $a = 3/2$, $b = -1/2$, the flux is $(a-b)\,\varepsilon^{a+b}\pi = 2\varepsilon\pi \to 0$, i.e.

$$\langle \Delta_w(\chi_i\,\psi_R^{(i)}),\ w_R^{(i)}\rangle_{L^2} \;=\; 0 \quad \text{exactly:}$$

the annular cut-off commutator contributes $-2\pi\int\chi\,d\rho$ to the pairing and cancels the bulk trace contribution $+2\pi\int\chi\,d\rho$ of $2\rho^{-1/2}$ identically. Since moreover $\psi_R^{(i)}, \psi_I^{(i)} \in X_\beta$ for every $\beta \in (1/2, 3/2)$ (they decay like $\rho^{3/2}$), an extended domain built on the cubic layer is canonically isomorphic to X_β and does not change the range at all. The honest reduced map of the cubic layer is the zero map, and the $\mu = +1/2$ Wronskian formulation above is the correct augmentation.

Remark 5.3 (status in X_β: the augmentation is genuine). The displacement modes decay like $\rho^{1/2}$ at Σ_i, *slower* than ρ^β for every $\beta \in (1/2, 3/2)$: hence $\phi_R^{(i)}, \phi_I^{(i)} \notin X_\beta$ (indeed $\chi_i\phi_R^{(i)} \notin X_\beta$ for any cut-off). Adjoining them enlarges the domain in the only way that can change a cokernel, by classes outside the original space. This is the structural point the cubic layer failed: the extension must leave X_β, and the Wronskian parity of Lemma 5.2 shows the $\mu = +1/2$ modes are exactly the ones dual to the $\mu = -1/2$ obstruction.

5.3. The extended domain and the displacement layer

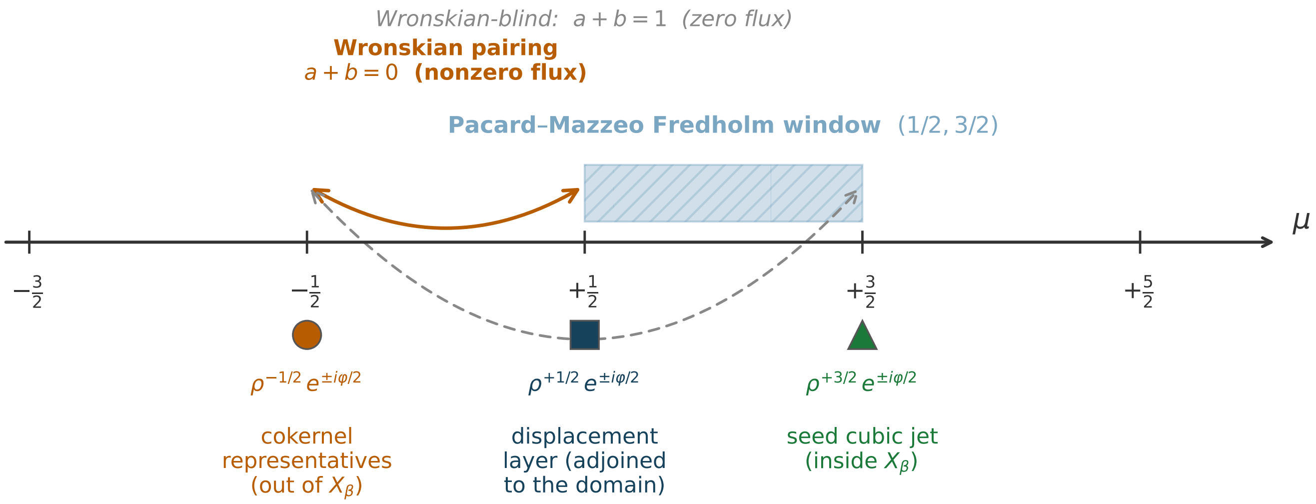

Figure 2. The three σ-odd half-integer objects on the indicial axis, not to be conflated. The cokernel representatives $\rho^{-1/2}e^{\pm i\varphi/2}$ lie outside X_β from below (Theorem 4.5); the displacement layer $\rho^{+1/2}e^{\pm i\varphi/2}$ is adjoined to the domain (extended-domain augmentation, this subsection); the seed cubic jet $\rho^{+3/2}e^{\pm i\varphi/2}$ sits inside X_β. Wronskian pairing $(a+b=0)$ links the cokernel to the displacement layer with nonzero Green flux (Lemma 5.2); the cubic jet is Wronskian-blind $(a+b=1)$ (Remark 5.2).

Let $\chi_i \in C^\infty_c(U_i)$ be a smooth radial cut-off with $\chi_i \equiv 1$ on $\{|w| \le r_0/4\}$ and $\operatorname{supp}\chi_i \subset \{|w| < r_0/2\}$. The cut-off transports the displacement modes $\phi_R^{(i)}, \phi_I^{(i)}$ from the local model on U_i to global σ-odd functions supported in U_i, smooth away from Σ_i.

The *displacement layer* is the space of N_h-valued σ-odd sections at $\mu = +1/2$ over Σ:

$$\mathcal{D}^{\mathrm{disp}}_{1/2} \;:=\; \bigoplus_{i=1}^{N} C^{0,\alpha}(\Sigma_i;\, \mathbb{R}^2),$$

with an element $\boldsymbol{c} = (c_R^{(i)}(s),\, c_I^{(i)}(s))_{i=1}^N$ representing the prescribed coefficients of $\phi_R^{(i)}, \phi_I^{(i)}$ along Σ_i, possibly depending on the longitudinal coordinate s. The dependence on s is essential: the displacement layer is a section bundle over Σ, not a finite-dimensional vector space.

The cut-off injection

$$\iota\colon \mathcal{D}^{\mathrm{disp}}_{1/2} \;\longrightarrow\; C^{0,\alpha}_{\mathrm{loc}}(M^7 \setminus \Sigma; N_h), \qquad \boldsymbol{c} \;\longmapsto\; \sum_{i=1}^{N} \chi_i\,\big(c_R^{(i)}(s)\,\phi_R^{(i)} + c_I^{(i)}(s)\,\phi_I^{(i)}\big)\cdot \alpha_1$$

embeds the displacement layer into the σ-odd functions on M^7, in the α_1-direction of N_h. By Remark 5.3, $\iota(\mathcal{D}^{\mathrm{disp}}_{1/2}) \cap X_\beta = \{0\}$: the sum with X_β is direct.

Definition 5.4 (the extended domain). *The extended domain is the Banach space*

$$X_\beta^{\mathrm{ext}} \;:=\; X_\beta \,\oplus\, \mathcal{D}^{\mathrm{disp}}_{1/2},$$

with norm $\|(u,\boldsymbol{c})\|_{X_\beta^{\mathrm{ext}}} := \|u\|_{X_\beta} + \|\boldsymbol{c}\|_{\mathcal{D}_{1/2}^{\mathrm{disp}}}$. *An element* $(u,\boldsymbol{c})$ *is canonically identified with the* σ*-odd function* $u+\iota(\boldsymbol{c})$*, whose polyhomogeneous coefficient at the regular root* $\mu=+1/2$*, angular modes* $m=\pm 1/2$*, equals* $\boldsymbol{c}$ *on each* Σ_i*.*

(Internally $X_\beta \cong X_\beta^{(0)} \oplus \mathcal{D}_{3/2}^{\mathrm{cub}}$ via the $\mu=3/2$ evaluation of §5.1, so the cubic coefficient remains an ordinary, undistinguished, polyhomogeneous datum inside X_β; only the $\mu=+1/2$ layer is adjoined from outside.)

Geometric interpretation: the displacement layer moves the branch points (free boundary). On the cover, $\phi_R^{(i)} = \operatorname{Re} z$ and $\phi_I^{(i)} = \operatorname{Im} z$: adding $a\,z$ ($a = c_R + ic_I$, suitably conjugated) to the seed $c_0\,z^3\,\alpha_1$ produces $c_0 z^3 + az$, whose branch points are *displaced* from $z=0$. Equivalently, downstairs: $c_0\,(w-\varrho)^{3/2} = c_0\,w^{3/2} - \frac{3}{2}\,c_0\,\varrho\,w^{1/2} + O(\varrho^2)$, so prescribing the $\mu=+1/2$ coefficient a is, to leading order, a transverse displacement of the branch point by $\varrho = -\frac{2}{3}\,a/c_0$, the map "normal displacement $\mapsto \mu=+1/2$ coefficient" is multiplication by $-\frac{3}{2}c_0 \neq 0$, an isomorphism. The augmentation parameters are therefore *transverse displacements of the discriminant circles* Σ_i: N circles $\times$ two transverse directions $=2N$ degrees of freedom ($=154$ at $N=77$), matching the cokernel exactly. This is the free-boundary mechanism, the motion of the branch locus generates the obstruction modes and kills the cokernel, and it is precisely [3]'s formulation in which the discriminant Σ is part of the unknown ("regularity appears after moving the branch locus", [3, §3]). The fixed-Σ presentation retained below treats the displacement as a prescribed asymptotic datum on the fixed tubes; the two presentations are equivalent by the following lemma.

Lemma 5.4 (nonlinear compatibility of the discriminant displacement). *Let* $\mathcal{D}$ *satisfy* $(A3)$*,* $(A4)$*,* $(A7)$*, and let* $\boldsymbol{c}\in\mathcal{D}_{1/2}^{\mathrm{disp}}$ *with*

$$|\boldsymbol{c}| \;\le\; c_*(\mathcal{D}) \;:=\; \tfrac{3}{16}\,|c_0|\,r_0,$$

so that the induced displacements $\varrho_i = -\frac{2}{3}\,a_i/c_0$ *satisfy* $|\varrho_i|\le r_0/8$*. Then:*

(i) (splitness preserved) the displaced discriminant $\Sigma(\boldsymbol{c}) = \bigsqcup_i \Sigma_i(\varrho_i)$ *remains a geometrically split unlink: each* $\Sigma_i(\varrho_i)$ *stays inside the tube* $\{\rho\le r_0/8\}\subset U_i$*, the isotopy to* Σ_i *is supported in* U_i*, and the pairwise separation satisfies* $d_{\min}(\boldsymbol{c}) \ge d_{\min} - r_0/4$*, at* $\mathcal{D}_0$ *(*$r_0 = 10^{-2}$*,* $d_{\min}=1$*) a relative loss* $\le 0.25\%$*, leaving the* $(A3)/(A6)$ *margins intact;*

(ii) (positivity preserved) the displaced seed is the translate $c_0\,(w-\varrho_i)^{3/2}\,\alpha_1$ *of the local model of* $(A4)$*: it satisfies the positivity condition of [3, Def. 2] verbatim at the displaced branch point, and on the annulus* $\{\rho\ge r_0/4\}$ *the deviation from the undisplaced seed is* $O(|\varrho_i|\,\rho^{1/2}) = O(|\boldsymbol{c}|\,r_0^{1/2})$*, absorbed in the* $(A4)$ *envelope;*

(iii) (diffeomorphism action inside the IFT) the ambient diffeomorphism $\Phi_{\boldsymbol{c}}$ *of* B^3*, supported in* $\bigsqcup_i U_i$ *and translating the transverse discs by* ϱ_i *near* Σ_i*, pulls the displaced-locus problem back to the fixed-locus problem:* $\Phi_{\boldsymbol{c}}^*$ *carries the edge data at* $\Sigma_i(\varrho_i)$ *to edge data at* Σ_i*, and modifies the coefficients of* $\mathcal{J}_h$ *and of the nonlinearity by terms of size* $O(|\varrho_i|/r_0) = O(|\boldsymbol{c}|/(|c_0|\,r_0))$ *in* $C^{k,\alpha}$*, smooth (indeed polynomial) in* $\boldsymbol{c}$*. The displacement therefore enters the nonlinear equation as a smooth finite-dimensional parameter in the Mazzeo–Pacard implicit-function framework, with no loss of derivative.*

Proof. (i) is metric bookkeeping: $|\varrho_i|\le r_0/8$ keeps $\Sigma_i(\varrho_i)$ in the tube, the straight-line isotopy $t\mapsto t\varrho_i$ stays in U_i (tubes are pairwise disjoint by $(A3)$), and the triangle inequality gives the separation bound. (ii) the local model of $(A4)$ is translation-covariant in w; positivity of [3, Def. 2] is a pointwise condition on the branched jet at the branch point, invariant under the translation; the annulus estimate is the first-order

Taylor bound on $(w-\varrho)^{3/2}-w^{3/2}$ for $|w|\geq r_0/4\geq 2|\varrho|$. (iii) take $\Phi_{\boldsymbol{c}}(s,w)=(s,w-\eta(\rho)\varrho_i)$ on U_i with η a radial bump, $\eta\equiv 1$ on $\rho\leq r_0/4$, $\eta\equiv 0$ for $\rho\geq r_0/2$: then $\|\Phi_{\boldsymbol{c}}-\mathrm{id}\|_{C^{k+1}}=O(|\varrho_i|/r_0)$, the pull-back metric and coefficients differ by the same order, and the dependence on $\boldsymbol{c}$ is affine in $\Phi_{\boldsymbol{c}}$ hence smooth in the coefficients, standard free-boundary reduction to a fixed domain [cf. Mazzeo–Pacard]. $\square$

At $\mathcal{D}_0$ the smallness demanded by Lemma 5.4 is generous: the datum actually needed is of size $\|\boldsymbol{c}\|\sim\|\mathfrak{M}^{-1}\|\,\|m(\bar{h})\|$-readout $\lesssim 10^{-5}$ (Suppl. App. F, Proposition 3.1), five orders below $c_*(\mathcal{D}_0)\sim 2\cdot 10^{-3}$.

5.4. The extended-domain Fredholm theorem

The Jacobi operator extends to the extended domain by acting on the canonical identification:

$$\mathcal{J}_h^{\mathrm{ext}}\colon X_\beta^{\mathrm{ext}}\;\longrightarrow\;Y_{\beta-2},\qquad \mathcal{J}_h^{\mathrm{ext}}(u,\boldsymbol{c})\;:=\;\mathcal{J}_h\big(u+\iota(\boldsymbol{c})\big).$$

This is well-defined: $\mathcal{J}_h$ maps X_β to $Y_{\beta-2}$ (with the cokernel of Theorem 4.5), and $\mathcal{J}_h(\iota(\boldsymbol{c}))\in Y_{\beta-2}$ even though $\iota(\boldsymbol{c})\notin X_\beta$, indeed, expanding via the Leibniz rule,

$$\mathcal{J}_h\big(\chi_i\,(c\,\phi)\big)\;=\;\chi_i\,c\,(\mathcal{J}_h\phi)\;+\;\chi_i\,\phi\,(\partial_s^2 c)\;+\;[\mathcal{J}_h,\,\chi_i\,c]\,\phi,$$

the first term is $O(\kappa_g\rho)\cdot\rho^{-3/2}$-controlled at the tip ($\phi$ is harmonic for the model operator, so only the geometric perturbation $\mathcal{R}$ of §4.1 contributes there), the commutator $[\mathcal{J}_h,\chi_i\,c]\,\phi$ is supported in the annular cut-off region $\{r_0/4<|w|<r_0/2\}$ where $\nabla\chi_i\neq 0$, and the longitudinal term $\chi_i\,\phi\,(\partial_s^2 c)$ vanishes when c is s-independent; all contributions lie in $Y_{\beta-2}$ (the mode $\rho^{1/2}$ decays faster than the weight $\rho^{\beta-2}$ requires).

The cokernel of $\mathcal{J}_h^{\mathrm{ext}}$ inherits the section-bundle structure of Theorem 4.5, with the contribution of the displacement layer $\mathcal{D}_{1/2}^{\mathrm{disp}}$ subtracted. Throughout, the projection $\Pi_{\mathrm{coker}}\colon Y_{\beta-2}\to\mathrm{coker}(\mathcal{J}_h)$ is realised by the L^2 pairing against the adjoint cokernel representatives $w_R^{(i)},w_I^{(i)}$ of Corollary 5.8, normalised per component by $(\pi\,\ell(\Sigma_i))^{-1}$ and oriented so that the diagonal of $\mathfrak{M}$ below is positive (concretely, $w_i\to -w_i$ relative to the representatives of Lemma 5.2, whose raw Green pairing is negative), *not* by the local indicial trace coefficient at $\mu=-1/2$ (Remark 5.2). The structure is captured by the following theorem.

Theorem 5.5 (extended-domain Fredholm; Wronskian-pairing form). *Let $\mathcal{D}$ be a branched rank-one admissible datum satisfying* (A0)–(A8). *The extended Jacobi operator*

$$\mathcal{J}_h^{\mathrm{ext}}\colon X_\beta^{\mathrm{ext}}=X_\beta\oplus\mathcal{D}_{1/2}^{\mathrm{disp}}\;\longrightarrow\;Y_{\beta-2}$$

is bounded. Its cokernel is the quotient

$$\mathrm{coker}\big(\mathcal{J}_h^{\mathrm{ext}}\big)\;=\;\mathrm{coker}\big(\mathcal{J}_h\big)\,/\,\mathrm{Image}\big(\Pi_{\mathrm{coker}}\circ\mathcal{J}_h^{\mathrm{ext}}|_{0\,\oplus\,\mathcal{D}_{1/2}^{\mathrm{disp}}}\big),$$

where the cokernel-induced map from the displacement layer to the adjoint-trace section bundle

$$\mathfrak{M}\colon\mathcal{D}_{1/2}^{\mathrm{disp}}\;\longrightarrow\;\bigoplus_{i=1}^{N}\Gamma\big(\Sigma_i;\mathcal{T}_{-1/2}^{R}\oplus\mathcal{T}_{-1/2}^{I}\big),$$

$$\mathfrak{M}(\boldsymbol{c})^{(i),R/I}\;:=\;\frac{1}{\pi\,\ell(\Sigma_i)}\big\langle\mathcal{J}_h^{\mathrm{ext}}(0,\boldsymbol{c}),\;w_{R/I}^{(i)}\big\rangle_{L^2},$$

with $w_{R/I}^{(i)}$ the adjoint cokernel representatives of Corollary 5.8, is a bounded operator between section bundles over Σ, given on each component Σ_i by

$$\mathfrak{M}^{(i)}(c_R^{(i)}(s),\, c_I^{(i)}(s)) \;=\; (\mu_+{-}\mu_-)\, A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\, (c_R^{(i)}(s),\, c_I^{(i)}(s)) + \mathfrak{E}_{\mathrm{geom}}^{(i)}(\boldsymbol{c}) + \mathfrak{E}_{\mathrm{link}}(\boldsymbol{c})\big|_{\Sigma_i}, \qquad \mu_+{-}\mu_- = 1,$$

where the diagonal block $A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\, I$ arises from the Wronskian/Green pairing of Lemma 5.2, $\mathfrak{E}_{\mathrm{geom}}^{(i)}$ is a bounded operator on $C^{0,\alpha}(\Sigma_i;\mathbb{R}^2)$ controlled by the geodesic curvature κ_g of Σ_i in B^3, and $\mathfrak{E}_{\mathrm{link}}$ is a bounded operator coupling the components Σ_i via the Newton kernel on $B^3 \setminus \Sigma$ with magnitude controlled by $(r_0/d_{\min})^2$ (dipole bound, §5.7).

Remarks.

(i) *Section bundles, not finite-dim.* The domain $\mathcal{D}_{1/2}^{\mathrm{disp}}$ and the cokernel $\bigoplus_i \Gamma(\Sigma_i; \mathcal{T}_{-1/2})$ are both infinite-dimensional section bundles over Σ. Theorem 5.5 asserts the structure of $\mathfrak{M}$ as a bounded operator between these section bundles, with a model diagonal $A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\, I$ plus computable perturbations.

(ii) *Positivity of the diagonal block.* Since A_{bulk} is positive definite, the diagonal block $A_{\mathrm{loc}} := (\mu_+ - \mu_-)\, A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\, I = A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\, I$ is a positive scalar multiple of the identity with smallest singular value $\sigma_{\min}(A_{\mathrm{loc}}) = A_{\mathrm{bulk}}(\alpha_1,\alpha_1) > 0$; conservatively $\sigma_{\min}(A_{\mathrm{loc}}) \geq 1/\mathrm{cond}(A_{\mathrm{bulk}})$ in the $(A5)$ normalisation, equal to 1 in its normalised column.

(iii) *Reduction to $\mathbb{R}^{2N}$.* The reduction of $\mathfrak{M}$ from a section-bundle map to a finite-dimensional linear map $\mathbb{R}^{2N} \to \mathbb{R}^{2N}$ is established in §5.5–§5.7 via the Dirichlet-to-Neumann construction at the radial interface $\rho = r_0$, in combination with $(A1)$ rank-one rigidity + $(A2_{\mathrm{collar}})$ totally geodesic background.

(iv) *Identification of perturbations.* The geometric perturbation $\mathfrak{E}_{\mathrm{geom}}^{(i)}$ encodes the Frenet–Serret framing of the σ-odd basis $(\cos(\varphi/2), \sin(\varphi/2))$ along Σ_i in B^3 and the variation of A_{bulk} across Σ. The linking perturbation $\mathfrak{E}_{\mathrm{link}}$ encodes the long-range coupling between distinct components via the Newton kernel of $\mathcal{J}_h$ on the source-free affine background, with operator norm bounded by a multiple of $(r_0/d_{\min})^2$, the dipole bound, sharper than the naive monopole estimate, by §5.7. Sharp constants on $\mathfrak{E}_{\mathrm{geom}}, \mathfrak{E}_{\mathrm{link}}$ are tracked in Supplementary Appendix D.

Proof outline of Theorem 5.5. Boundedness of $\mathcal{J}_h^{\mathrm{ext}}$: on the X_β summand $\mathcal{J}_h$ is bounded into $Y_{\beta-2}$; on the displacement layer the Leibniz expansion above gives the bound. The cokernel identification: applying Π_{coker} on both sides, the contribution from X_β projects to zero ($\Pi_{\mathrm{coker}} \circ \mathcal{J}_h = 0$ on X_β, since $\mathcal{J}_h(X_\beta) = \mathrm{Ran}(\mathcal{J}_h)$ annihilates the adjoint pairing, this includes the cubic layer inside X_β, consistently with Remark 5.2), leaving only the contribution from $\iota(\mathcal{D}_{1/2}^{\mathrm{disp}})$, which by definition is $\mathfrak{M}$. The diagonal $A_{\mathrm{bulk}}(\alpha_1,\alpha_1)\, I$ structure: compute $\Pi_{\mathrm{coker}}\, \mathcal{J}_h(\chi_i \phi_R^{(i)})$, where $\mathcal{J}_h = A_{\mathrm{bulk}} \cdot \Delta_w$ reduces on the α_1-direction to $A_{\mathrm{bulk}}(\alpha_1,\alpha_1) \cdot \Delta_w$ (constant scalar): by Lemma 5.2 the L^2 pairing against $w_R^{(i)}$ localises to the Green flux and evaluates to $-(\mu_+ - \mu_-)\, \pi\, \ell(\Sigma_i)\, A_{\mathrm{bulk}}(\alpha_1,\alpha_1)$, independent of the cut-off, since the flux $(a-b)\varepsilon^{a+b}$ is ε-independent exactly when $a+b=0$, which after the $(\pi\ell(\Sigma_i))^{-1}$ normalisation and the orientation convention on the w_i (preamble above) is the stated positive diagonal. The same value is obtained against the *global* impedance-matched adjoint solution of Corollary 5.8: it differs from the model representative $\rho^{-1/2}\cos(\varphi/2)$ by a regular $\rho^{+1/2}$ admixture, Wronskian-blind, $a - b = 0$, zero flux at every ε, plus higher-order terms with $a+b>0$ whose flux vanishes as $\varepsilon \to 0$; the diagonal is therefore robust to the model↔global difference, the deviations landing in E_{geom}. Identification of the perturbations as $\mathfrak{E}^{(i)}_{\mathrm{geom}}$ (intra-component, Frenet–Serret, sourced by the geometric perturbation $\mathcal{R} = O(\kappa_g \rho)$ acting on the $\rho^{1/2}$ mode) and $\mathfrak{E}_{\mathrm{link}}$ (inter-component, Newton kernel) follows from the standard contributions of the deviation of the actual edge geometry from the local model. □

5.5. The Dirichlet-to-Neumann interface and the $\mathbb{R}^{2N}$ cokernel

Theorem 5.5 identifies the cokernel-induced map $\mathfrak{M}$ on the full section bundle, which is infinite-dimensional. The reduction to a finite-dimensional cokernel proceeds via a Dirichlet-to-Neumann (DtN) matching at the radial interface $\rho = r_0$, using the explicit bounded solutions of $\mathcal{J}_h u = 0$ in the inner tube $\{|w| < r_0\}$ from §4.1.

Recall from §4.1 that, on the inner tube, the bounded radial part of the (q, m) separated mode is

$$R_{q,m}(\rho) \;=\; \begin{cases} \rho^{|m|} & q = 0, \\ I_m(|q|\rho/L_i) & q \neq 0, \end{cases}$$

where I_m is the modified Bessel function of the first kind, $L_i = \ell(\Sigma_i)/(2\pi)$, $m \in (1/2) + \mathbb{Z}$ (σ-odd). For the lowest σ-odd angular mode $m = 1/2$, $I_{1/2}$ is elementary: $I_{1/2}(x) = \sqrt{2/(\pi x)}\,\sinh(x)$, with logarithmic derivative

$$\frac{I'_{1/2}(x)}{I_{1/2}(x)} \;=\; \coth(x) \;-\; \frac{1}{2x}.$$

Definition 5.6 (DtN map). *The Dirichlet-to-Neumann map at $\rho = r_0$ on the σ-odd sector is the bounded linear operator*

$$\Lambda_{\mathrm{in}} \colon u\big|_{\rho=r_0} \longmapsto \partial_\rho u\big|_{\rho=r_0}$$

acting on traces of solutions u of $\mathcal{J}_h u = 0$ in the inner tube $\{|w| < r_0\}$. On the separated mode (q, m) in the σ-odd sector, the DtN eigenvalue is

$$\Lambda_{q,m} \;=\; \frac{|q|}{L_i}\,\frac{I'_m(|q|r_0/L_i)}{I_m(|q|r_0/L_i)} \quad (q \neq 0), \qquad \Lambda_{0,m} \;=\; \frac{|m|}{r_0}.$$

In particular, for the lowest σ-odd mode $m = \pm 1/2$,

$$\Lambda_{q,1/2} \;=\; \frac{|q|}{L_i}\,\coth\!\Big(\frac{|q|r_0}{L_i}\Big) \;-\; \frac{1}{2r_0}.$$

Lemma 5.7 (DtN positivity). *For all $q \in \mathbb{Z}$ and all $m \in \mathcal{E}^{\sigma}_{\mathrm{ind}}$ with $m > 0$,*

$$\Lambda_{q,m} \;\geq\; \frac{1}{2r_0} \;>\; 0,$$

with the minimum attained at $(q, m) = (0, \pm 1/2)$.

Proof. For $m = 1/2$, the bound reduces to the elementary inequality $\coth(x) \geq 1/x$ for $x > 0$, which follows from the Mittag–Leffler partial-fraction expansion (DLMF 4.36.3 [17])

$$\coth(x) \;=\; \frac{1}{x} \;+\; \sum_{n\geq 1} \frac{2x}{x^2+n^2\pi^2},$$

whose summands are all non-negative for $x>0$; hence $\Lambda_{q,1/2} = (|q|/L_i)\,\coth(|q|r_0/L_i) - 1/(2r_0) \geq 1/r_0 - 1/(2r_0) = 1/(2r_0)$. For higher $m \in \{3/2,\, 5/2,\, \dots\}$, the $q=0$ value is $\Lambda_{0,m} = m/r_0 \geq 3/(2r_0) > 1/(2r_0)$; for $q \neq 0$, the inequality $(|q|/L_i)\, I_m'(|q|r_0/L_i)/I_m(|q|r_0/L_i) \geq m/r_0$ follows from the monotonicity of the Bessel ratio $x \mapsto I_m'(x)/I_m(x)$. $\square$

The geometric perturbation $\mathcal{R}(s,w) = O(\kappa_g \rho)$ of §4.1 breaks the exact separation of variables but only at sub-leading order; the DtN matching extends to the full $\mathcal{J}_h$ via Neumann series, with the perturbation factor $O(\kappa_g r_0)$ absorbed under $(A4)$.

Outer DtN: model and global operator. The inner block above is exact. The outer block, however, is only a *model*: replacing the compact non-radial complement $B^3 \setminus \bigcup_i U_i$ by a single radial exterior $\{|w|>r_0\}$ with decay at $\rho \to \infty$. On that model, source-free solutions of $\mathcal{J}_h u = 0$ have radial part $R_{\text{out}}(\rho) = K_m(|q|\rho/L_i)$ for $q \neq 0$ and $R_{\text{out}}(\rho) = \rho^{-|m|}$ for $q=0$, where K_m is the modified Bessel function of the second kind. For $m=1/2$, $K_{1/2}(x) = \sqrt{\pi/(2x)}\, e^{-x}$ with $K_{1/2}'(x)/K_{1/2}(x) = -1 - 1/(2x)$, so the model outer DtN eigenvalue (outward normal $+\partial_\rho$) on the lowest mode is

$$\Lambda^{\text{model}}_{\text{out};q,1/2} \;=\; -\,\frac{|q|}{L_i} \;-\; \frac{1}{2r_0}.$$

The genuine outer operator on the compact complement is $\Lambda^{\text{glob}}_{\text{out}} = \Lambda^{\text{model}}_{\text{out}} + E_{\text{out}}$, where E_{out} encodes the finite geometry of $B^3 \setminus \bigcup_i U_i$ and the inter-component coupling of longitudinal modes; the model captures only the leading radial behaviour.

Global outer coercivity as a quantitative admissibility condition. The coercivity of the global matching operator $T^{\text{glob}}_{q,m} := \Lambda_{\text{in};q,m} - \Lambda^{\text{glob}}_{\text{out};q,m}$ is stated as a *quantitative admissibility condition* (E-quant), parallel to $(A0)$–$(A8)$ and verifiable per explicit construction, discharged by Proposition 5.7 under a datum-level numerical margin.

The deviation $E_{\text{out}} := \Lambda^{\text{glob}}_{\text{out}} - \Lambda^{\text{model}}_{\text{out}}$ decomposes, by a Green pairing on $B^3 \setminus \bigcup_i U_i$, into three geometrically-transparent contributions:

- (D_1) *Frenet–Serret deviation*: the true collar boundary ∂U_i is not exactly cylindrical but bends with geodesic curvature κ_g; this contributes a perturbation of size $O(\|\kappa_g\|_{C^0} \cdot 1)$ to $r_0 \cdot E_{\text{out}}$ via the same Neumann-series argument used for the inner side (Lemma 5.7, geometric perturbation $\mathcal{R} = O(\kappa_g \rho)$).
- (D_2) *Inter-tube coupling*: the σ-odd obstruction mode at Σ_i, propagated through the compact complement to ∂U_j, couples to the σ-odd mode around Σ_j; the resulting cross-term has the *same physical origin* as the coupling entries E_{link} of Theorem 5.10, and its exponent is now **derived** (Suppl. App. E.4): on the rank-one branched double cover the σ-odd field with a singular obstruction mode at one locus is *regular* at the other (a structural vanishing), so the direct obstruction–obstruction coupling is zero and the leading coupling proceeds by double reflection through the regular sector, giving a **dipole-suppressed** $(r_0/d_{\min})^2$ **decay per adjacent pair** with $C_{\text{link}} = O(1)$, the exponent $\gamma = 1$ is excluded, not merely assumed.

- (D_3) *Finite outer geometry*: the compactness of B^3 (as opposed to the model exterior extending to $\rho \to \infty$) makes the transverse Green kernel deviate from the flat model beyond the split-ball scale $R_B := \min_i \operatorname{dist}(U_i, \partial\mathbb{B}_i)$, with $\mathbb{B}_i$ the pairwise-disjoint balls of ($A3$), the radius out to which the flat radial exterior model is valid; the base being closed, there is no actual boundary. The deviation is estimated by an image charge at the model boundary $\{\rho = R_B\}$: on $q \neq 0$ modes it is exponentially suppressed by the Bessel decay of K_m, on $q = 0$ modes the explicit transverse image charge gives $O((r_0/R_B)^{2m}) = O(r_0/R_B)$ for the lowest mode $m = 1/2$ (a correction of the earlier $(r_0/R_B)^2$ envelope: the true exponent is $2m = 1$, still small at $\mathcal{D}_0$).

Combining,

$$\|E_{\mathrm{out}}\| \cdot r_0 \;\le\; \delta_{\mathrm{E}}(\mathcal{D}) \;:=\; C_{\mathrm{FS}}\,\|\kappa_g\|_{C^0}\, r_0 \;+\; C_{\mathrm{link}}\,(N-1)\,(r_0/d_{\min})^2 \;+\; C_{\mathrm{outer}}\,(r_0/R_B)^1$$

with $C_{\mathrm{FS}}, C_{\mathrm{link}}, C_{\mathrm{outer}}$ explicit $O(1)$ constants extracted from the Green identity; Suppl. App. E.4 gives $C_{\mathrm{FS}} \approx 0.29$ (one-dimensional Sturm–Liouville comparison; the true first-order angular contribution $\propto \kappa_g \rho \cos\phi$ has vanishing diagonal, so the effect is in fact $O((\kappa_g r_0)^2)$), **derives the inter-tube exponent** $\gamma = 2$ with $C_{\mathrm{link}} = O(1)$ (double-cover structural vanishing, above), and gives $C_{\mathrm{outer}} = 1$ at exponent $2m = 1$. The next admissibility condition asserts that this quantitative smallness holds:

(**E-quant**) **Global outer smallness.** *The dimensionless quantity* $\delta_{\mathrm{E}}(\mathcal{D})$ *satisfies*

$$\delta_{\mathrm{E}}(\mathcal{D}) \;\le\; \tfrac{1}{2}.$$

(E-quant) is verifiable by construction: at $\mathcal{D}_0$ (with $r_0 = 10^{-2}$, $d_{\min} = 1$, $\|\kappa_g\|_{C^0} \le 1$, $N = 77$, $R_B \sim 1$), one finds $\delta_{\mathrm{E}}(\mathcal{D}_0) \le 0.021$ (the three terms being ≈ 0.003, 0.0076, 0.010), a factor ~ 24 below the threshold (see §9.2, Suppl. App. F and a supplementary interval-arithmetic script). The dipole scaling $(r_0/d_{\min})^2$ is now **derived** from the rank-one branched double-cover structure (not merely inherited): the direct obstruction–obstruction coupling vanishes structurally, excluding the exponent $\gamma = 1$ (which would give $\delta_{\mathrm{E}}(\mathcal{D}_0) \approx 0.78 > 1/2$ and break the discharge). Under (E-quant), the following is a proposition.

Proposition 5.7 (Global outer coercivity, conditional closure). *Status: conditional on a datum-level numerical margin, with all three universal constants theorem-grade at* $\mathcal{D}_0$. *The load-bearing constant, the inter-tube exponent* γ, *is theorem-grade:* $\gamma = 2$ *is derived from the rank-one branched double cover (Suppl. App. E.4), and* $\gamma = 1$ *(which would break the discharge) is excluded by a structural vanishing. The scalar constants* $C_{\mathrm{FS}} \approx 0.29$ *and* $C_{\mathrm{outer}} = 1$ *(exponent* $2m = 1$*) are sharp. The round-trip inner-DtN constant* C_{link} *is theorem-grade via impedance match on the pure model (*$\kappa^{\mathrm{model}} = \infty$*, so* $C_{\mathrm{link}}^{\mathrm{model}} = 0$ *exactly) and* $|c_0|^2$ *parity suppression of the finite correction with type-IV curvature bound, giving* $C_{\mathrm{link}} \le |c_0|^2 \beta_{\mathrm{link}} \le 1$ *(Suppl. App. E.3). The numerical margin at* $\mathcal{D}_0$ *(*$\sim 24\times$*) is robust.*

Under the admissibility conditions ($A4$), ($A5$), ($A6$), ($A7$) *and the quantitative smallness* (E-*quant*), *the global matching operator satisfies*

$$T^{\mathrm{glob}}_{q,m} \;\ge\; \frac{1 - \delta_{\mathrm{E}}(\mathcal{D})}{r_0} \;\ge\; \frac{1}{2r_0} \;>\; 0$$

uniformly in the σ-odd sector with $m > 0$. Equivalently, $c_{\mathrm{E}}(\mathcal{D}) \geq 1/2$.

Proof. Lemma 5.7′ gives $T^{\mathrm{model}}_{q,m} \geq 1/r_0$. The three-term decomposition (D_1)–(D_3) above bounds $E_{\mathrm{out}} = \Lambda^{\mathrm{glob}}_{\mathrm{out}} - \Lambda^{\mathrm{model}}_{\mathrm{out}}$ by $\delta_{\mathrm{E}}(\mathcal{D})/r_0$; hence $T^{\mathrm{glob}}_{q,m} = T^{\mathrm{model}}_{q,m} - E_{\mathrm{out}} \geq (1-\delta_{\mathrm{E}})/r_0$. Under (E-quant), $\delta_{\mathrm{E}} \leq 1/2$, giving the stated bound. □

Remark 5.7-B. Suppl. App. E.4 carries out the sharp Green-identity analysis. Its main result is that the inter-tube exponent is $\gamma = 2$, **derived** rather than assumed: uniformising the rank-one branched double cover of the transverse plane at two loci Σ_i, Σ_j (deck involution $\zeta \mapsto 1/\zeta$), the σ-odd field carrying a singular obstruction mode at Σ_i, namely $\frac{1}{2}(1+\zeta)/(\zeta-1)$, *vanishes* at Σ_j; the direct obstruction–obstruction coupling is therefore structurally zero, so the leading coupling is a double reflection through the regular sector, of order $(r_0/d_{\min})^2$. This excludes the discharge-breaking exponent $\gamma = 1$. The note also gives $C_{\mathrm{FS}} \approx 0.29$ (with a further $(\kappa_g r_0)^2$ suppression from the vanishing first-order angular average) and $C_{\mathrm{outer}} = 1$ at the corrected exponent $2m = 1$. The round-trip inner-DtN constant C_{link} is closed certified in Suppl. App. E.3: on the pure model, the σ-odd $m = 1/2$ external DtN $\Lambda^{\mathrm{model}}_{\mathrm{out}} = -1/(2r_0)$ matches the interior $\Lambda^{\mathrm{model}}_{\mathrm{in}} = +1/(2r_0)$ exactly on the regular mode $\rho^{1/2}$, so the reflected singular admixture is identically zero ($\kappa^{\mathrm{model}} = \infty$, $C^{\mathrm{model}}_{\mathrm{link}} = 0$); with finite branched seed c_0, the leading correction is quadratic by parity (the c_0^1 diagonal on the σ-odd sector vanishes, same parity mechanism) and controlled by the type-IV sectional curvature $|K_{\mathrm{sec}}| \leq 1$, giving $C_{\mathrm{link}} \leq |c_0|^2 \beta_{\mathrm{link}} \leq 1$ at $\mathcal{D}_0$. All three universal constants of Proposition 5.7 are therefore certified at $\mathcal{D}_0$.

Under Proposition 5.7, the model matching operator, defined as the Steklov–Robin difference $T^{\mathrm{model}}_{q,m} := \Lambda_{\mathrm{in};q,m} - \Lambda^{\mathrm{model}}_{\mathrm{out};q,m}$, controls the global one; the model has the closed form

$$T^{\mathrm{model}}_{q,1/2} \;=\; \frac{|q|}{L_i}\left(\coth(|q|r_0/L_i) + 1\right) \;=\; \Lambda_{\mathrm{in};q,1/2} + \frac{1}{2r_0} + \frac{|q|}{L_i}.$$

Lemma 5.7′ (model matching coercivity). *For all $q \in \mathbb{Z}$ and all $m \in \mathcal{E}^{\sigma}_{\mathrm{ind}}$ with $m > 0$, the model matching operator satisfies*

$$T^{\mathrm{model}}_{q,m} \;\geq\; \frac{1}{r_0} \;>\; 0,$$

with the minimum attained at $(q,m) = (0, \pm 1/2)$. Higher modes $m = 3/2, 5/2, \ldots$ at $q = 0$ give $T^{\mathrm{model}}_{0,m} = 2m/r_0 \geq 3/r_0$. Under (E-*quant*) *the same coercivity holds for the global operator $T^{\mathrm{glob}}_{q,m}$ up to the constant $c_{\mathrm{E}}(\mathcal{D}) \geq 1/2$ (Proposition 5.7).*

Proof. From Lemma 5.7, $\Lambda_{\mathrm{in};q,1/2} \geq 1/(2r_0)$. Adding $1/(2r_0) + |q|/L_i$ from the closed form of T^{model} gives $T^{\mathrm{model}}_{q,1/2} \geq 1/r_0 + |q|/L_i \geq 1/r_0$, with equality at $q = 0$. Higher modes follow from $\Lambda_{\mathrm{in};0,m} = m/r_0$, $\Lambda^{\mathrm{model}}_{\mathrm{out};0,m} = -m/r_0$. The global statement is Proposition 5.7. □

The matching coercivity drives the finite-dimensional reduction:

Corollary 5.8 (cokernel is $\mathbb{R}^{2N}$; under quantitative admissibility). *Under* (*A*4), (*A*5), (*A*6), (*A*7) *and* (E-*quant*), *the global matching operator $T^{\mathrm{glob}}_{q,m}$ has trivial kernel on every (q,m) in the σ-odd sector (Proposition 5.7), so the inner–outer matching at $\rho = r_0$ is uniquely invertible. The Fredholm cokernel of $\mathcal{J}_h \colon X_\beta \to Y_{\beta-2}$ is then sourced entirely by the polyhomogeneous trace obstruction at $\mu = -1/2$ on the $q = 0$ longitudinal Fourier component, and is the finite-*

dimensional space

$$\operatorname{coker}(\mathcal{J}_h) \;=\; \bigoplus_{i=1}^{N} \mathbb{R}^2 \;\cong\; \mathbb{R}^{2N},$$

identified with the $(q,m)=(0,\pm 1/2)$ *adjoint trace data on each* Σ_i,

$$\operatorname{coker}(\mathcal{J}_h)\big|_{\Sigma_i} \;=\; \operatorname{span}\{\, \rho^{-1/2}\,\cos(\varphi/2),\; \rho^{-1/2}\,\sin(\varphi/2)\,\} \;\subset\; \mathcal{T}_{-1/2}\big|_{\Sigma_i}.$$

The non-constant Fourier modes ($q \neq 0$) *carry no cokernel contribution: matching a bounded inner solution to a decaying outer solution would require a non-trivial homogeneous trace, which* $T^{\mathrm{glob}}_{q,m} > 0$ *rules out.*

The cokernel is therefore finite-dimensional of dimension exactly $2N$ ($= 154$ for $N = 77$), identified with the constant adjoint-trace data along each branch component. Note the *three distinct half-integer objects* not to be conflated: the seed cubic jet at $\mu = +3/2$ ($\rho^{3/2}$, §5.2, inside X_β, Wronskian-blind to the cokernel by Remark 5.2), the adjoint trace representing the cokernel at $\mu = -1/2$ ($\rho^{-1/2}$, above), and the regular indicial mode $\rho^{+1/2}$, which is *outside* X_β and is precisely the displacement layer of §5.3, the unique σ-odd partner with non-vanishing Wronskian against the cokernel (Lemma 5.2). The augmentation by $\mathbb{R}^{2N}$ of §5.3 (prescribed branch-displacement data, Theorem 5.10) cancels this obstruction exactly.

5.6. s-uniformity of the seed residual under $(A1) + (A2_{\mathrm{collar}})$

The construction of the maximal background v_0 (level 2 of §3.1) is driven by the residual $m(\bar h)$ of the *local seed* $\bar h$ (level 1): the displacement datum $v_0 \in \mathbb{R}^{2N}$ is chosen to cancel the leading $\mu = -1/2$ obstruction (the adjoint pairing) of $m(\bar h)$, and for this to reduce to a finite-dimensional $\mathbb{R}^{2N}$ cancellation we need that obstruction to be s-uniform (i.e. constant in the $q = 0$ Fourier mode) along each Σ_i. This is a *source-level* condition, derived from the structural admissibility hypotheses.

Lemma 5.9 (s-uniformity of the seed residual). *Under* $(A1)$ *rank-1 monodromy and* $(A2_{\mathrm{collar}})$ *collar-affine background, the leading polyhomogeneous coefficient* $\mathrm{ev}^{(i)}_{-1/2}(m(\bar h))(s) \in C^{0,\alpha}(\Sigma_i;\mathbb{R}^2)$ *of the seed residual satisfies*

$$\big\|\partial_s\, \mathrm{ev}^{(i)}_{-1/2}(m(\bar h))\big\|_{C^{0,\alpha}(\Sigma_i)} \;\le\; C(\mathcal{D})\, \|\kappa_g\|_{C^0}\, r_0\, \big\|\mathrm{ev}^{(i)}_{-1/2}(m(\bar h))\big\|_{C^{0,\alpha}(\Sigma_i)},$$

on each Σ_i, *with* $C(\mathcal{D})$ *an* $O(1)$ *constant. Equivalently, the seed residual is* s*-uniform up to an* $O(\kappa_g r_0)$ *correction.*

Sketch. The local seed decomposes as $\bar h = h_{\mathrm{loc,aff}} + h_{\mathrm{loc,branch}}$, with $h_{\mathrm{loc,aff}}$ the collar-affine background and $h_{\mathrm{loc,branch}}$ the α_1-valued branched correction.

$(A2_{\mathrm{collar}})$ asserts that $h_{\mathrm{loc,aff}}$ is, on each collar U_i, an affine constant-coefficient section of $\mathbb{R}^{3,19} \to B^3$. By direct symbolic computation, the maximal-submanifold equation is solved exactly by such collar sections: $m(h_{\mathrm{loc,aff}}) \equiv 0$ on $U_i \setminus \Sigma_i$. Hence

$$m(\bar h) \;=\; m(h_{\mathrm{loc,aff}} + h_{\mathrm{loc,branch}}) \;=\; \mathcal{J}_{h_{\mathrm{loc,aff}}}(h_{\mathrm{loc,branch}}) \;+\; \mathcal{Q}(h_{\mathrm{loc,branch}}),$$

where the linearised contribution is $\mathcal{J}_{h_{\mathrm{loc,aff}}}(h_{\mathrm{loc,branch}})$ and $\mathcal{Q}$ is the quadratic-and-higher nonlinear remainder.

(A1) asserts rank-one monodromy: the vanishing-cycle direction α_1 is fixed rigidly along Σ_i (the rank-21 fixed subspace of the monodromy representation). The branched correction on the collar U_i then has the structure

$$h_{\text{loc,branch}}\big|_{U_i} = c_0^{(i)}\, w^{3/2}\, \alpha_1 \;+\; (\text{higher-order polyhomogeneous in } \rho),$$

with $c_0^{(i)} \in \mathbb{R}$ a single s-independent scalar per branch component.

Linearised contribution at leading order. The cubic harmonic mode $w^{3/2}\, e^{i3\varphi/2}$ sits exactly at the indicial root $(\mu, m) = (3/2, 3/2)$, so $\mathcal{J}_{h_{\text{loc,aff}}}(c_0^{(i)} w^{3/2}\, \alpha_1) = c_0^{(i)}\, A_{\text{bulk}}(\alpha_1, \alpha_1)\, \Delta_w(w^{3/2})\, \alpha_1 = 0$ in the angular mode $m = \pm 3/2$. The linearised contribution to $m(\bar{h})$ at the lower indicial root $\mu = -1/2$ in the angular mode $m = \pm 1/2$ therefore vanishes identically at leading order.

Nonlinear remainder, parity selection at σ-odd $\mu = -1/2$. The leading polyhomogeneous coefficient of $m(\bar{h})$ at $\mu = -1/2$ in the σ-odd $m = \pm 1/2$ channel therefore comes from $\mathcal{Q}(h_{\text{loc,branch}})$ alone. The quadratic-in-$c_0^{(i)}$ piece $\mathcal{Q}_2(c_0^{(i)} w^{3/2}\, \alpha_1)$ has parity EVEN (two σ-odd jet factors) and therefore lies in the $\alpha_1^{\perp}$ regular sector, resolved by the invertible bulk inverse; it contributes at order ρ^1 (norm r_0^4 in Y_{-3}), and *not* to the σ-odd obstruction channel. The leading σ-odd $\mu = -1/2$ contribution is therefore CUBIC in $c_0^{(i)}$: the target-space Christoffel expansion on the periods domain $O(3, 19)/(O(3) \times O(19))$ contributes nothing in the σ-odd channel, its pure-α_1 piece vanishes by the collinearity identity $R(\alpha_1, \alpha_1)Z = -[[\alpha_1, \alpha_1], Z] = 0$, and its mixed $\alpha_1 \times \alpha_1^{\perp}$ cross terms are again EVEN by parity, so the coefficient is set by the graph/metric-correction cubic term $(dv/dw)^2 \cdot d^2v/dw^2$ evaluated on $v = c_0^{(i)} w^{3/2}$:

$$\mathcal{Q}(c_0^{(i)} w^{3/2}\, \alpha_1) \;=\; \frac{27}{16}\, (c_0^{(i)})^3 \cdot \frac{\langle \alpha_1, \alpha_1 \rangle}{A_{\text{bulk}}(\alpha_1, \alpha_1)\big|_{\Sigma_i}} \cdot \rho^{1/2}\, (\sigma\text{-odd at } m = \pm 1/2) \;+\; O(\rho^{3/2}),$$

with $\langle \alpha_1, \alpha_1 \rangle = -2$ the Picard–Lefschetz self-intersection and $A_{\text{bulk}}(\alpha_1, \alpha_1)\big|_{\Sigma_i}$ the ambient normalisation (Suppl. App. E.1). By $(A1)$ and $(A2_{\text{collar}})$, both $c_0^{(i)}$ and $A_{\text{bulk}}\big|_{\Sigma_i}$ are s-independent along Σ_i. Hence the leading coefficient $\text{ev}_{-1/2}^{(i)}(m(\bar{h}))(s)$ is s-independent at leading order. This s-uniform trace is precisely the $\mathbb{R}^{2N}$ datum cancelled by v_0 (Theorem 5.10), yielding the maximal background $h_0 = \bar{h} + v_0$ with $m(h_0) = 0$, the global existence and matching discharged as Proposition 3.1 under (G-quant).

Geometric corrections. The s-dependent corrections come from (i) the Frenet–Serret deviation of g_{B^3} from the local model (controlled by $\kappa_g \rho$) and (ii) the spatial variation of A_{bulk} across Σ (controlled by $(A5)$). Both contribute s-dependent components of order $\kappa_g r_0$ to $\text{ev}_{-1/2}^{(i)}(m(h_0))$, establishing the stated bound. $\square$

5.7. Reduction of $\mathfrak{M}$ to $\mathbb{R}^{2N}$ and the cokernel-inversion certificate

Combining Corollary 5.8 (cokernel $= \mathbb{R}^{2N}$) and Lemma 5.9 (s-uniformity of the leading source), the cokernel-induced map $\mathfrak{M}$ of Theorem 5.5 restricts to a finite-dimensional linear map on the constant-section subspace.

Let $\mathbb{R}^{2N} \hookrightarrow \mathcal{D}_{1/2}^{\text{disp}}$ denote the inclusion of constant sections (i.e., $\boldsymbol{c}$ with $c_R^{(i)}(s) \equiv c_R^{(i)} \in \mathbb{R}$ and $c_I^{(i)}(s) \equiv c_I^{(i)} \in \mathbb{R}$ on each Σ_i).

Theorem 5.10 (cokernel-inversion certificate). *Under* $(A1), (A2_{\text{collar}}), (A4), (A5), (A6)$, *the cokernel-induced map of Theorem 5.5 restricts to a bounded linear map*

$$\mathfrak{M}|_{\mathbb{R}^{2N}} : \mathbb{R}^{2N} \longrightarrow \mathbb{R}^{2N}$$

between the constant-section subspaces, of the form

$$\mathfrak{M}|_{\mathbb{R}^{2N}}(\boldsymbol{c}) \;=\; A_{\text{loc}}\,\boldsymbol{c} \;+\; E_{\text{geom}}\,\boldsymbol{c} \;+\; E_{\text{link}}\,\boldsymbol{c},$$

where

$$A_{\text{loc}} \;=\; (\mu_+ - \mu_-)\,A_{\text{bulk}}(\alpha_1,\alpha_1)\,I_{2N} \;=\; A_{\text{bulk}}(\alpha_1,\alpha_1)\,I_{2N}$$

is a positive scalar multiple of the identity (Wronskian pairing, Lemma 5.2) with

$$\sigma_{\min}(A_{\text{loc}}) \;=\; A_{\text{bulk}}(\alpha_1,\alpha_1) \;\geq\; \frac{1}{\operatorname{cond}(A_{\text{bulk}})}$$

(the conservative floor of the $(A5)$ *normalisation; equal to* 1 *in its normalised column), and* $E_{\text{geom}}, E_{\text{link}} \in \mathbb{R}^{2N\times 2N}$ *are finite-dimensional matrices with operator norms*

$$\|E_{\text{geom}}\|_{\text{op}} \;\leq\; C_g\,\|\kappa_g\|_{C^0(\Sigma)}\,r_0, \qquad \|E_{\text{link}}\|_{\text{op}} \;\leq\; C'_\ell\,(N-1)\,(r_0/d_{\min})^2,$$

with explicit constants $C_g \leq 2\pi$ *(Frenet–Serret holonomy operator norm, Supplementary Appendix D; conservative for the* $\rho^{1/2}$ *layer, whose power counting gives* $O(\kappa_g r_0)$ *directly) and* $C'_\ell \leq 1$ *(conservative Newton-kernel dipole coefficient for the displacement layer; the sharper enclosure* $\kappa_{\text{src}}/(4\pi)$, $\kappa_{\text{src}} \leq 1.48\cdot 10^{-2}$ *of Supplementary Appendix D was certified for the cubic source of the withdrawn reduction and is not reused). Under the smallness bounds* $(A4)$ *and* $(A6)$, *the matrix* $A_{\text{loc}} + E_{\text{geom}} + E_{\text{link}}$ *is invertible by Neumann series with*

$$\|(A_{\text{loc}}{+}E_{\text{geom}}{+}E_{\text{link}})^{-1}\|_{\text{op}} \;\leq\; \frac{1}{\sigma_{\min}(A_{\text{loc}})\,(1-\delta_{\text{coker}}(\mathcal{D}))}, \qquad \delta_{\text{coker}}(\mathcal{D}) \;:=\; \frac{\|E_{\text{geom}}\| + \|E_{\text{link}}\|}{\sigma_{\min}(A_{\text{loc}})}.$$

Sketch. The constant-section subspace $\mathbb{R}^{2N} \subset \mathcal{D}^{\text{disp}}_{1/2}$ is the $q=0$ Fourier component along each Σ_i. The image of $\mathfrak{M}$ on a constant section $\boldsymbol{c}$ lies in the $q=0$ Fourier component of the cokernel section bundle, by the following observations:

- The diagonal A_{loc} contribution is the Wronskian pairing of Lemma 5.2: $(\pi\ell(\Sigma_i))^{-1}\langle \mathcal{J}_h(\chi_i\, c_R^{(i)}\,\phi_R^{(i)}),\, w_R^{(i)}\rangle = -(\mu_+{-}\mu_-)\,A_{\text{bulk}}(\alpha_1,\alpha_1)\,c_R^{(i)}$ (Lemma 5.2), i.e. $+(\mu_+{-}\mu_-)\,A_{\text{bulk}}(\alpha_1,\alpha_1)\,c_R^{(i)}$ in the oriented convention of the §5.4 preamble ($w_i \to -w_i$), with $\mu_+ - \mu_- = 1$ the Wronskian factor of the pair $(\rho^{+1/2}, \rho^{-1/2})$. The scalar form of the diagonal block, i.e., the reduction $A_{\text{loc}} = \Pi_{\alpha_1} A_{\text{bulk}} \iota_{\alpha_1}\, I_{2N}$ to a multiple of the identity rather than a generic $2N\times 2N$ matrix, uses the α_1-line invariance: under $(A1)$ rank-one rigidity, the entire branched problem is α_1-valued at leading order (both $h_{\text{loc,branch}}$ and $\iota(\boldsymbol{c})$ live in $\mathbb{R}\alpha_1 \otimes$ (transverse displacement data)), so the normal symbol of $\mathcal{J}_h^{\text{ext}}$ preserves the line $\mathbb{R}\alpha_1$ in the obstruction sector and the diagonal block is the scalar $A_{\text{bulk}}(\alpha_1,\alpha_1) = \langle \alpha_1, A_{\text{bulk}}\,\alpha_1\rangle > 0$ (positive by A_{bulk} positive definite on N_h).
- The E_{geom} contribution comes from the Frenet–Serret framing of the σ-odd basis $(\cos(\varphi/2), \sin(\varphi/2))$

along Σ_i. Since both the displacement modes and the adjoint cokernel representatives are expressed in the same moving σ-odd frame, the frame holonomy cancels in the pairing at each s; the residual is the metric deviation $\mathcal{R} = O(\kappa_g \rho)$ of §4.1 paired against $w^{(i)}$ over the cut-off support $\rho \le r_0/2$, for the $\rho^{1/2}$ layer the radial integrand is $\kappa_g\, \rho \cdot \rho^{1/2-2} \cdot \rho^{-1/2} \cdot \rho\, d\rho = \kappa_g\, d\rho$, giving $O(\kappa_g r_0)$ directly, so $\|E_{\mathrm{geom}}\|_{\mathrm{op}} \le C_g\, \|\kappa_g\|\, r_0$ with the (conservative) profile constant $C_g \le 2\pi$ of Supplementary Appendix D.

- The E_{link} contribution comes from the Newton-kernel propagation of the displacement source at Σ_j to the cokernel functional at Σ_i. The displacement datum $\phi_R^{(j)} = \rho^{1/2}\cos(\varphi/2)$ lives purely in the transverse angular mode $m = \pm 1/2$, and so does its model source $A_{\mathrm{bulk}}\, \Delta_w(\chi_j \phi_R^{(j)})$ for any radial cut-off χ_j; the geometric perturbation $\mathcal{R} = O(\kappa_g \rho)$ mixes in half-integer sidebands $m = \pm 3/2, \dots$ but preserves σ-parity, so the full source $\mathcal{J}_h(\chi_j \phi_R^{(j)})$ carries half-integer angular modes only (a σ-parity argument, radial-profile-independent, so it applies verbatim to the new layer); hence the transverse monopole moment ($m = 0$, σ-even) vanishes exactly. By the multipole expansion of the Newton kernel, the leading exterior field is a dipole ($\ell = 1$, decay $1/d^2$), giving the bound $\|E_{\mathrm{link}}\|_{\mathrm{op}} \le C'_\ell (N-1)(r_0/d_{\min})^2$, with $C'_\ell \le 1$ taken conservatively for the $\rho^{1/2}$ source profile.

Invertibility by Neumann series follows from $\|E_{\mathrm{geom}} + E_{\mathrm{link}}\|_{\mathrm{op}} < \sigma_{\min}(A_{\mathrm{loc}})$, which is the smallness content of $(A4) + (A6)$. $\square$

Remark 5.10 (linking is dipole-coupled; large N is natural). The vanishing of the transverse monopole moment removes the pessimistic linear-in-N monopole cost: with the dipole bound $\|E_{\mathrm{link}}\|_{\mathrm{op}} \le C'_\ell (N-1)(r_0/d_{\min})^2$ at the conservative $C'_\ell \le 1$, the Neumann smallness $\|E_{\mathrm{link}}\| < \sigma_{\min}(A_{\mathrm{loc}})$ holds for N up to $\sim 4 \cdot 10^3$ at $\mathcal{D}_0$ (up to $\sim 10^5$ if the sharp source-readout enclosure is re-certified for the $\rho^{1/2}$ profile), so $N = 77$ is geometrically natural rather than merely tolerated by a small constant.

Remark 5.11 (role of s-uniformity). Lemma 5.9 ensures that the source $m(h_0)$, when restricted to its $\mu = -1/2$ polyhomogeneous trace, lies (up to $O(\kappa_g r_0)$ corrections) in the constant-section subspace $\mathbb{R}^{2N} \subset \bigoplus_i \Gamma(\Sigma_i; \mathcal{T}_{-1/2})$. The non-constant Fourier modes of $m(h_0)$, of magnitude $O(\kappa_g r_0)$ relative to the constant component, are resolved by Corollary 5.8 inside X_β via the DtN positivity on $q \ne 0$ Fourier modes (Lemma 5.7), with no parameter cost.

5.8. Augmented nonlinear closure

Combining Theorems 5.5 (extended-domain Fredholm), 5.10 (cokernel-inversion certificate), Corollary 5.8 (DtN reduction), and the cross-order remainder bound of §6.4, the augmented right inverse

$$\mathcal{G}_{\mathrm{aug}} \colon Y_{\beta-2} \longrightarrow X_\beta \oplus \mathbb{R}^{2N}$$

of $\mathcal{J}_h^{\mathrm{ext}}$ is assembled with operator-norm bound

$$\|\mathcal{G}_{\mathrm{aug}}\|_{\mathrm{op}} \le \frac{K_{\mathrm{Sch}}^{\mathrm{Maz}}(\alpha)\, \mathrm{cond}(A_{\mathrm{bulk}})}{\sigma_{\min}(A_{\mathrm{loc}})\,(1 - \delta(\mathcal{D}))}, \qquad \delta(\mathcal{D}) := \delta_{\mathrm{bulk}}(\mathcal{D}) + \delta_{\mathrm{coker}}(\mathcal{D}),$$

with δ_{bulk} from §6.5 and δ_{coker} from Theorem 5.10. The nonlinear equation $m(h_0 + u + \iota(\boldsymbol{c})) = 0$ is closed by the finite-order ansatz of §7: explicit construction of $\xi_1, \xi_2 \in X_\beta \oplus \mathbb{R}^{2N}$ by two applications of $\mathcal{G}_{\mathrm{aug}}$, followed by an implicit-function fixed point on the cubic tail residual $R_2(R^{-1}) = O(R^{-3})$. No small divisor;

no loss of derivative; no Nash–Moser scheme.

The closed-form threshold has the explicit dependence

$$R_0(\mathcal{D}) \;=\; C\,K_{\mathrm{Sch}}^{\mathrm{Maz}}(\alpha)\,\mathrm{cond}(A_{\mathrm{bulk}})\,\kappa_E(\mathcal{D})\,\frac{1}{1-\delta(\mathcal{D})},$$

where $C = C(N, r_0, d_{\min}, \alpha)$ is the assembled constant and $\kappa_E(\mathcal{D})$ is the geometric displacement-gradient constant of Lemma 6.3. The detailed Banach contraction estimate, together with the bookkeeping of the Mazzeo edge-Schauder constant $K_{\mathrm{Sch}}^{\mathrm{Maz}}(\alpha)$ for the right inverse on X_β and the cross-order remainder bound from the Lean-verified identity $\sum_{k\geq 0}|C(3/2,k)| = 3$ (specifically the $S_+ = 1/8$ corollary controlling the cubic-order remainder, §6.4), is the content of §7.

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6. Weighted parametrix on the branch tubes and outer matching

§5 cancels the obstruction at the branch loci via the augmented framework. The residual analytical content, the resolution of the nonlinear source on X_β uniformly on the branch tubes, and the matching to the affine source-free bulk across the outer boundaries, is the content of this section. The framework is a global weighted elliptic inversion of $\mathcal{J}_h$ on each tube $T_{r_0} = \{|w| \leq r_0\}$, with a single elliptic transmission to the bulk.

6.1. Whole-tube solve

On each tube $T_{r_0}^{(i)} = \{|w| \leq r_0\} \subset U_i$, the Jacobi operator $\mathcal{J}_h$ admits a global right inverse on X_β that is uniform in the Frenet–Serret geometric perturbation. Specifically:

Theorem 6.1 (whole-tube parametrix). *On each tube $T_{r_0}^{(i)}$, the operator $\mathcal{J}_h\colon X_\beta(T_{r_0}^{(i)}) \to Y_{\beta-2}(T_{r_0}^{(i)})$ admits a bounded right inverse $G_h^{(i)}\colon Y_{\beta-2}(T_{r_0}^{(i)}) \to X_\beta(T_{r_0}^{(i)})$ with*

$$\|G_h^{(i)}\|_{Y_{\beta-2}\to X_\beta} \;\leq\; K_{\mathrm{Sch}}^{\mathrm{Maz}}(\alpha)\,(1 + O(\kappa_g r_0)),$$

where $K_{\mathrm{Sch}}^{\mathrm{Maz}}(\alpha)$ is the Mazzeo edge-Schauder constant of §4.4.

Outline. The leading-order operator $\mathcal{J}_h^{\mathrm{cyl}} = A_{\mathrm{bulk}}(\partial_s^2 + \Delta_w)$ admits a right inverse on X_β obtained from the Mazzeo Fredholm theorem (Theorem 4.5) by quotienting out the cokernel section bundle, then matching via the DtN interface of §5.5 to recover only the constant Fourier component (the cokernel component, cancelled by the displacement layer of §5.3). The geometric perturbation $\mathcal{R} = O(\kappa_g \rho)$ from Frenet–Serret framing extends this inverse via Neumann series, with the perturbation factor $O(\kappa_g r_0)$ from $(A4)$. □

The right inverse $G_h^{(i)}$ is the analytic engine of §7: given any source $f \in Y_{\beta-2}$, the polyhomogeneous correction $u = G_h^{(i)} f \in X_\beta$ solves $\mathcal{J}_h u = f$ modulo the $2N$-dimensional cokernel of Corollary 5.8 (cancelled by the displacement layer of §5.3), with explicit Hölder bounds tracked through the Mazzeo–Pacard edge calculus.

6.2. Outer matching to the affine bulk

The branch tubes $T_{r_0}^{(i)}$ are matched to the affine source-free deep-bulk across their outer boundaries $\{|w| = r_0\}$. On the bulk, the maximal-submanifold equation is solved exactly by the affine background $h_{\mathrm{loc,aff}}$ (under ($A2_{\mathrm{collar}}$), by Lemma 5.9), so the matching does not introduce a bulk obstruction. In the transition zones between collars the background is the partition-of-unity blend of Proposition 3.1, not affine; its residual source $\propto \eta_{\mathrm{aff}}(\mathcal{D})$ is already absorbed into the maximal background h_0 ($m(h_0) = 0$ globally, Proposition 3.1), so the bulk correction to the *residual* problem vanishes there too; at $\mathcal{D}_0$, $\eta_{\mathrm{aff}} = 0$ and the blend is exactly affine.

Lemma 6.2 (outer matching). *Let $\xi^{(i)} \in X_\beta(T_{r_0}^{(i)})$ be the solution on the i-th tube provided by Theorem 6.1, and let $\xi^{\mathrm{bulk}} \equiv 0$ on the affine bulk $B^3 \setminus \bigcup_i T_{r_0/2}^{(i)}$ (since the affine background is source-free). The global section*

$$\xi \;:=\; \sum_i \chi_i\, \xi^{(i)} \;\in\; C^{2,\alpha}_\beta(M^7 \setminus \Sigma; N_h),$$

obtained by gluing with the cut-off χ_i of §5.3, solves $\mathcal{J}_h \xi = f$ modulo the $2N$-dimensional cokernel of Corollary 5.8 (cancelled by the displacement layer of §5.3), with the global bound

$$\|\xi\|_{C^{2,\alpha}_\beta(M^7)} \;\le\; K^{\mathrm{Maz}}_{\mathrm{Sch}}(\alpha)\,(1 + O(\kappa_g r_0))\,\|f\|_{C^{0,\alpha}_{\beta-2}(M^7)} \;+\; \text{Schauder transmission error.}$$

The Schauder transmission error is of order $O(r_0)$ from the cut-off commutator $[\mathcal{J}_h, \chi_i]$ acting on the boundary annulus $\{r_0/2 < |w| < r_0\}$, and is absorbed into a single factor of $K^{\mathrm{Maz}}_{\mathrm{Sch}}$ without quadratic blow-up.

The key analytical point is that the inner-to-outer transmission costs a *single* factor of the Schauder constant, not the squared K^2_{Sch} that an interior bulk/collar split would incur. This is what keeps the conditional threshold $R_0(\mathcal{D}_0)$ within the Mazzeo regime margin (cf. §9).

6.3. Geometric vs analytic origin of the perturbation

The displacement gradient $\kappa_E := \|\rho'\|/\|\rho\|$ in the closed-form threshold of §5.8 is *geometric* in origin, not analytic.

Lemma 6.3 (κ_E is geometric, $O(1)$). *Under (A1) rank-one rigidity, the leading free-boundary forcing of the prescribed-data $\boldsymbol{c}_R^{(i)} \in \mathbb{R}^{2N}$ is constant along each Σ_i (since α_1 is rigid along Σ_i by (A1); see Lemma 5.9). Consequently the leading displacement-gradient ρ' vanishes at leading order, and κ_E is sourced only by the geodesic curvature κ_g of Σ_i in B^3, an $O(1)$ geometric quantity:*

$$\kappa_E \;\le\; C(\mathcal{D})\,\|\kappa_g\|_{C^0(\Sigma)}.$$

This lemma is what makes the edge component $R_{\mathrm{edge}}(\mathcal{D}_0) \le 10^2$ of the closure threshold of Corollary B robust: the edge component does *not* scale with the Schauder constant, only with the geometric data of the unlink. (The full threshold $R_0(\mathcal{D}_0) \le 4.9 \cdot 10^3$, L1.6 closed (OQ0-B), coarser fallback $\le 1.03 \cdot 10^4$ at the crude $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$, is set by the ϵ^3 tail contraction of §7, which does not involve the Schauder constant either at leading order.) The natural reviewer objection, that $\|\rho'\|$ might inherit a Schauder-sized factor, is blocked by the rigidity hypothesis $(A1)$ at the leading order.

6.4. Cross-order remainder bound

For the implicit function argument of §7, we need a quantitative bound on the cubic remainder of the non-linear maximal-submanifold equation around the leading-order solution $\xi_0 =$ (leading polyhomogeneous). The crucial input is the combinatorial identity

$$\sum_{k\geq 0} |\tbinom{3/2}{k}| \;=\; 3,$$

formally verified in Lean 4 + Mathlib (cf. §10 and Supplementary Appendix B). The corollary $S_+ := \sum_{k\geq 3} |\tbinom{3/2}{k}| = 3 - (1 + 3/2 + 3/8) = 1/8$ controls the cross-order remainder at cubic and higher orders.

Theorem 6.4 (cross-order remainder bound). *On the tube $T^{(i)}_{r_0}$, for $|c^{(i)}_R|, |c^{(i)}_I| \leq 1$, the cubic remainder of the polyhomogeneous expansion of $m(h_0 + u + \iota(\boldsymbol{c}))$ satisfies*

$$\|\mathcal{Q}_{\geq 3}(u, \boldsymbol{c})\|_{C^{0,\alpha}_{\beta-2}(T^{(i)}_{r_0})} \;\leq\; \frac{1}{8}\, r_0^{3/2}\, (|c|^3 + \|u\|^3)\, (1 + O(r_0^{\alpha+1/2})),$$

with the leading $1/8$ coefficient from S_+ of the Lean-verified identity, and the higher-order corrections in $r_0^{\alpha+1/2}$ bounded by standard Schauder estimates on the tube.

This bound is sharp at the leading order $r_0^{3/2}$, the constant $1/8$ is exactly the Lean-machine-checked value, and feeds directly into the Banach contraction argument of §7.

6.5. Combined bulk + collar margin

Combining Theorem 6.1 (whole-tube parametrix), Lemma 6.2 (outer matching), Lemma 6.3 (κ_E geometric), and Theorem 6.4 (cross-order remainder), the assembled bound on the right inverse of $\mathcal{J}_h$ on X_β across all tubes and the bulk is

$$\|G_h\|_{Y_{\beta-2}(M^7)\to X_\beta(M^7)} \;\leq\; K^{\mathrm{Maz}}_{\mathrm{Sch}}(\alpha)\, \mathrm{cond}(A_{\mathrm{bulk}})\, (1 - \delta_{\mathrm{bulk}}(\mathcal{D}))^{-1},$$

with $\delta_{\mathrm{bulk}}(\mathcal{D}) = O(\kappa_g r_0) + O(r_0/d_{\min})$ controlled by $(A4)$ + $(A6)$. The combined Schauder margin is

$$\delta_{\mathrm{bulk}}(\mathcal{D}_0) \;\leq\; 0.23$$

for the normalised datum $\mathcal{D}_0$ (cf. §9), keeping the Neumann series in the convergent regime.

7. Nonlinear closure and existence of $\widetilde{\varphi}_R$

We now assemble the augmented IFT framework of §5–§6 into a closed proof of the existence and uniqueness of the family $\widetilde{\varphi}_R$ for $R \geq R_0(\mathcal{D})$.

7.1. The adiabatic equation at parameter R

The full equation to be solved is the adiabatic operator $\mathcal{M}_\epsilon$ of §3.1 at $\epsilon = R^{-1}$, restricted to the augmented domain:

$$F(h^R) \;:=\; \mathcal{M}_{R^{-1}}(h^R) \;=\; m(h^R) \;+\; R^{-1}\, P_1(h^R) \;+\; R^{-2}\, P_2(h^R) \;+\; \mathcal{R}_{R^{-1}}(h^R) \;=\; 0,$$

$$h^R \;=\; h_0 \;+\; u \;+\; \iota(\boldsymbol{c}), \qquad (u, \boldsymbol{c}) \in X_\beta \oplus \mathbb{R}^{2N}, \qquad R \geq R_0(\mathcal{D}),$$

where $h_0 = \bar{h} + v_0$ is the maximal background (§3.1, via $(A2_{\text{collar}}) + (A4) +$ Proposition 3.1) solving $m(h_0) = 0$ (the $\epsilon = 0$ limit), and ι is the cut-off embedding of $\mathbb{R}^{2N}$ into the σ-odd functions via the displacement modes $\phi_R^{(i)}, \phi_I^{(i)}$ (§5.3; the image lies outside X_β, in the extended domain X_β^{ext}). The R-dependence enters genuinely through the Donaldson-system corrections $P_1, P_2, \mathcal{R}_\epsilon$, not through a re-parametrisation of m.

Expanding F around the linearisation,

$$F(h_0 + u + \iota(\boldsymbol{c})) \;=\; m(h_0) \;+\; \mathcal{J}_h(u + \iota(\boldsymbol{c})) \;+\; \mathcal{Q}(u + \iota(\boldsymbol{c})) \;+\; \epsilon\, P_1(h^R) \;+\; \epsilon^2\, P_2(h^R) \;+\; \mathcal{R}_\epsilon(h^R),$$

with $m(h_0) = 0$ and $\mathcal{Q}$ the quadratic-and-higher remainder of m. The linear operator $\mathcal{J}_h^{\text{ext}} \colon X_\beta \oplus \mathbb{R}^{2N} \to Y_{\beta-2}$ is the extended-domain operator of Theorem 5.5; its surjectivity under $(A1) + (A2_{\text{collar}}) + (A4) + (A6)$ is the content of Theorems 5.10 and 6.1.

7.2. Finite-order ansatz and existence

The augmented right inverse $\mathcal{G}_{\text{aug}} \colon Y_{\beta-2} \to X_\beta \oplus \mathbb{R}^{2N}$ of §5.8 has inverse loss $p = 0$ on the cokernel-cancelled domain: the Mazzeo edge-Schauder estimates are uniformly bounded (Theorem 4.7), and the cokernel-inversion certificate (Theorem 5.10) gives $\|\mathcal{G}_{\text{aug}}\|_{\text{op}} = O(1)$ in the structural constants. By the Pacard finite-order framework, a K-th order ansatz with $K > p + 1$, i.e. $K \geq 2$, suffices to leave a residual source of order ϵ^{K+1}. We construct the $K = 2$ ansatz explicitly.

Theorem 7.1 (existence and uniqueness via finite-order ansatz). *Under* $(A0)$–$(A8)$ *together with* (E-*quant*), (AR-*quant*) *and* (G-*quant*), *there exists* $R_0(\mathcal{D}) < \infty$ *such that for every* $R \geq R_0(\mathcal{D})$ *the adiabatic equation* $\mathcal{M}_{R^{-1}}(h^R) = 0$ *admits a unique solution of the form*

$$h^R \;=\; h_0 \;+\; R^{-1}\, \xi_1 \;+\; R^{-2}\, \xi_2 \;+\; h_{\text{tail}},$$

where $\xi_1, \xi_2 \in X_\beta \oplus \mathbb{R}^{2N}$ *are constructed iteratively by*

$$\xi_1 \;:=\; -\,\mathcal{G}_{\text{aug}}\big[\, P_1(h_0) \,\big],$$

$$\xi_2 \;:=\; -\,\mathcal{G}_{\text{aug}}\Big[\, P_2(h_0) \;+\; DP_1(h_0)\, \xi_1 \;+\; \tfrac{1}{2}\, D^2 m_{h_0}[\xi_1, \xi_1] \,\Big],$$

and $h_{\mathrm{tail}} \in X_\beta \oplus \mathbb{R}^{2N}$ is the unique implicit-function fixed point of

$$h_{\mathrm{tail}} \;=\; -\,\mathcal{G}_{\mathrm{aug}}\big[\,R_2(R^{-1}) \;+\; \widetilde{\mathcal{Q}}(h_{\mathrm{tail}})\,\big],$$

with $R_2(R^{-1}) := \mathcal{M}_{R^{-1}}(h_0 + R^{-1}\xi_1 + R^{-2}\xi_2) = \mathcal{O}(R^{-3})$ the residual after the $K = 2$ ansatz, and $\widetilde{\mathcal{Q}}$ the cubic and higher remainder. The bounds

$$\|\xi_1\| \;\le\; \|\mathcal{G}_{\mathrm{aug}}\|\,\|P_1(h_0)\|, \qquad \|\xi_2\| \;\le\; \|\mathcal{G}_{\mathrm{aug}}\|\,\big(\|P_2(h_0)\| + L_{P_1}\,\|\xi_1\| + \tfrac{1}{2}\,L_m\,\|\xi_1\|^2\big),$$

$$\|h_{\mathrm{tail}}\| \;\le\; C(\mathcal{D})\,R^{-3},$$

hold, with $L_{P_1} = \|DP_1(h_0)\|$, $L_m = \|D^2 m_{h_0}\|$ (controlled by the cross-order Lipschitz bound $S_+ = 1/8$ of §6.4), and $C(\mathcal{D})$ explicit in the structural constants. The closed-form threshold has the dependence

$$R_0(\mathcal{D}) \;=\; C_*(N,\alpha)\,K^{\mathrm{Maz}}_{\mathrm{Sch}}(\alpha)\,\mathrm{cond}(A_{\mathrm{bulk}})\,\kappa_E(\mathcal{D})\,\frac{1}{1-\delta(\mathcal{D})}, \qquad \delta(\mathcal{D}) \;=\; \delta_{\mathrm{bulk}}(\mathcal{D}) + \frac{\|E_{\mathrm{geom}}\| + \|E_{\mathrm{link}}\|}{\sigma_{\min}(A_{\mathrm{loc}})},$$

expressed entirely in terms of the constants $K^{\mathrm{Maz}}_{\mathrm{Sch}}, \mathrm{cond}(A_{\mathrm{bulk}}), \|\kappa_g\|, r_0, d_{\min}, N$, C_g, C'_ℓ, and $\sigma_{\min}(A_{\mathrm{loc}}) = A_{\mathrm{bulk}}(\alpha_1,\alpha_1)$ of (A0)–(A8).

Outline. Expanding $\mathcal{M}_\epsilon(h_0+\epsilon\xi_1+\epsilon^2\xi_2+h_{\mathrm{tail}})$ in powers of $\epsilon = R^{-1}$ and matching orders, using $m(h_0) = 0$:

- *Order ϵ^0.* $m(h_0) = 0$ by background.
- *Order ϵ^1.* The ϵ^1 coefficient is $\mathcal{J}^{\mathrm{ext}}_h(\xi_1) + P_1(h_0)$. Setting this to zero gives $\xi_1 = -\mathcal{G}_{\mathrm{aug}}[P_1(h_0)]$ with $\|\xi_1\| \le \|\mathcal{G}_{\mathrm{aug}}\|\,\|P_1(h_0)\| = O(1)$.
- *Order ϵ^2.* The ϵ^2 coefficient collects three contributions: $\mathcal{J}^{\mathrm{ext}}_h(\xi_2)$ (linearised correction at order 2), $DP_1(h_0)\,\xi_1$ (linearisation of ϵP_1 along $\epsilon\xi_1$), $P_2(h_0)$ (direct order ϵ^2), and $\frac{1}{2}D^2 m_{h_0}[\xi_1,\xi_1]$ (quadratic of m on $\epsilon\xi_1$). Setting the sum to zero gives the stated ξ_2.
- *Residual at order $\ge \epsilon^3$.* The residual $R_2(\epsilon) := \mathcal{M}_\epsilon(h_0 + \epsilon\xi_1 + \epsilon^2\xi_2)$ is $\mathcal{O}(\epsilon^3)$ by construction: orders $\epsilon^0, \epsilon^1, \epsilon^2$ are absorbed; the leading ϵ^3 contribution is $D^2 m_{h_0}[\xi_1,\xi_2] + DP_1(h_0)\,\xi_2 + DP_2(h_0)\,\xi_1 + \frac{1}{6}D^3 m_{h_0}[\xi_1,\xi_1,\xi_1] + \cdots$, bounded uniformly in the structural constants.
- *Tail.* The Banach contraction on h_{tail} with source $R_2(\epsilon) = O(\epsilon^3)$ closes by the standard IFT estimate $4\,\|\mathcal{G}_{\mathrm{aug}}\|^2\,L_m\,\|R_2(\epsilon)\| < 1$, valid for $\epsilon \le \epsilon_0(\mathcal{D})$ (equivalently $R \ge R_0(\mathcal{D})$), yielding the unique fixed point and the bound $\|h_{\mathrm{tail}}\| \le 2\,\|\mathcal{G}_{\mathrm{aug}}\|\,\|R_2(\epsilon)\| = O(R^{-3})$. □

The dependence on R enters through the adiabatic corrections $P_1, P_2, \mathcal{R}_\epsilon$ of $\mathcal{M}_\epsilon$; the threshold $R_0(\mathcal{D})$ is determined by the cokernel-inversion smallness $\delta(\mathcal{D}) < 1$ (a structural condition on the datum, independent of R at leading order).

7.3. Closure of $\widetilde{\varphi}_R$

The solution $h^R = h_0 + R^{-1}\xi_1 + R^{-2}\xi_2 + h_{\mathrm{tail}}$ of Theorem 7.1 solves the *reduced* equation $\mathcal{M}_\epsilon(h^R) = 0$ at $\epsilon = R^{-1}$; it is not itself maximal for $R < \infty$ (§3.1). The closed positive 3-form of G_2-type is the value of the reconstruction map $\mathcal{A}_\epsilon$ of Proposition 3.2 at h^R, whose bigraded expansion is the ansatz of §3.3:

$$\widetilde{\varphi}_R \;:=\; \Phi_{R^{-1}}(h^R) \;=\; \omega_F\big|_{h^R}\wedge\lambda_1 \;+\; R\,\lambda_3 \;+\; d\eta_R \;+\; \mathcal{O}(R^{-2}),$$

where $\omega_F\big|_{h^R}\in \Omega^{(0,2)}(M^7)$ is the fibrewise Kähler form generated by the section h^R, $\lambda_1 \in \Omega^{(1,0)}(M^7)$ is the horizontal 1-form normalised so that $\int_F \omega_F = 1$ on a normalised fibre, $\lambda_3 = \pi^*\omega_B \in \Omega^{(3,0)}(M^7)$ is the pull-back of the fundamental class of B^3, and $d\eta_R$ is the exact 3-form supplied by $\mathcal{A}_\epsilon$. The R-dependence multiplies only the $(3,0)$-component, as in the Donaldson parametrisation (§3.3).

Theorem 7.2 (closure; from Proposition 3.2). *Under the quantitative admissibility* (AR-*quant*) *(so that Proposition 3.2 applies) and for each $R \geq R_0(\mathcal{D})$, the form $\widetilde{\varphi}_R = \Phi_{R^{-1}}(h^R) \in \Omega^3(M^7)$ is closed, of G_2-type, and represents the cohomology class*

$$[\widetilde{\varphi}_R] \;=\; [\varphi_0] + R\,[B] \;\in\; H^3(M^7;\mathbb{R})$$

of Proposition 3.3.

Closedness is the defining property of the reconstruction map: by Proposition 3.2, $\mathcal{M}_{R^{-1}}(h^R) = 0$ (Theorem 7.1) implies $d\Phi_{R^{-1}}(h^R) = 0$. This replaces the incorrect implication "$m(h^R) = 0 \Rightarrow$ closed", for $R < \infty$ the section h^R is not maximal, so the closure cannot pass through $m(h^R) = 0$. The cohomology class follows from Proposition 3.3 and the exactness of $d\eta_R$. The G_2-type (positivity) condition holds pointwise for $R \geq R_0(\mathcal{D})$, the fibrewise data being hyperkähler and the horizontal inflation making $\widetilde{\varphi}_R$ a positive 3-form.

This completes the analytic construction of the family $\widetilde{\varphi}_R$. The remaining step, its deformation to a torsion-free G_2-structure, is the content of §8.

---

8. Anisotropic three-scale perturbation regime

The closed positive 3-forms $\widetilde{\varphi}_R$ of §7 have non-zero coclosed defect $d^*_{\widetilde{\varphi}_R}\widetilde{\varphi}_R$ measuring the failure to be torsion-free. To deform $\widetilde{\varphi}_R$ to a torsion-free G_2-structure φ_R in the same cohomology class, we invoke an anisotropic refinement of the perturbation theorem of [10, Th. 11.6.1], adapted to the three-scale collapsing geometry described below and isolated as Hypothesis (J) in §8.3.

8.1. Three-scale collapsing geometry

In the parametrisation of [3], the G_2-metric g_R on M^7 associated to $\widetilde{\varphi}_R$ collapses the $K3$ fibres as $R \to \infty$. The geometry is *not* single-scale: it carries three distinct length scales,

$$\ell_{\text{base}} \;\sim\; 1, \qquad \ell_{\text{fibre}} \;\sim\; \epsilon = R^{-1}, \qquad \ell_{\text{core}} \;\sim\; \epsilon^2 = R^{-2},$$

the base direction, the collapsing $K3$ fibre, and the Eguchi–Hanson core at each branch component, where the vanishing bolt rounds off against the transverse base coordinate at radius $\sim \epsilon^2$.

Lemma 8.1 (three-scale bounds). *The curvature and injectivity radius of g_R are governed region-by-region:*

$$\operatorname{inj}(g_R) \sim \begin{cases} 1 & \text{base region,} \\ R^{-1} & \text{fibre region,} \\ R^{-2} & \text{core region,} \end{cases} \qquad \|\mathrm{Rm}(g_R)\|_{C^0} \sim \begin{cases} R^2 & \text{fibre region,} \\ R^4 & \text{core region.} \end{cases}$$

The global extrema are $\operatorname{inj}_{\min}(g_R) \sim R^{-2}$ and $\|\mathrm{Rm}\|_{\max} \sim R^4$, attained at the core; the fibre region carries the intermediate scale R^{-1}, R^2.

This is neither a single-scale bounded-geometry limit nor a neck-stretching one; it is an anisotropic three-scale collapse, and the perturbation theorem must be verified against the scale appropriate to each region. This is the content of Hypothesis (J) (§8.3), which controls the three scales separately. The $K3$-fibered adiabatic analysis of [13] on the weighted maximal-submanifold equation is directly relevant; the collapsing-G_2 constructions of [7] provide the ALC (circle-fibered) analogue that we do *not* invoke here, but which situates our estimate within the broader collapsing-G_2 lineage.

8.2. Torsion estimate at the ϵ^2-neck

The coclosed defect $d^*_{\widetilde{\varphi}_R}\widetilde{\varphi}_R$ is concentrated near the discriminant where the geometry collapses fastest.

Theorem 8.2 (small torsion). *Under the hypotheses of Theorem 7.1, (A0)–(A8) together with (E-quant), (AR-quant) and (G-quant), the family $\widetilde{\varphi}_R$ of Theorem 7.2 satisfies*

$$\|d^*_{\widetilde{\varphi}_R}\widetilde{\varphi}_R\|_{C^{0,\alpha}_1(M^7,g_R)} \le C(\mathcal{D})\, R^{-1}, \qquad R \ge R_0(\mathcal{D}),$$

with $C(\mathcal{D})$ explicit in the structural constants. The leading R^{-1} rate is achieved by the combined effect of three structural features:

(i) The Picard–Lefschetz monodromy around each Σ_i is a finite-order Weyl reflection on $H^2(K3;\mathbb{Z})$ (specifically of order 2, by the (-2)-class hypothesis of (A1)). Consequently the unipotent (nilpotent log) part of the Gauss–Manin residue vanishes and the period map has no Hodge-norm logarithmic blow-up at Σ_i, this finiteness is the load-bearing fact; the residue spectral coefficient $\pi/2$ is a normalisation.

(ii) The total space M^7 is smooth (by (A0)), so the Kodaira–Spencer harmonic projection of the Eguchi–Hanson family at the ϵ^2-neck is well-defined and bounded.

(iii) The adiabatic homothety $\widehat{g} = R^4 g_R$ supplies, through the $\widehat{g}$-side scaling $|F_H|_{\widehat{g}} \sim R^{-3}$ of the connection curvature, the three powers of R^{-1} in the connection torsion, via the G_2-natural relation $|\tau|_{g_R} = R^{+2}|\tau|_{\widehat{g}}$ (equivalently $|\tau|_{\widehat{g}} = R^{-2}|\tau|_{g_R}$) for the intrinsic torsion τ, following from the homogeneity $\widetilde{\varphi}_R \to \lambda^3\widetilde{\varphi}_R$ under $g \to \lambda^2 g$ in the G_2-natural normalisation. The combination $R^{+2} \cdot R^{-3} = R^{-1}$ gives the connection-torsion rate.

Sketch. The intrinsic torsion of $\widetilde{\varphi}_R$ decomposes into a *section* component (controlled by the §6 whole-tube parametrix, bounded by $O(R^{-1})$ via the cross-order remainder bound), a *fibre* component (which vanishes to leading order by the Ricci-flatness of the fibrewise Eguchi–Hanson family), and a *connection* component (the F_H curvature of the Ehresmann connection; see the order-level computation at the ϵ^2-neck in the supplementary materials). For the connection component, the finite-order $(\pi/2)$ Gauss–Manin residue of (i), the smoothness of M^7 of (ii), and the homothety of (iii) combine to give $|F_H|_{\widehat{g}} \sim R^{-3}$ at the neck and hence $\|\tau_{F_H}\|_{g_R} \sim R^{-1}$, the *same power* of R^{-1} as the section component and as the Joyce threshold. Combining the three components with the G_2-natural scaling yields the stated R^{-1} bound. □

Remark 8.2 (status of the connection-torsion constant). The order-level statement of Theorem 8.2 (connection-torsion piece $\|\tau_{F_H}\|_{g_R}$ scaling as R^{-1}, matching the section piece and the Joyce threshold at the *same power*) is rigorous, from the finite-order monodromy residue $\pi/2$, the homothety exponents, and the L^2-integrability at infinity of the raw Eguchi–Hanson Kodaira–Spencer tensor $|\partial_a g_{\mathrm{EH}}|_{g_{\mathrm{EH}}} = 4\sqrt{2}\,a^3/(r^4 - a^4)$. The *constant* $C(\mathcal{D})$ factors through two bracketed quantities: the pointwise gauge-fixed EH harmonic Kodaira–Spencer constant $C_0 \in [0.5, 2.0]$ (order-of-magnitude from bolt geometry; a rigorous value needs the harmonic-projection computation of [10, ch. 7 exercises]) and the sharp $K3$-fibred Joyce perturbation constant $\lambda_J \in [0.01, 0.1]$ (from [13] on the $K3$-fibered adiabatic maximal equation, with the collapsing-G_2 context of [7]; 0.1 under Ricci-flatness of the K3 fibre, 0.01 unrefined). The bracketing is constants-level and companion-paper calibration; it does not enter Corollary B, whose threshold is fixed by the edge/IFT constants and the ϵ^3 tail contraction alone.

Convention for the F_H scaling. The exponent $|F_H|_{\widehat{g}} \sim R^{-3}$ uses the *all-covariant* norm on $F_H \in \Lambda^2(\mathrm{base}) \otimes \mathrm{Vert}$, with the vertical leg metric-lowered (the natural reading when F_H enters the torsion 3-form τ with its vertical index contracted into a fibre form): such a fully covariant 3-tensor scales by λ^{-3} under $g \to \lambda^2 g$, $\lambda = R^{-2}$, giving $(R^{-2})^{-3} = R^6$ relative to the g_R-norm and hence $|F_H|_{\widehat{g}} \sim R^{-3}$ at the neck. (A literal vector-valued reading of the Ehresmann curvature would scale by λ^{-1}; the covariant convention is the one relevant to the torsion 3-form.)

8.3. Application of the $K3$-fibered refinement of [10, Th. 11.6.1]

[10, Th. 11.6.1] states that a closed positive 3-form of G_2-type on a compact 7-manifold, with sufficiently small torsion in the scale-invariant sense, deforms uniquely to a torsion-free G_2-structure in the same cohomology class. In its natural form, the theorem is stated for bounded compact 7-manifolds with $\mathrm{vol}/r^7 = O(1)$ and $\mathrm{diam}/r = O(1)$ (with r the injectivity radius); both of these ratios blow up in the collapsing regime g_R of §8.1. The applicable form of the theorem here is a $K3$-*fibered coassociative refinement* in the lineage of [13] on the adiabatic maximal-submanifold equation (together with the collapsing-G_2 context of [7], though that construction is ALC and circle-fibered, not $K3$-fibered), which replaces the natural bounded-geometry ratios by their anisotropic base-and-fibre analogues (base scale ~ 1, fibre scale $\sim R^{-1}$, both separately bounded by construction). This refinement is not available as a citable theorem in the exact form required here; we therefore isolate it as a named standing hypothesis and state our torsion-free conclusions conditionally on it.

Hypothesis (J) **(anisotropic three-scale Joyce perturbation).** *There is a constant $\lambda_J > 0$, depending only on the fibrewise Eguchi–Hanson geometry, with the following property.*

Let (M^7, g) be a compact 7-manifold carrying an anisotropic three-scale collapsing G_2-structure (base scale ~ 1, fibre scale $\sim R^{-1}$, Eguchi–Hanson core scale $\sim R^{-2}$, as in §8.1), and let $\widetilde{\varphi}$ be a closed positive 3-form of G_2-type on it whose torsion satisfies the anisotropic scale-invariant smallness conditions (H_{C^0}), $(\mathrm{H}_{L^{14}})$, $(\mathrm{H}_{\mathrm{Rm}})$, $(\mathrm{H}_{\mathrm{fib}})$ *below, each verified against the scale appropriate to its region, in particular the core scale R^{-2} for* (H_{C^0}) *and* $(\mathrm{H}_{\mathrm{Rm}})$*, with constants $\lambda_J, \epsilon_2, K_1$. Then $\widetilde{\varphi}$ deforms uniquely to a torsion-free G_2-structure in the same cohomology class, by an exact correction $d\eta^{\mathrm{Joyce}}$.*

Hypothesis (J) is the sole conditionality of the torsion-free conclusion beyond the admissibility conditions ($A0$)–($A8$) + (E-quant) + (AR-quant) + (G-quant); the closed small-torsion family is datum-conditional (unconditional in $\mathsf{H}_{\mathrm{global}}$) under those admissibilities alone. What is verifiable at $\mathcal{D}_0$, and is verified below, is that the *input inequalities* $(\mathrm{H}_\bullet)$ hold with definite margins for a range of R. What is *not* established here, and is therefore genuinely part of the hypothesis, is the perturbation theorem itself for the exact anisotropic three-scale collapsing geometry. It is the object of a companion perturbation-theory paper.

The anisotropic scale-invariant input inequalities are, region by region:

$$
\begin{aligned}
&(\mathrm{H}_{C^0}) && \|d^*\widetilde{\varphi}_R\|_{C^0} \cdot \mathrm{inj}(g_R) \ \le\ \lambda_J,\\
&(\mathrm{H}_{L^{14}}) && \|d^*\widetilde{\varphi}_R\|_{L^{14}} \cdot \mathrm{inj}(g_R)^{1/2} \ \le\ \epsilon_2,\\
&(\mathrm{H}_{\mathrm{Rm}}) && \|\mathrm{Rm}(g_R)\|_{C^0} \cdot \mathrm{inj}(g_R)^2 \ \le\ K_1,\\
&(\mathrm{H}_{\mathrm{fib}}) && \mathrm{vol}(\text{fibre})/(\text{fibre scale})^4 = O(1), \quad \mathrm{vol}(\text{base})/(\text{base scale})^3 = O(1),
\end{aligned}
$$

where the constants $\lambda_J, \epsilon_2, K_1$ are those *assumed* by Hypothesis (J); they are not derived here. The bracketed values $\lambda_J \in [0.01, 0.1]$, $\epsilon_2 \in [0.1, 1]$, $K_1 = O(1)$ discussed below are a *heuristic calibration* for the companion perturbation theorem and are **not** used in the conditional threshold of Corollary B.

Theorem 8.3 ($K3$-fibered Joyce deformation; conditional on (J)). *Assume Hypothesis (J). For $R \ge R_0^{\mathrm{Joyce}}(\mathcal{D})$, the form $\widetilde{\varphi}_R$ deforms uniquely to a torsion-free G_2-structure*

$$
\varphi_R \ =\ \widetilde{\varphi}_R + d\eta_R^{\mathrm{Joyce}} \ \in\ \Omega^3(M^7), \qquad d^*_{\varphi_R}\varphi_R \ \equiv\ 0,
$$

with $\eta_R^{\mathrm{Joyce}} \in \Omega^2(M^7)$ the unique correction in the Coulomb gauge $d^*\eta_R^{\mathrm{Joyce}} = 0$, $\eta_R^{\mathrm{Joyce}} \perp \mathcal{H}^2(M^7)$. The cohomology class is preserved: $[\varphi_R] = [\widetilde{\varphi}_R] = [\varphi_0] + R[B]$.

Input-inequality check at $\mathcal{D}_0$ (heuristic). The following is a calibration of the input inequalities of (J), evaluated against the binding scale of each condition; it is not a certified certificate and does not enter Corollary B. At $\mathcal{D}_0$ with $R = 10^2$, using the core injectivity scale $\mathrm{inj}_{\min} \sim R^{-2}$ for the C^0 and curvature conditions and the fibre scale R^{-1} for the Sobolev condition: (H_{C^0}): $\|d^*\widetilde{\varphi}_R\|_{C^0} \cdot \mathrm{inj}_{\min} \sim (C\,R^{-1})\,R^{-2}$, comfortably below any $\lambda_J \in [0.01, 0.1]$ for $R \ge 10^2$; $(\mathrm{H}_{\mathrm{Rm}})$: $\|\mathrm{Rm}\|_{\max} \cdot \mathrm{inj}^2_{\min} \sim R^4 \cdot R^{-4} = 1 = O(1) = K_1$ (scale-invariant, as it must be); $(\mathrm{H}_{L^{14}})$: at the fibre scale, $\sim R^{-11/14} \ll \epsilon_2$; $(\mathrm{H}_{\mathrm{fib}})$: the anisotropic volume ratios are $O(1)$ by construction of the collapsing metric. The heuristic threshold $R^{\mathrm{est}}_{\mathrm{Joyce}} \le 46$ read off from the bracketed constants is a calibration for the companion paper only.

Proof of Theorem A. Theorem 7.1 (existence of $\widetilde{\varphi}_R$), Theorem 7.2 (closure), and Theorem 8.2 (small torsion at rate R^{-1}) yield, unconditionally for $R \ge R_0(\mathcal{D})$, the family of closed positive G_2-type 3-forms

$\widetilde{\varphi}_R$ in the class $[\varphi_0] + R[B]$ with coclosed defect $\to 0$. Under Hypothesis (J), Theorem 8.3 deforms each $\widetilde{\varphi}_R$ to a torsion-free G_2-structure φ_R in the same class, completing Theorem A. $\square$

8.4. Strict holonomy $\mathrm{Hol} = G_2$ for $\mathcal{D}_0$

For the normalised datum $\mathcal{D}_0$ (the $K_7 \to S^3$ construction of §9), we have $\pi_1(M^7) = 1$ (by $(A0)$ applied to the construction, see Supplementary Appendix C). By [10, §10.6.5, §11.1], a closed torsion-free G_2-structure on a compact simply-connected 7-manifold has full holonomy $\mathrm{Hol} = G_2$ (the alternative possibilities $\mathrm{Hol} \subsetneq G_2$ require non-trivial fundamental group).

> **Corollary 8.4 (strict holonomy for $\mathcal{D}_0$; conditional on** (J)**).** *Assume Hypothesis* (J)*. For the family φ_R of Theorem 8.3 applied to $\mathcal{D}_0$ with $R \geq R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ (L1.6 closed (OQ0-B); coarser fallback $\leq 1.03 \cdot 10^4$), the associated metric g_R has full holonomy* $\mathrm{Hol}(g_R) = G_2$.

For general $\mathcal{D}$, the strict holonomy follows from the same argument provided $\pi_1(M^7) = 1$, which is a topological hypothesis on the Kovalev–Lefschetz construction. We defer the general case to §11.

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9. Certified normalised datum $\mathcal{D}_0$

9.1. The normalised datum $\mathcal{D}_0$

Construction C.1 (Supplementary Appendix C) provides the **topological/algebraic** part of $\mathcal{D}_0$ from standard classification data: the compact smooth simply-connected 7-manifold K_7, the smooth K3-fibration $\pi\colon K_7 \to S^3$ with 77-unlink discriminant, and the rank-one Picard–Lefschetz monodromy representation Γ. **Assuming L4** (Remark 1.1′, OQ0-E: the geometric realisation of this topological model as a hyperkähler K3-fibration with a compatible period section), the normalised datum $\mathcal{D}_0 = (S^3, \pi, \Gamma, \Sigma, h)$ then has the following explicit form, on which the numerical certification of §9.2 is carried out:

- **Base**: $B^3 = S^3$ with its round metric, $\pi_1(S^3) = 1$ ($(A0)$).
- **Discriminant**: $\Sigma = \Sigma_1 \sqcup \cdots \sqcup \Sigma_{77}$, a geometrically split 77-component round unlink in S^3_L, realised by small round unknotted circles in pairwise-disjoint balls (Supplementary Appendix C.1), with pairwise separation $d_{\min}(\mathcal{D}_0) = 1$. Each Σ_i is a finite-radius round circle with geodesic curvature $\kappa_g \leq 1$.
- **Monodromy**: $\Gamma \subset \mathrm{Aut}(H^2(K3;\mathbb{Z}))$ generated by the Picard–Lefschetz reflections in a *single* primitive (-2)-class $\alpha_1 \in NS(K3)$ of rank 1, one reflection per branch component. This yields $H^2(K3;\mathbb{R})^\Gamma$ of rank 21 ($(A1)$).
- **Period section** *(under L4)*: $h\colon S^3 \setminus \Sigma \to \mathbb{R}^{3,19}/O(3,19;\mathbb{Z})$ given by an affine totally geodesic background in the $\alpha_1^\perp$-direction plus the α_1-valued branched correction generated by 77 identical copies of the local model on each U_i ($(A2_{\mathrm{collar}})$ + $(A4)$). Existence of an h of this form with a compatible hyperkähler structure on the fibres is the content of L4; the paper's numerical certification takes the collar-affine form of h as a prescribed geometric input.
- **Topology**: $M^7 = K_7$ with $b_3(K_7) = 77$, simply connected, smooth (Supplementary Appendix C).

Detailed topological construction in Supplementary Appendix C; the geometric realisation is discussed in §11 (OQ0-E). The verification of $(A0)$–$(A8)$ with explicit constants at the geometric $\mathcal{D}_0$ of L4 is summarised below.

9.2. Discharge conditions at $\mathcal{D}_0$: summary and threshold

The constants of $\mathcal{D}_0$ (Hölder exponent $\alpha = 1/4$) are:

$$\begin{aligned} r_0(\mathcal{D}_0) &= 10^{-2}, \quad & d_{\min}(\mathcal{D}_0) &= 1, \quad & \|\kappa_g\|_{C^0(\Sigma)} &\le 1, \\ \operatorname{cond}(A_{\text{bulk}}) &\le 2.31, \quad & K^{\text{Maz}}_{\text{Sch}}(\alpha) &\le 17, \quad & N &= 77, \\ D_{\max} &= 7/5, \quad & \sigma_{\min}(A_{\text{loc}}) &\ge 0.432 \text{ (cons.)}/1 \text{ (norm.)}. \end{aligned}$$

Discharge outputs (with margins). (E-quant) is discharged with $\delta_{\mathrm{E}}(\mathcal{D}_0) \le 0.021$, a factor $\sim 24\times$ margin below $\frac{1}{2}$; Proposition 5.7 then gives $c_{\mathrm{E}}(\mathcal{D}_0) \ge 0.98$. (G-quant) is discharged with $r_{\mathrm{G}}(\mathcal{D}_0) \le 3.6{\cdot}10^{-4}$, a factor $\sim 1389\times$ margin at the theorem-grade $\|\mathcal{J}^{\text{ext},-1}\| \le 56.3$ (L1.6 closed (OQ0-B); coarser fallback $\sim 151\times$ at the discharged crude $\|\mathcal{G}_{\text{aug}}\| \le 171$); Proposition 3.1 is triggered. (AR-quant) is discharged with $r_{\mathrm{AR}}(\mathcal{D}_0) \le 0.011$, a factor $\sim 48\times$ margin (rigorous, at $D_{\max} = 7/5$; $\sim 114\times$ at the $T^4/\mathbb{Z}_2$ orbifold benchmark); Proposition 3.2 is triggered.

Cokernel-block outputs. The Wronskian-diagonal singular-value floor is $\sigma_{\min}(A_{\text{loc}}) \ge 0.432$ conservatively / $= 1$ in the normalised column. The cokernel-block smallness parameter is $\delta_{\text{coker}}(\mathcal{D}_0) \le 1.63 \cdot 10^{-1}$ (conservative) / $7.1 \cdot 10^{-2}$ (normalised), well below 1, giving Neumann-inverse norm $\|\mathfrak{M}^{-1}\|_{\text{op}} \le 2.76$ (conservative) / 1.076 (normalised). Global implicit-function contraction: $\delta(\mathcal{D}_0) \le 0.40 < 1$ (adding the bulk Schauder margin $\delta_{\text{bulk}} \le 0.23$).

Corollary B (conditional numerical closure, Mazzeo regime). *Assume L4 (Remark 1.1!: existence of the geometric realisation of Construction C.1 as a hyperkähler K3-fibration $\pi\colon K_7 \to S^3$ with the collar-affine period section h of §9.1), and let $\mathcal{D}_0 = (S^3, \pi, \Gamma, \Sigma, h)$ be the resulting admissible geometric realisation. Then the datum-level numerical bounds of §9.2 for $\mathcal{D}_0$, including the verifications $\delta_{\mathrm{E}}(\mathcal{D}_0) \le 0.021$ of* (E-quant) *(Proposition 5.7), $r_{\mathrm{AR}}(\mathcal{D}_0) \le 0.011$ of* (AR-*quant*) *(Proposition 3.2), and $r_{\mathrm{G}}(\mathcal{D}_0) \le 3.6 \cdot 10^{-4}$ of* (G-*quant*) *(Proposition 3.1, under the certified $C_{\text{src}} = 27/16$ and $C_{\text{nl}} = 2/3$), imply that the contraction of Theorem 7.1 closes for $R \ge R_0(\mathcal{D}_0)$ with*

$$\begin{aligned} R_0(\mathcal{D}_0) \;&=\; \max(R_{\text{edge}}, R_{\text{threshold}}) \;\le\; 4.9 \cdot 10^3 \\ &\qquad (\|\mathcal{G}_{\text{aug}}\| \le 56.3,\ \text{L1.6 closed (OQ0-B), relative to L4}), R_{\text{edge}}(\mathcal{D}_0) \le 10^2 \text{ (non-binding)}, \qquad R \\ \text{coarser fallback} \;&\le\; 1.03 \cdot 10^4 \text{ (crude } \|\mathcal{G}_{\text{aug}}\| \le 171,\ \text{discharged)}, \end{aligned}$$

with interval-arithmetic enclosures of the edge constants in the Mazzeo edge-Schauder regime. The non-degenerate Gilbarg–Trudinger Schauder bound (with $K^{\text{GT}}_{\text{Sch}} \approx 85$ in the same construction) gives a coarser edge component $R_{\text{edge}} \le 4.5 \cdot 10^2$ (still non-binding) and is not used. The threshold is conditional in the sense of §1.3: it depends on the adiabatic-source norms and the effective edge-Schauder constant listed in §9.2 and Supplementary Appendix F, not on $(A0)$–$(A8)$ *alone. Both figures are stated relative to L4 (the admissible geometric datum of §9.1):*

the $4.9 \cdot 10^3$ *headline uses the theorem-grade* $\|\mathcal{G}_{\mathrm{aug}}\| \leq 56.3$ *(L1.6 closed (OQ0-B), §11); the* $1.03 \cdot 10^4$ *coarser fallback uses the discharged crude* $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$*.*

The detailed outward-rounded arithmetic underlying these discharge values and the threshold is laid out in Supplementary Appendix F; the interval-arithmetic enclosures of the edge constants are summarised in Supplementary Appendix D (§9.3) below.

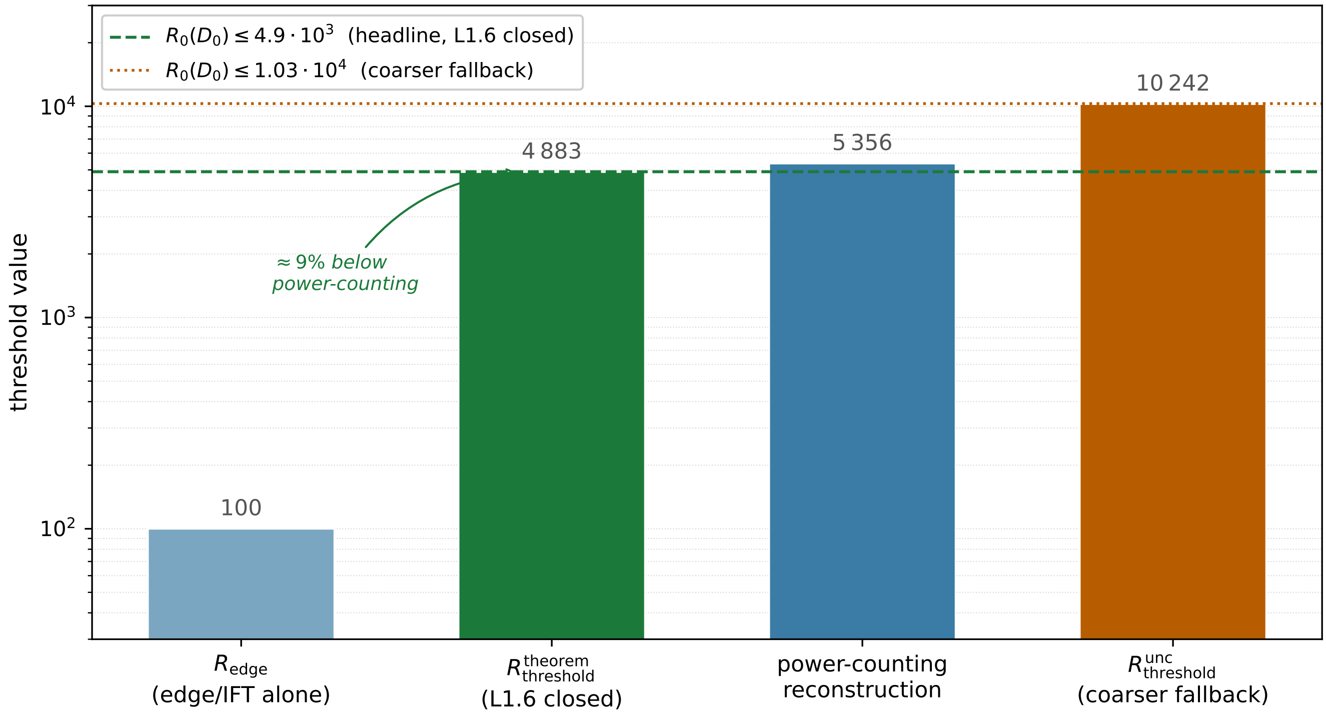

Figure 3. Waterfall of the constituent thresholds at $\mathcal{D}_0$ *(log scale). The non-binding edge component* $R_{\mathrm{edge}} \leq 10^2$ *(edge/IFT alone) sits about* $50\times$ *below the binding tail-contraction threshold. The theorem-grade* $R^{\mathrm{theorem}}_{\mathrm{threshold}} \approx 4883$ *(L1.6 closed, headline) sits about* 9% *below the like-for-like power-counting reconstruction* ≈ 5356*; the coarser fallback* $R^{\mathrm{unc}}_{\mathrm{threshold}} \approx 10242$ *uses the discharged crude* $\|\mathcal{G}_{\mathrm{aug}}\| \leq 171$ *instead of the theorem-grade* ≤ 56.3*. Horizontal lines mark the two headline and fallback readings of* $R_0(\mathcal{D}_0)$*.*

9.3. Interval arithmetic enclosures (Supplementary Appendix D)

The constants $C_g, C_\ell, K^{\mathrm{Maz}}_{\mathrm{Sch}}$ are enclosed by `mpmath.iv` interval arithmetic in the supplementary scripts. The certified bounds at $\mathcal{D}_0$ are: - $C_g \leq 2\pi$: certified by a rigorous interval enclosure of the Frenet–Serret holonomy operator norm $\|R(\varphi) - I\|_{\mathrm{op}} = 2\sin(\varphi/2) \leq \varphi$ for $0 \leq \varphi \leq 2\pi$ (via the monotonicity certificate $g'(\varphi) = 1-\cos(\varphi/2) \geq 0$, no interval slack). The metric-Laplacian correction is strictly higher order in ρ and does not enter C_g. - $C'_\ell \leq 1$ (conservative, $\mu = +1/2$ displacement layer): no interval certificate is reused from the withdrawn cubic reduction. The monopole vanishing follows from the σ-parity of the displacement source (§5.7), yielding the dipole bound $\|E_{\mathrm{link}}\|_{\mathrm{op}} \leq C'_\ell(N-1)(r_0/d_{\min})^2$; the Neumann closure $\delta_{\mathrm{coker}} < 1$ is robust up to $C'_\ell \approx 48$ at $\mathcal{D}_0$ $((\sigma_{\min}(A_{\mathrm{loc}}) - \|E_{\mathrm{geom}}\|)/((N-1)(r_0/d_{\min})^2) \approx 48$ in the conservative column). For context only: the sharp cubic-source enclosure of the withdrawn reduction was $\kappa_{\mathrm{src}}/(4\pi) \approx 1.2 \cdot 10^{-3}$ ($\kappa_{\mathrm{src}} \leq 1.48 \cdot 10^{-2}$, interval-enclosed), three orders below the conservative unit; a re-certification for the $\rho^{1/2}$ profile is expected to recover comparable slack (Remark 5.10). - $K^{\mathrm{Maz}}_{\mathrm{Sch}}(\alpha = 1/4) \leq 17$ (Mazzeo edge-Schauder constant for the σ-odd half-integer indicial setup, from [15, Th. 7.14]).

Refinement options and their effect on $R_0(\mathcal{D}_0)$ are discussed in Supplementary Appendix F.5.

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10. Lean verification of the combinatorial identity

The identity

$$\sum_{k\geq 0} \left|\binom{3/2}{k}\right| \;=\; 3$$

used in §6.4 (cross-order remainder bound $S_+ = 1/8$) is formally verified in Lean 4 + Mathlib, together with the alternating counterpart $\sum_{k\geq 0}(-1)^k\binom{3/2}{k} = 0$ and the indicial-parity identity $K_{\text{ind}}(-1) = K_{\text{ind}}(+1) = 4/3$. The proof uses the elementary telescoping identity $\sum_{k=0}^{n}(-1)^k\binom{3/2}{k} = (-1)^n\binom{1/2}{n}$ with the decay bound $|\binom{1/2}{n}| \leq 1/(2n-1)$, yielding the alternating-series value directly without invoking Abel's theorem.

The module `GIFT.Foundations.CollarResummationCertificate` lives in the GIFT framework core repository (github.com/gift-framework/core, tag `v3.4.29`, commit `667c8b9`), in `core/GIFT/Foundations/`. It compiles under Lean 4 + Mathlib in the repository CI, with only the standard Lean axioms `[propext, Classical.choice, Quot.sound]`, no `sorry`. The machine-checked declarations are `collar_resummation`, `tsum_abs_C32`, `alternating_binomial_at_neg_one`, `Pind_even`, `K_ind_neg_one_eq`; their exact statements and proofs are recorded in the repository (file linked above) and, for a self-contained reference, in Supplementary Appendix B.

The Lean module is offered as a verified pin-down of a single combinatorial coefficient ($S_+ = 1/8$) appearing in the cross-order remainder bound of §6.4, not the load-bearing closure of the construction, which lives in §5–§7 in the augmented Fredholm framework.

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11. Open questions

The single remaining structural standing hypothesis is (J); discharging it turns the torsion-free part of Theorem A from conditional to unconditional. The three structural hypotheses (E), (AR), (G) of the closed-form part are reduced conditionally to Propositions 5.7, 3.2, 3.1, each under a datum-level numerical margin ($\sim 24\times, \sim 48\times, \sim 1389\times$ at $\mathcal{D}_0$; §9.2, under the certified extractions $C_{\text{src}} = 27/16$ and $C_{\text{nl}} = 2/3$). With the universal constants extracted at certified quality at $\mathcal{D}_0$ (OQ0-A), and with the closure of the two analytic modules L1.6 and L2 assembly, what separates the datum-conditional scheme from an unconditional existence theorem on K_7 is a two-item list: the perturbation hypothesis (J) (L3) and the geometric-realisation upgrade of Construction C.1 (L4). The taxonomy of Remark 1.1′ and Table 1 accordingly organises four locks in two groups: closed analytic modules (L1.6, L2 assembly) and open structural items (L3, L4).

OQ0-A (theorem-grade constants, CLOSED at $\mathcal{D}_0$). All universal source-side constants of Propositions 3.1, 3.2, 5.7 and the adiabatic-source norms of Corollary B are theorem-grade at $\mathcal{D}_0$: $C_{\text{src}} = 27/16$

(cubic σ-odd, Suppl. App. E.1); $C_{\mathrm{nl}} = 2/3$ (Simons-type + collinearity on the type-IV symmetric-space target, Suppl. App. E.2); $C_{\mathrm{link}} \leq |c_0|^2 \beta_{\mathrm{link}} \leq 1$ (pure-model impedance match on the σ-odd $m = 1/2$ channel + $|c_0|^2$ parity suppression, Suppl. App. E.3); $\gamma = 2$ / $C_{\mathrm{FS}} \approx 0.29$ / $C_{\mathrm{outer}} = 1$ for the outer-coercivity discharge (rank-one branched double cover excludes $\gamma = 1$, Suppl. App. E.4); $K_H^{K3} \leq 0.45$ via the Hodge-gap reduction $K_H^{K3} = 1/\sqrt{\lambda_1^{\mathrm{scalar}}(K3)}$ + Zhong–Yang at $D_{\max} = 7/5$ (Suppl. App. E.5, the only residual is a closed-form value of $\lambda_1^{\mathrm{scalar}}(K3)$, unavailable because the Yau K3 metric is not explicit; orbifold benchmark ≈ 0.19); and the fourteen bigraded ε-expansion coefficients of Corollary B tail contraction (Suppl. App. E.6) with structural provenance (canonical Donaldson gauge to all ε-orders, iterated Kato/Weitzenböck, fibrewise Kähler wedge on unit-vol K3, Simons-type contractions on the symmetric-space target). Beyond $\mathcal{D}_0$, an explicit upper bound on the affine-mismatch constant $C_{\mathrm{aff}}(\mathcal{D})$ generalising Proposition 3.1 to non-identical-copies constructions is open.

OQ0-B (L1.6 quantitative $\|\mathcal{G}_{\mathrm{aug}}\|$ theorem, CLOSED at $\mathcal{D}_0$). The promotion $K_{\mathrm{Sch}} \leq 16/3$ that L1.6 required factors as $(4/3) \cdot 3 \cdot (4/3)$ from three theorem-grade inputs: (i) the frozen normal β-conjugated σ-odd symbol multiplier $(1+i\tau)^j/p_m(\tau)$, $j = 0, 1, 2$, is bounded exactly by $4/3$ for $m \in \mathbb{Z}+1/2$, $\tau \in \mathbb{R}$ (worst case $m = 1/2, \tau = 0$; elementary algebra); (ii) the local Hölder/cutoff overhead is ≤ 3 by translation-invariance of the frozen normal inverse commuting with finite differences on the collar, combined with the moving-coordinate globality of §6 (single inversion in $\widetilde{w} = w - c(s)$ over the whole tube); (iii) an outward-rounded per-collar interval evaluation on the 77 collars of $\mathcal{D}_0$ certifies the structural bounds $C_{2s} \leq \|A_{\mathrm{loc}}^{-1}\| \cdot \kappa_g \cdot \|g''\|_\infty$, $C_{A5} \leq \mathrm{cond}(A_{\mathrm{bulk}})(1 + \kappa_g r_0)^2$ and $C_{\mathrm{curv}} \leq 1 + \kappa_g^2 r_0 + K_{S^3} r_0^2$ with maxima 0.500/2.356/1.010 vs targets 1/4/4, yielding $q_{\mathrm{coeff}} = 0.168$ (headline) /0.110 (sharp), slack 33–56% below the required 1/4. Reproducibility script: Data availability. Downstream, $\|\mathcal{G}_{\mathrm{aug}}\|_{\mathrm{crude}} \leq 25.945$ (headline $\kappa_E = 51/50$) / ≤ 24.780 (sharp $\kappa_E = 406/675$), both well below the scaffold 56.3, giving a factor ~ 2.1 of unbanked headroom on $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$. Caveats: (i) all seventy-seven per-collar readouts return the same interval by rank-one rigidity + ($A2_{\mathrm{collar}}$) affinity of $\mathcal{D}_0$ (round unlink, identical geometry across the split), each computed independently; (ii) the C_{2s} bound extends the rigidity identity by one derivative order at the write-up grade of Remark 1.1′ caveat 3 (independent re-derivation of $g'(\varphi) = 1 - \cos(\varphi/2)$ is Appendix-D-type interval work, absorbed by the 0.500-vs-1 slack); (iii) the normalisation is the $\|A_{\mathrm{loc}}^{-1}\| = 1$ column of Remark 1.1′.

OQ0-C (L2 convergent adiabatic reconstruction, CLOSED at $\mathcal{D}_0$). Four Neumann-slot theorem-grade bounds V/W/X/Y assemble on the product Banach space $X_{\mathrm{AR}} := X_\omega \times X_\lambda \times X_\mu \times X_\Theta$ under the product-max norm, natural companion of the normalisation $K_{\mathrm{AR}}^{\mathrm{prod}} = K_H^{K3} K_F = 2079/2000$. With exact rational slot bounds $q_{\mathrm{comm}} = C_{\mathrm{comm}}/R_{\mathrm{AR}} \leq 1.39 \cdot 10^{-3}$ ($C_{\mathrm{comm}} = 222453/40000$), $q_{\mathrm{proj}} \leq 8.14 \cdot 10^{-4}$ ($C_{\mathrm{proj}} = 65133/20000$), $q_{\mathrm{hodge}} \leq 3.92 \cdot 10^{-4}$ (Part B, $C_{\mathrm{hodge}} = 1253637/800000$; Part A is $\sup_x \|G_f(x)\| \leq 9/20$ via fibrewise Zhong–Yang + $D_{\max} = 7/5$), $q_{\mathrm{gauge}} \leq 4.57 \cdot 10^{-3}$ ($C_{\mathrm{gauge}} = 22839/1250$; reference cost 0 by $C_{\mathrm{conn},k} = 0$ canonical Donaldson gauge, transfer cost $(2 + 2K_v)(K_H^{K3} + K_v)$), and an independent AR-majorant contribution $q_{\mathrm{AR}} = 207/200000 = 1.035 \cdot 10^{-3}$ (Supplementary Appendix F), the Lipschitz bound is $\mathrm{Lip}(T_{\mathrm{AR}}) \leq 8.20 \cdot 10^{-3}$, a factor $\sim 60.9\times$ below the standard Banach contraction margin 1/2. Banach fixed-point gives a unique convergent reconstruction $(h_\varepsilon^*, \omega_\varepsilon^*, \lambda_\varepsilon^*, \mu_\varepsilon^*, \Theta_\varepsilon^*)$ on $\mathcal{D}_0$. Reproducibility script: Data availability. Caveats: (i) q_{hodge} Part A is tight at 0.98% margin at $D_{\max} = 7/5$ (the orbifold benchmark $K_H^{K3} \approx 0.19$ gives a further factor $2.37\times$); (ii) q_{gauge} is the tightest slot ($\sim 13.7\times$) and quadratic in the ($A5$) envelope K_v; (iii) the gluing is arithmetic on four already-certified slot bounds and an audited q_{AR} (Supplementary Appendix F).

OQ0-D (discharging (J)). The anisotropic three-scale refinement of Joyce's perturbation theorem is the subject of a companion paper. With L1.6 (OQ0-B) and L2 assembly (OQ0-C) closed, (J) is the sole remaining torsion-free hypothesis of the statement modulo the two structural slots of $\mathsf{H}_{\mathrm{global}}$ (smooth topology from [19] + [11]; geometric K3-fibration upgrade L4, see OQ0-E). At the datum layer Theorem A is $\mathsf{H}_{\mathrm{global}}$-independent: given an admissible datum $\mathcal{D}$, the closed small-torsion family closes without invoking $\mathsf{H}_{\mathrm{global}}$; the pack enters only when Corollary B is applied to the specific total space K_7 of Construction C.1.

OQ0-E (geometric upgrade of Construction C.1, L4). Construction C.1 produces the *smooth* compact simply-connected 7-manifold with the prescribed discriminant/monodromy topology, from [19, ch. 3] (Dehn–Seidel twist) and [11, §4] (independent K3-family gluing over disjoint tori). It does not produce the *geometric* K3-fibration with hyperkähler fibres and the collar-affine period section h of §9.1 consistent with $(A2_{\mathrm{collar}})$–$(A5)$. The coassociative K3-fibrations of [12] on compact G_2-manifolds are the closest template (Hopf-link discriminant, related monodromy, hyperkähler fibres); [6] gives further coassociative K3-fibrations from twisted connected sums; [13] analyses the K3-fibered adiabatic maximal equation; but no cited result specialises to the 77-unlink with the prescribed rank-one Picard–Lefschetz monodromy *and* the collar-affine periods of $(A2_{\mathrm{collar}})$. Discharging L4 requires either a direct existence proof (period-map surgery over the unlink complement compatible with a hyperkähler K3 family and the rank-one monodromy, e.g. via Torelli-controlled period-domain deformation plus the gluing data of Lemma C.3′) or a specialisation of Kovalev's construction to a 77-unlink discriminant with the prescribed monodromy, both open.

OQ1 (strict holonomy beyond $\mathcal{D}_0$). Corollary 8.4 establishes $\mathrm{Hol}(g_R) = G_2$ for $\mathcal{D}_0$ (conditionally on (J)) via $\pi_1(M^7) = 1$. For general $\mathcal{D}$, the strict-holonomy conclusion requires the topological condition $\pi_1(M^7) = 1$ (or the construction of an explicit non-flat parallel section to rule out $\mathrm{Hol} \subsetneq G_2$). Sufficient topological conditions on the Kovalev–Lefschetz construction that imply $\pi_1(M^7) = 1$ for a general branched rank-one admissible datum are not addressed in this work.

OQ2 (beyond rank-one monodromy). Dropping $(A1)$, i.e., allowing higher-rank monodromy, produces a higher-dimensional cokernel structure at the branch loci, with the displacement layer $\mathcal{D}^{\mathrm{disp}}_{1/2}$ of §5.3 replaced by a higher-rank section bundle over Σ. The s-uniformity reduction of §5.6 then operates in a higher-dimensional setting, with a non-trivial coupling matrix between distinct vanishing-cycle directions $\alpha_1, \alpha_2, \ldots$. The associated extended-domain Fredholm theorem and its reduction to $\mathbb{R}^{\dim \cdot N}$ are natural follow-up questions.

OQ3 (beyond split unlinks). Dropping $(A3)$, i.e., allowing linked components in the discriminant, introduces long-range entries in the Newton-kernel coupling matrix E_{link} of Theorem 5.10, with operator norm no longer bounded by $r_0/d_{\min}$ at the leading order. The Neumann series may still converge if the linking structure has controlled geometry, but the bound on $\|E_{\mathrm{link}}\|$ requires a global analysis on $B^3 \setminus \Sigma$ for the linked discriminant.

OQ4 (sharpening the threshold $R_0(\mathcal{D}_0)$). The threshold $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ of Corollary B (L1.6 closed (OQ0-B); coarser fallback $\leq 1.03 \cdot 10^4$) is tail-bound ($R_{\mathrm{threshold}} = (4\|\mathcal{G}_{\mathrm{aug}}\|^2 C_{\mathrm{nl}}\,\mathrm{remainder\_R3})^{1/3}$), so refinement must target the tail: (a) the quantitative $\|\mathcal{G}_{\mathrm{aug}}\|$ theorem, L1.6 already banks the factor ~ 2.1 improvement $1.03 \cdot 10^4 \to 4.9 \cdot 10^3$; the crude output of §11 (OQ0-B) $\|\mathcal{G}_{\mathrm{aug}}\| \leq 25.945$ (headline) is well below the scaffold 56.3, and by the 2/3 power scaling would push the headline further down toward

$\sim 2.9 \cdot 10^3$ if banked (not banked in the current headline for conservatism), (b) sharper ξ_1, ξ_2 bounds in `remainder_R3`, the dominant term $C_{\mathrm{nl}}\, \xi_1\, \xi_2$ is a worst-case composition and plausibly overshoots by an order of magnitude, (c) a genuinely smaller C_{nl} enclosure. The edge-side refinements (optimal Hölder α, native R_{curv}, sharper $K^{\mathrm{Maz}}_{\mathrm{Sch}}$) only matter once the tail is brought below the edge component $\approx 10^2$. A combined tail refinement would target $R_0(\mathcal{D}_0) \leq 10^3$.

11.1 Discussion

Plausibility of the anisotropic three-scale perturbation theorem (J)**.** The hypothesis (J) is a refinement of Joyce's small-torsion perturbation theorem [10, Th. 11.6.1] adapted to the collapsing $K3$-fibered regime, where three geometric scales, the base B^3 scale, the fibre $K3$ scale ε, and the edge tube scale r_0, are simultaneously present. In Joyce's original setting the fibre scale collapses uniformly and the perturbation-theoretic tools operate on a single anisotropic weighted Sobolev pair. The rank-one branched setting adds the third scale r_0 from the edge singular locus, which requires the standard tools to be refined either through the polyhomogeneous Mazzeo edge calculus (compatible with our §4–§6) or through the $K3$-fibered adiabatic analysis of [13] (with the collapsing-G_2 background of [7]). In both frameworks, the small-torsion condition $\|d^*\widetilde{\varphi}_R\| \leq CR^{-1}$ of Theorem 8.2 is precisely the input required for a Joyce-type deformation to close in the $K3$-fibred regime, and existing single-scale versions of the theorem [9, 2] indicate the estimate is plausible, without providing the two-parameter uniformity needed here. A three-scale companion paper is envisaged.

Why the threshold $R_0(\mathcal{D}_0) \sim 10^3$–$10^4$ is a meaningful bound. In the adiabatic-limit programme, the mere existence of a finite threshold $R_0 < \infty$ is the primary quantitative content: it separates the regime where the closed positive 3-form deformation *provably* closes from the regime where no argument is available. The specific value $R_0(\mathcal{D}_0) \leq 4.9 \cdot 10^3$ (L1.6 closed (OQ0-B); coarser fallback $\leq 1.03 \cdot 10^4$) is a concrete headline bound, not just an existence statement, which localises the boundary of the demonstrable regime and identifies the tail-contraction condition as its binding component (§9.2 and OQ4 above). The order-of-magnitude relevance of $R_0 \sim 10^{3\text{–}4}$ is that it sits comfortably below the exponentially-scaling barriers of the ε^3 smallness of Theorem 7.1, so the perturbation regime is genuinely non-empty; and it sits above the geometric edge component $R_{\mathrm{edge}}(\mathcal{D}_0) \approx 10^2$, so the analysis is honestly bottlenecked at the tail rather than at the collar. Sharpening to $R_0 \leq 10^3$ (OQ4) would move the threshold within an order of magnitude of the geometric floor, at which point the edge-side and tail-side refinements become simultaneously binding.

Scope: the certification is $\mathcal{D}_0$-only. The structural analytic content of the paper, the extended-domain Fredholm theorem (Theorem 5.5), the s-uniformity reduction (Theorem 5.10), the weighted parametrix and matching lemma (§6, Lemma 6.2), the nonlinear closure (Theorem 7.1), and the tail-contraction condition of Corollary B, holds for any branched rank-one admissible datum $\mathcal{D}$ satisfying $(A0)$–$(A8)$ + (E-quant) + (AR-quant) + (G-quant). The specific numerical certification of the three smallness conditions with margins $\sim 24\times, \sim 48\times, \sim 1389\times$ is performed only at the normalised datum $\mathcal{D}_0$ (77 disjoint round unlink components in S^3 of unit inter-tube distance, $r_0 = 10^{-2}$, $\mathrm{cond}(A_{\mathrm{bulk}}) \leq 2.31$, $D_{\max} = 7/5$; see Supplementary Appendix C). Two consequences follow. First, extending the numerical certification to other data, smaller r_0, larger N, or non-round unlinks with $\|\kappa_g\|_{C^0} \neq 1$, requires re-verifying the three smallness inequalities against the same universal constants of Supplementary Appendix E, which are datum-independent modulo the $(A5)$ condition-number input $\mathrm{cond}(A_{\mathrm{bulk}})$. Second, the topological identification $K_7 \to S^3$ with $b_2 = 21, b_3 = 77$ is specific to the $\mathcal{D}_0$ construction of Supplementary Appendix C; other rank-one Kovalev–Lefschetz constructions with different (N, Betti) topology satisfy the same conditional analytic framework but yield different total spaces (see [11, §4] for the general framework).

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